LINEAR IC APPLICATIONS

UNIT-I :

INTEGRATED CIRCUITS : Differential Amplifier - DC and AC analysis of Dual input Balanced output Configuration, properties of other differential amplifier configuration (Dual Input unbalanced Output, Single Ended Input - Balanced/unbalanced output), DC coupling and cascade Differential Amplifier stages, level translator, Basic current mirror circuit, Improved version of current mirror circuit.

UNIT-II :

CHARACTERISTICS OF OP-AMPS : Integrated circuits - Types, classification, Package Types and Temperature ranges, power supplies, Op-amp Block Diagram, ideal and practical Op-amp specifications, DC and AC characteristics, 741 op-amp & its features, Op-Amp parameters & measurements, Input and output off set voltages & currents, slew rate, CMRR, PSRR, drift, frequency Compens-ation techniques.

UNIT-III :

LINEAR AND NON-LINEAR APPLICATIONS of OP-AMPS: Inverting and Non-inverting amplifier, Integrator and differentiator, Differences amplifier, Instru-mentation amplifier, AC amplifier, V to I, I to V converters, Buffers, Non-linear function generation, Comparators, multivibrators, Triangular and square wave generators, Log and Anti log amplifiers.

FILTERS, OSCILLATORS: Comparison between passive and active networks, Design and Analysis of Butterworth active filters - 1st order, 2nd order LPF, HPF filters, Band Pass, Band reject and all pass filters, sample & Hold circuits. Precision rectifier, RC Phase shift oscillator, wien bridge oscillator.

UNIT-IV:

TIMERS & PHASE LOCKED LOOPS: Introduction to 555 timer, functional diagram, Monostable and Astable operations and applications, schmitt Trigger, PLL-introduction, block schematic, principles, and description of individual blocks, 565 PLL, Applications of PLL - frequency multiplication frequency translation, A.M, F.M, & FSK demodulators, Applications of VCO, fixed and variable voltage regulators.

UNIT-V:

DIGITAL TO ANALOG AND ANALOG TO DIGITAL CONVERTERS: Introduction, basic DAC techniques weighted resistors DAC, R-2R ladder DAC, inverted R-2R DAC, and Monolithic DAC, Different types of ADCs - parallel comparator type ADC, Counter type ADC, Successive approximation ADC and dual slope ADC, DAC and ADC specifications.

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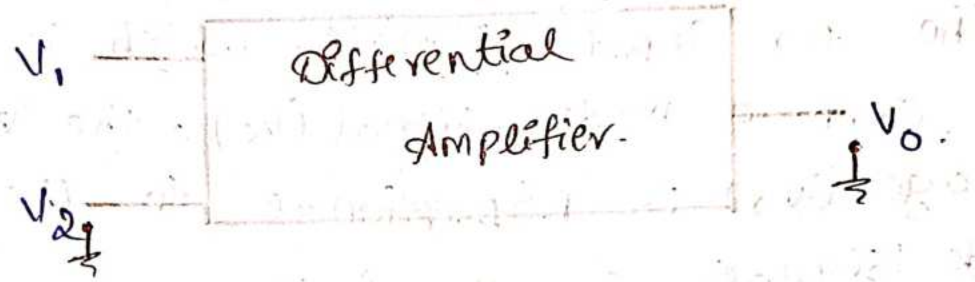
UNIT - I

Integrated Circuits

i, Differential Amplifier :- A differential amplifier is a two-input circuit that amplifies only the difference between its two inputs.
Ex :- operational amplifier (or) OP-amp.



$V_1, V_2 \rightarrow$ inputs signals.
$V_o \rightarrow$ output signal.

$V_o \propto (V_1 - V_2)$

ii, Differential gain (Ad) :- A differential gain is the gain with which a differential amplifier amplifies the difference between two input signals. It is denoted by Ad, which is equal to the output voltage (V_o) divided by the difference in voltage (V_d) between the two inputs (V_1 and V_2):

$V_o \propto (V_1 - V_2)$

$V_o = A_d (V_1 - V_2)$

$V_d = V_1 - V_2$

$V_o = A_d \cdot V_d$.

$$\boxed{A_d = \frac{V_o}{V_d}}$$

Ex:- $V_1 = 2V$
$V_2 = 1V$

$V_o = A_d (V_1 - V_2)$

$V_o = A_d (1V)$

$\because A_d = 10$

$V_o = 10 (1V)$

$V_o = 10V$.

$$A_d = 20 \log_{10} \left(\frac{V_o}{V_d}\right)$$

$A_d \rightarrow$ differential gain.
$V_d \rightarrow$ differential voltage.

iii, Common Mode Gain (A_c):- A common mode voltage gain is the amplification of signals that appear on both inputs of an operational amplifier (op- amp) relative to a common reference, usually ground. It is the average of the two input signals, which is called the common mode signal (V_c). the output voltage (V_o) is proportional to the common mode signal, so, $V_o = A_c \cdot V_c$.

$$V_1 = V_2 \Rightarrow V_o = V_1 - V_2 \Rightarrow V_o = 0.$$

$$V_c \propto \left(\frac{V_1 + V_2}{2}\right)$$

$$V_o = A_c\left(\frac{V_1 + V_2}{2}\right)$$

$$V_c = \frac{V_1 + V_2}{2}$$

$$\boxed{\therefore V_o = A_c \cdot V_c}$$

$$V_o = A_d \cdot V_d + A_c \cdot V_c.$$

iv, Common Mode Rejection Ratio (CMRR):- The ability of differential amplifier to reject a common mode signal is expressed by a ratio is called common mode rejection ratio. It is denoted by CMRR.

CMRR is defined as ratio of differential voltag gain to common mode Votage gain.

$$CMRR = \frac{A_d}{A_C}.$$

$$\boxed{\therefore \rho = \frac{A_d}{A_C}}$$

<u>Ideal case.</u>

$A_C = 0$

$$\rho = \frac{A_d}{A_C}$$

$$= \frac{A_d}{0}$$

$$= \infty$$

$\therefore CMRR = \infty.$

<u>Practical case</u>

A_C = small

A_d = High

$$\rho = \frac{A_d}{A_C}$$

$$= \frac{\text{High}}{\text{small}}$$

CMRR = High.

$$V_O = A_d V_d + A_C V_C.$$

$$= A_d \cdot V_d \left[1 + \frac{A_C V_C}{A_d V_d}\right]$$

$$= A_d V_d \left[1 + \frac{A_C}{A_d} \cdot \frac{V_C}{V_d}\right]$$

$$= A_d V_d \left[1 + \frac{1}{\frac{A_d}{A_C}} \cdot \frac{V_C}{V_d}\right] \quad \therefore \frac{A_C}{A_d} = \frac{1}{\frac{A_d}{A_C}}$$

$$= A_d V_d \left[1 + \frac{1}{CMRR} \cdot \frac{V_C}{V_d}\right]$$

* Features of Differential Amplifier :-

1. High differential voltage gain.
2. Low common mode gain.
3. High CMRR.
4. Two input terminals.
5. High input Impedance.

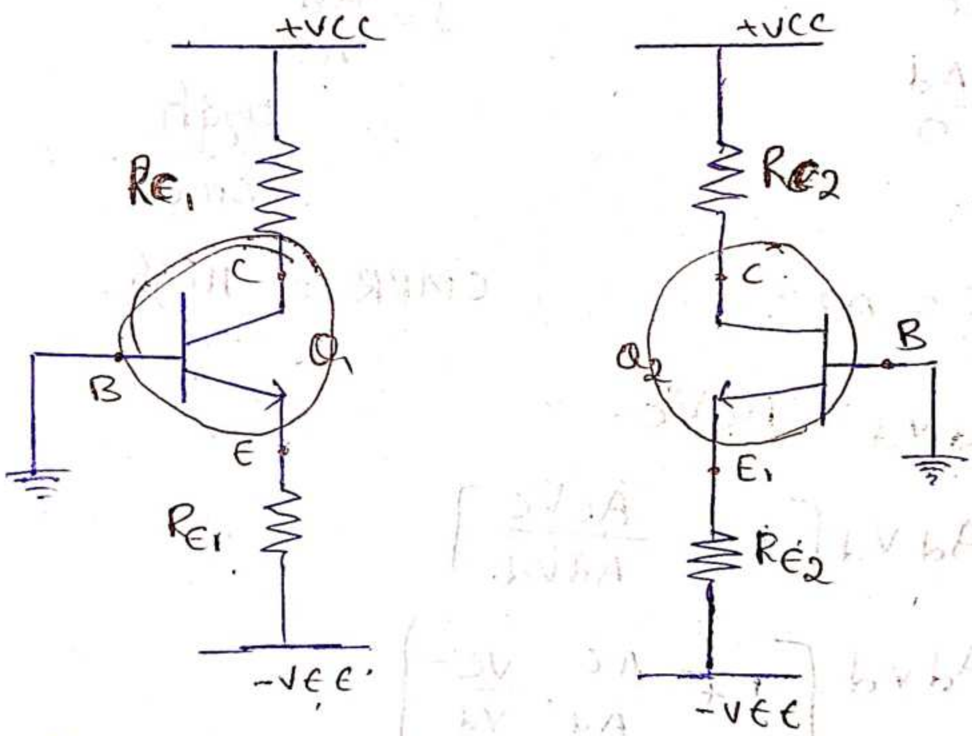
6. Low - output Impedance.

7. Large Bandwidth.

8. Low offset voltage and Current.

* If the input Signal, is high, and the output is low in Impedances, loading effect will be reduce.

* Construction of Differential amplifier.



Same properties.

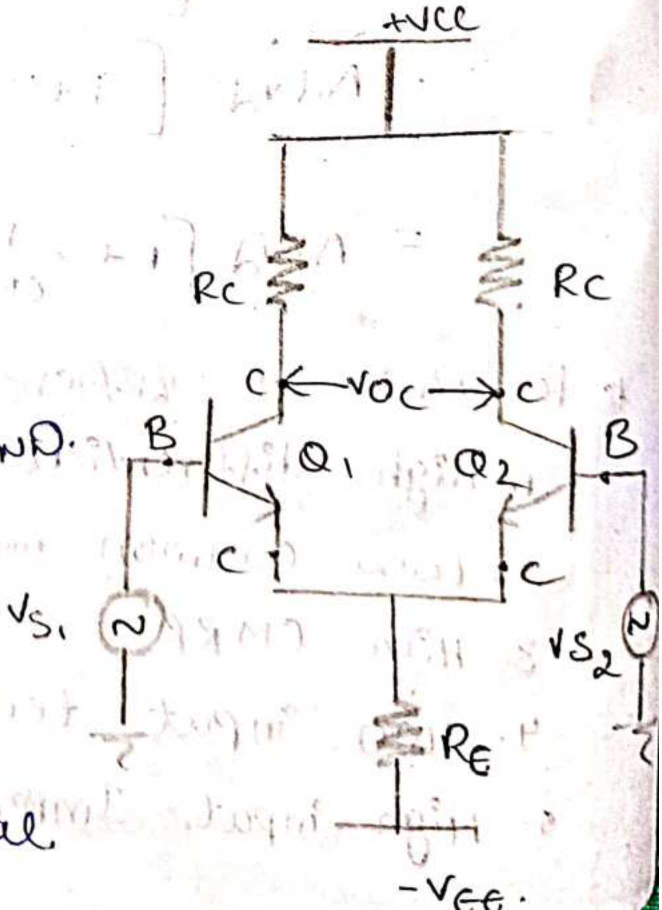
$Q_1 \parallel Q_2$.

$R_{E_1} = R_{E_2}$.

$R_{C_1} = R_{C_2}$.

$\underbrace{VCC, VEE}$ are same

with respect to GND.

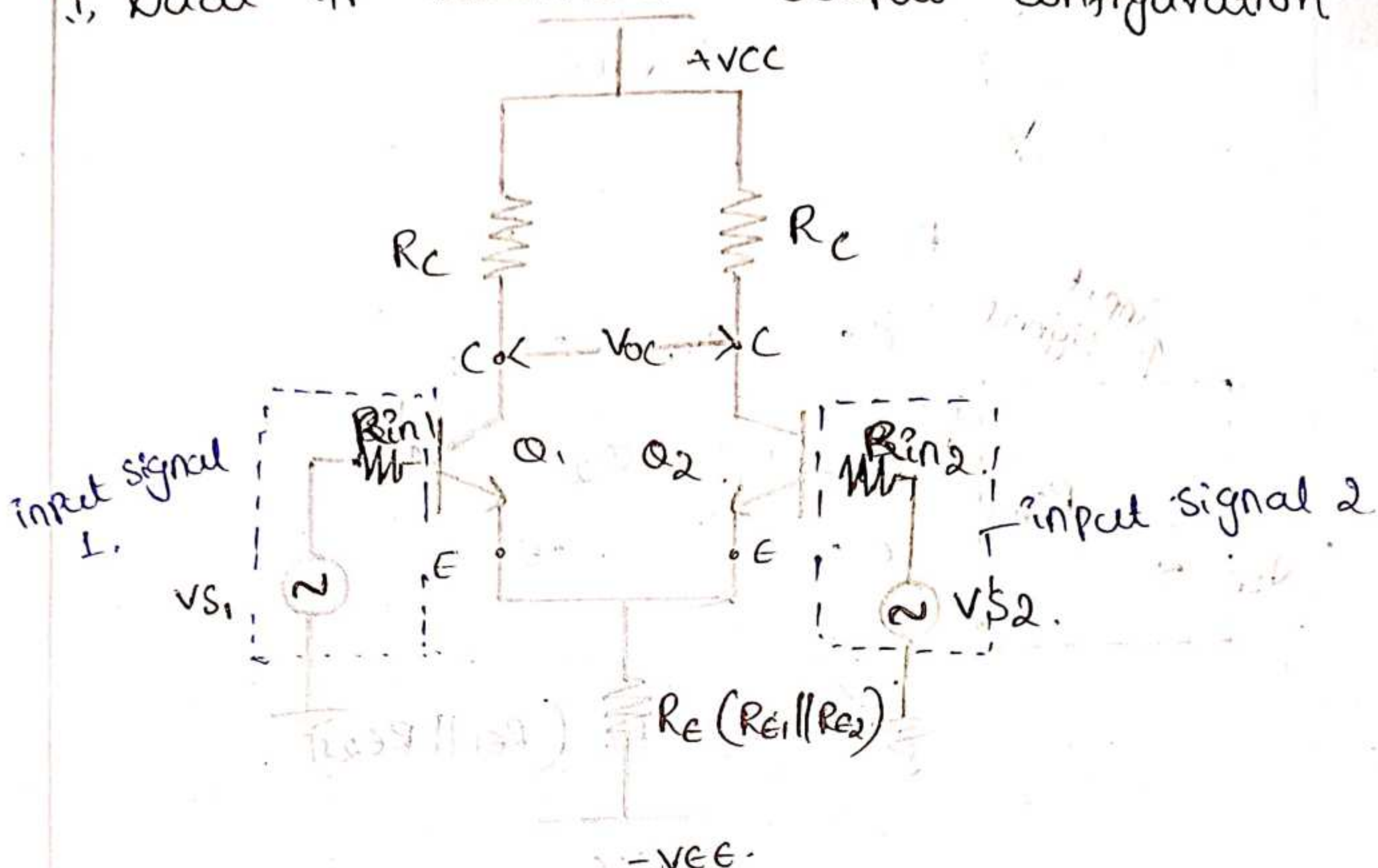


* V_{S_1}, V_{S_2} are the input signals.

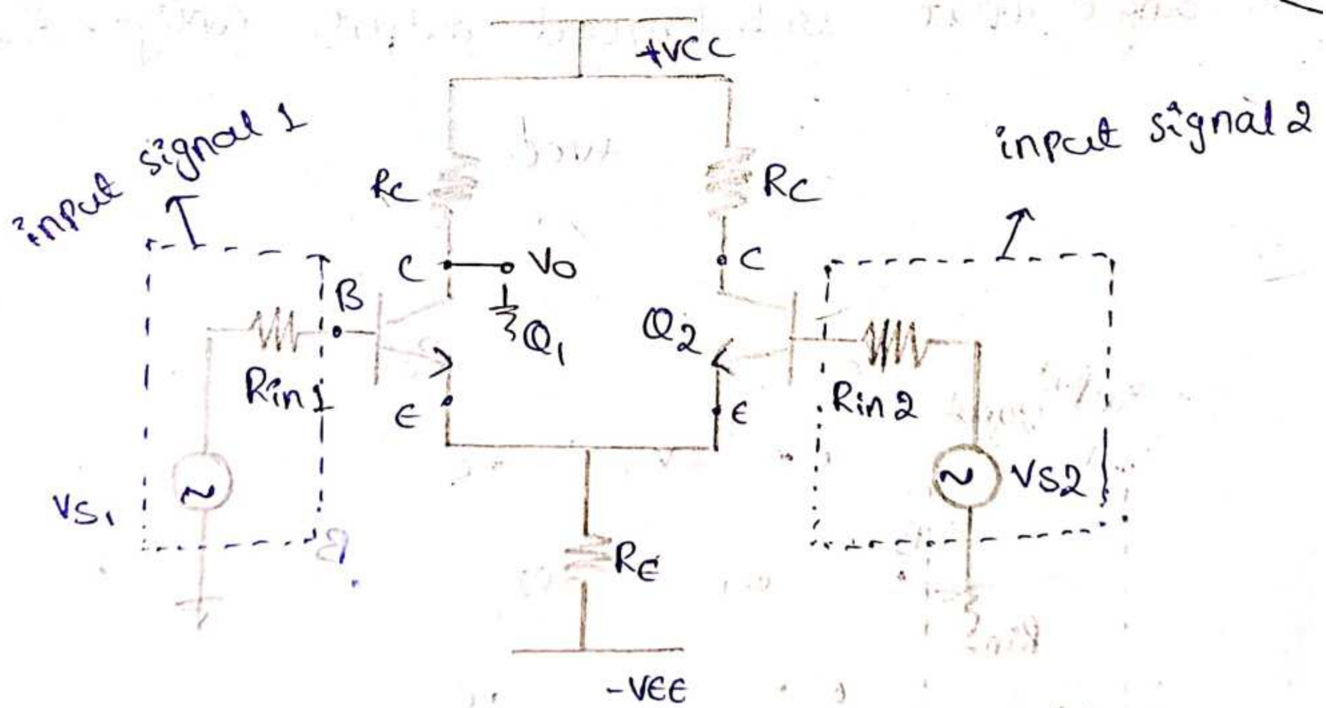
* Voc is the output signal.

* Configuration of Differential Amplifier.

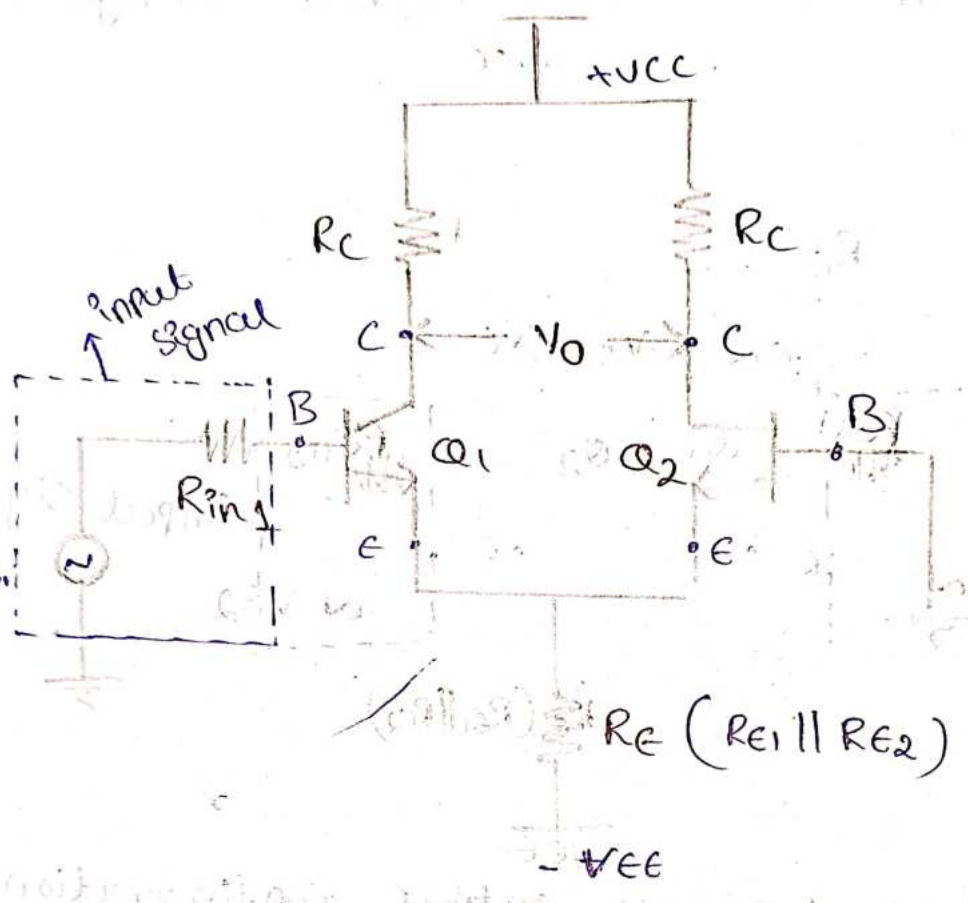
i, Dual I/P Balanced output configuration



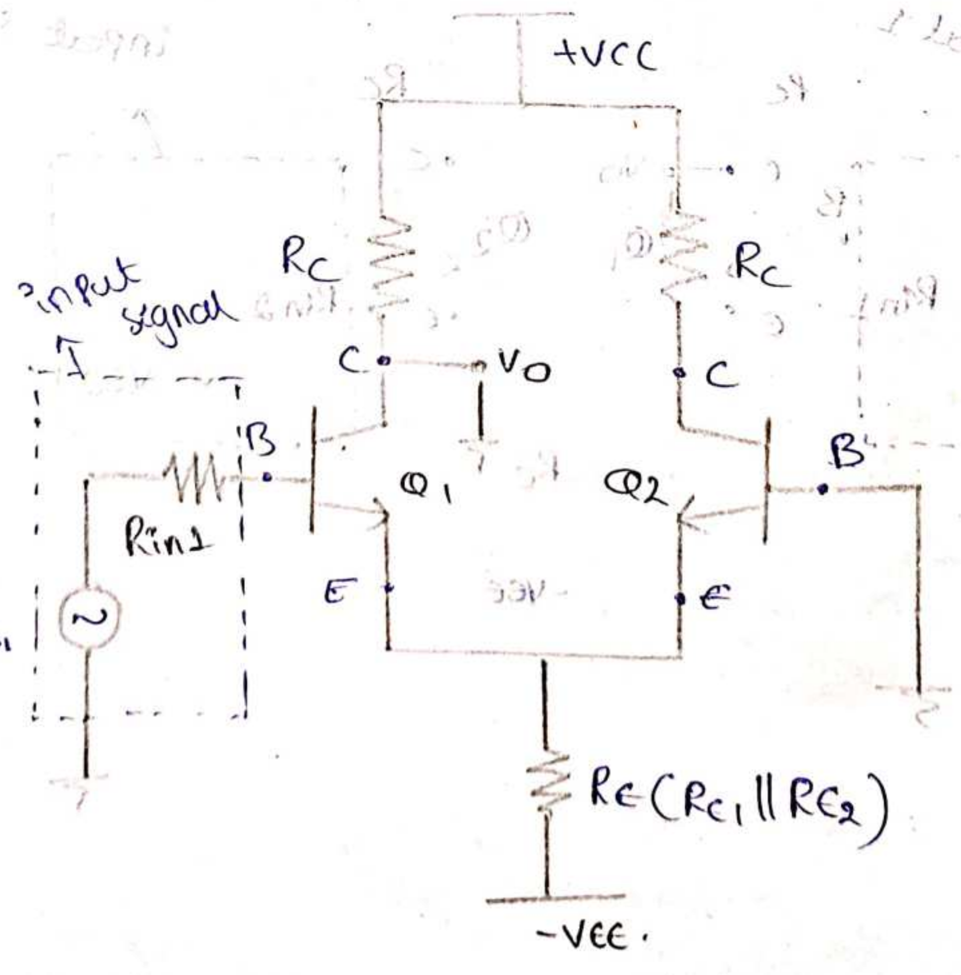
ii, Dual I/P unbalanced output configuration



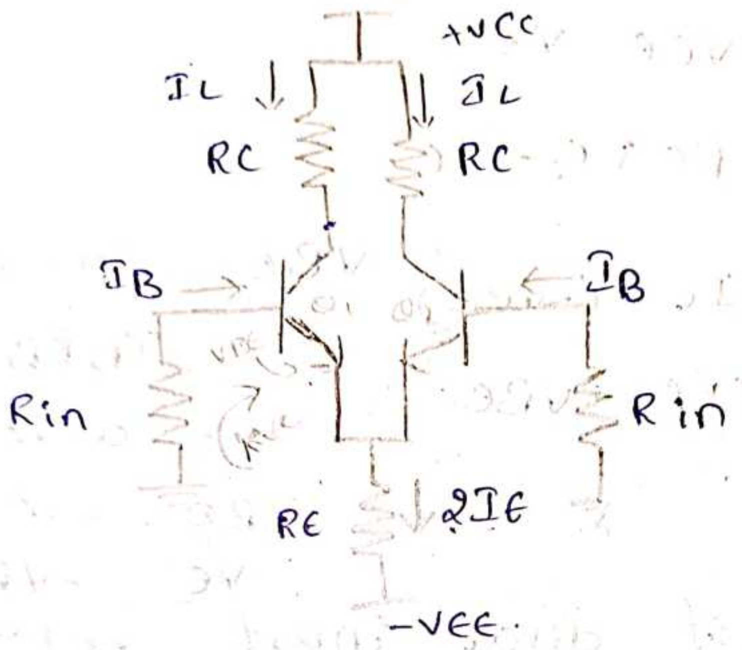
iii, Single input balanced output configuration



iv, Single input unbalanced output configuration



26/7/24 DC Analysis of dual input balanced output configuration.



Operating Point [VCEQ, ICQ]

$I_B R_{in} + V_{BE} + R_E 2I_E - V_{EE} = 0$ — ①

$\because I_B = \frac{I_C}{\beta}$ $\quad I_C \simeq I_E$

$I_B = \frac{I_E}{\beta}$.

Sub I_B value in ①.

$\frac{I_E}{\beta} R_{in} + V_{BE} + R_E \cdot 2I_E = V_{EE}$

$\frac{I_E}{\beta} R_{in} + 2I_E R_E = V_{EE} - V_{BE}$.

$I_E \left[\frac{R_{in}}{\beta} + 2R_E \right] = V_{EE} - V_{BE}$.

$I_E = \frac{V_{EE} - V_{BE}}{\frac{R_{in}}{\beta} + 2R_E}$.

input resistance is small, so neglect

$$\boxed{I_E = \frac{V_{EE} - V_{BE}}{2R_E}}$$

$V_C = V_{CC} - R_C I_C$

$\because V_{CE} = V_C - V_E$

$V_C = V_{CE} - V_E$

$V_{CE} = V_{CC} - R_C I_C - V_E$

$V_{CE} = V_{CC} - R_C I_C - (-V_{BE})$

$= V_{CC} - I_C R_C + V_{BE}$

$\because V_{BE} = V_B - V_E$

$= I_B R_B - V_E$

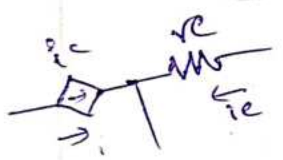
$= 0 - V_E$

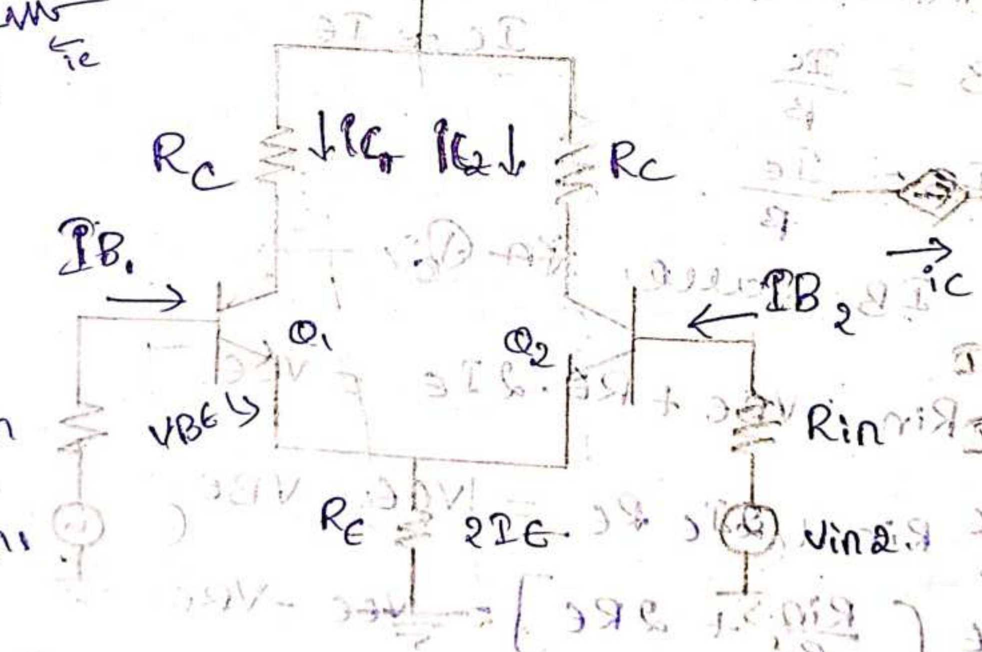
$V_{BE} = -V_E$

$V_E = -V_{BE}$

* AC analysis of dual input balanced output configuration.

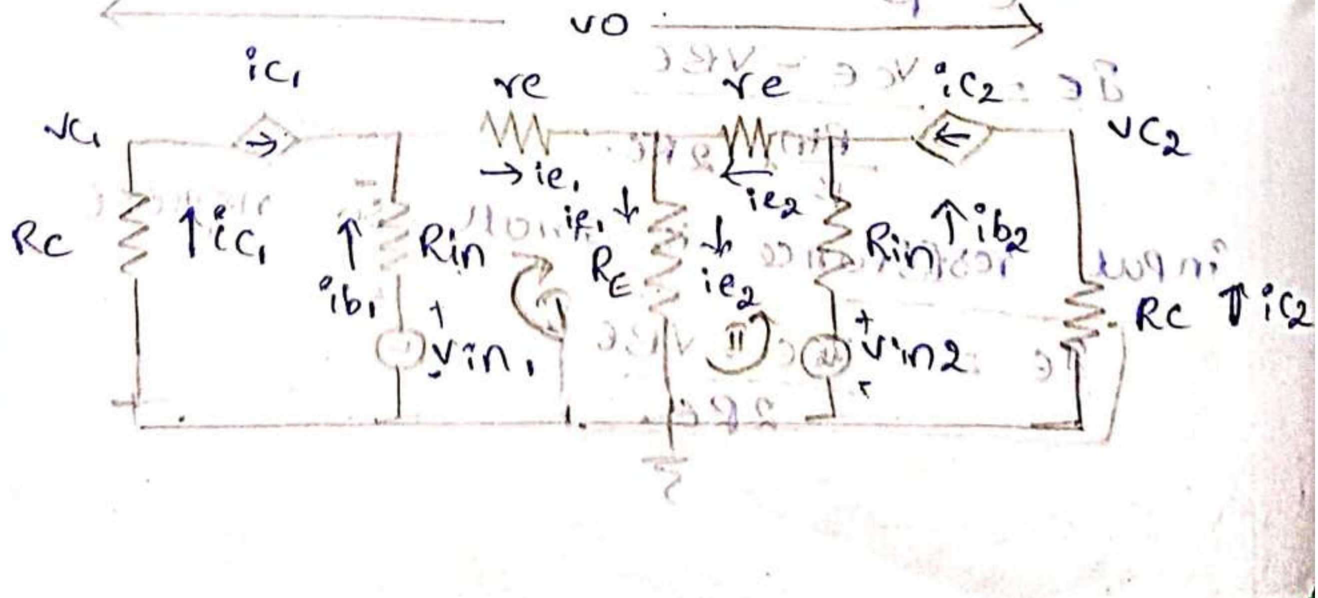
$V_{CC}, V_{EE} \Rightarrow$ Short / GND.





$I_B = I_C / \beta$

$I_C = I_B \cdot \beta$



Apply KVL at loop 1 :-

$-V_{in1} + R_{in} i_{b1} + r_e i_{e1} + R_E (i_{e1} + i_{e2}) = 0 \rightarrow ①$

Apply KVL at loop 2 :-

$-V_{in2} + R_{in} i_{b2} + r_e i_{e2} + R_E (i_{e1} + i_{e2}) = 0 \rightarrow ②$

Replace i_b with i_c/β.

from ①

$-V_{in1} + R_{in} \frac{i_{c1}}{\beta} + r_e i_{e1} + R_E (i_{e1} + i_{e2}) = 0 \rightarrow ③$

from ②

$-V_{in2} + R_{in} \frac{i_{c2}}{\beta} + r_e i_{e2} + R_E (i_{e1} + i_{e2}) = 0 \rightarrow ④$

$R_{in}/\beta \rightarrow$ very small value, so neglect.

3 & 4 Equations can be written as.

$-V_{in1} + r_e i_{e1} + R_E (i_{e1} + i_{e2}) = 0$

$\Rightarrow (r_e + R_E) i_{e1} + R_E i_{e2} = V_{in1} \rightarrow ⑤$

$-V_{in2} + r_e i_{e2} + R_E (i_{e1} + i_{e2}) = 0$

$\Rightarrow R_E i_{e1} + (r_e + R_E) i_{e2} = V_{in2} \rightarrow ⑥$

Cramers Rule.

$a_1 x + b_1 y = d_1$

$a_2 x + b_2 y = d_2$

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} ; \quad X = \frac{\begin{vmatrix} d_1 & b_1 \\ d_2 & b_2 \end{vmatrix}}{D} ; \quad Y = \frac{\begin{vmatrix} a_1 & d_1 \\ a_2 & d_2 \end{vmatrix}}{D}$$

i_{e1}, i_{e2} can be solved by using crammers rule.

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$$ie_1 = \frac{\begin{vmatrix} v_{in1} & R_E \\ v_{in2} & R_E + r_e \end{vmatrix}}{\begin{vmatrix} r_e + R_E & R_E \\ R_E & r_e + R_E \end{vmatrix}}$$

$$ie_1 = \frac{(v_{in1})(r_e + R_E) - (v_{in2})(R_E)}{(r_e + R_E)^2 - (R_E)^2}.$$

$$ie_2 = \frac{\begin{vmatrix} r_e + R_E & v_{in1} \\ R_E & v_{in2} \end{vmatrix}}{\begin{vmatrix} r_e + R_E & R_E \\ R_E & r_e + R_E \end{vmatrix}}$$

$$ie_2 = \frac{(r_e + R_E)\, v_{in2} - R_E\, v_{in1}}{(r_e + R_E)^2 - (R_E)^2}.$$

$$V_o = V_{C2} - V_{C1}.$$

$$= -i_{C2} R_C - (-i_{C1} R_C)$$

$$= i_{C1} R_C - i_{C2} R_C. \qquad V = IR.$$

$$I_C \simeq i_e$$

$$v_o = R_C\left[i_{C1} - i_{C2}\right].$$

$$= R_C\left[ie_1 - ie_2\right]$$

$$v_o = R_C\left[\frac{v_{in1}(r_e + R_E) - v_{in2}(R_E)}{(r_e + R_E)^2 - (R_E)^2} - \frac{(r_e + R_E)v_{in2} + R}{(r_e + R_E)^2 - (R}\right.$$

$$v_o = R_C \left[\frac{(r_e + R_E)(v_{in1} - v_{in2}) + R_E (v_{in1} - v_{in2})}{(r_e + R_E)^2 - R_E^2} \right]$$

$$= R_C \left[\frac{(v_{in1} - v_{in2})[r_e + R_E + R_E]}{(r_e + R_E)^2 - R_E^2} \right]$$

$$= R_C \left[\frac{(v_{in1} - v_{in2})(r_e + 2R_E)}{r_e^2 + \cancel{R_E^2} + 2 r_e R_E - \cancel{R_E^2}} \right]$$

$$= R_C \left[\frac{(v_{in1} - v_{in2})(r_e + 2R_E)}{r_e^2 + 2 r_e R_E} \right]$$

$$= R_C \left[\frac{(v_{in1} - v_{in2})\cancel{(r_e + 2R_E)}}{r_e \cancel{(r_e + 2R_E)}} \right]$$

$$= R_C \left[\frac{v_{in1} - v_{in2}}{r_e} \right]$$

$$= v_{in1} - v_{in2} \left(\frac{R_C}{r_e} \right)$$

$$v_o = V_d \left(\frac{R_C}{r_e} \right)$$

$$\frac{v_o}{V_d} = \frac{R_C}{r_e} \qquad v_o = A_d \cdot V_d.$$

$$\boxed{A_d = \frac{R_C}{r_e}}$$

↳ gain.

Input Resistance (R_{i1}, R_{i2}):

It is defined as equivalent resistance that measured that either of input terminals with other terminal grounded.

$$R_{i1} = \left| \frac{v_{in1}}{i_{b1}} \right|_{v_{in2}=0}$$

$$R_{i2} = \left| \frac{v_{in2}}{i_{b2}} \right|_{v_{in1}=0}$$

$$R_{i1} = \left| \frac{v_{in1}}{\frac{i_{e1}}{\beta_{ac}}} \right|_{v_{in2}=0}$$

$$R_{i1} = \left| \frac{v_{in1} \times \beta_{ac}}{i_{e1}} \right|_{v_{in2}=0}$$

$$R_{i1} = \left| \frac{v_{in1} \times \beta_{ac}}{\frac{v_{in1}(r_e+R_E) - R_E \cdot v_{in2}}{(r_e+R_E)^2 - R_E^2}} \right|_{v_{in2}=0}$$

$$= \left| \frac{(v_{in1} \times \beta_{ac})((r_e+R_E)^2 - R_E^2)}{v_{in1}(r_e+R_E) - R_E v_{in2}} \right|_{v_{in2}=0}$$

$$= \frac{(\cancel{v_{in1}} \beta_{ac})((r_e+R_E)^2 - R_E^2)}{\cancel{v_{in1}}(r_e+R_E)}$$

$$= \frac{\beta_{ac}(r_e^2 + \cancel{R_E^2} + 2 r_e R_E - \cancel{R_E^2})}{r_e + R_E}$$

$$= \frac{\beta_{ac}(r_e^2 + 2 r_e R_E)}{r_e + R_E}$$

$\because r_e << R_E$; $r_e << 2R_E$

$$= \frac{\beta_{ac}\, r_e (r_e + 2R_E)}{r_e + R_E}.$$

$$= \frac{\beta_{ac}\ r_e (2\cancel{R_E})}{\cancel{R_E}}$$

$$= \beta_{ac}\ r_e (2)$$

$$\boxed{R_{i_1} = 2 r_e \beta_{ac}} \; ; \; \boxed{R_{i_2} = 2 r_e \beta_{ac}}$$

output Resistance (R_{o_1}, R_{o_2}) :-

It is defined as the equivalent resistance that measured at either of 2 output terminals with other terminal is grounded.

$$R_{o_1} = \left| \frac{v_{o_1}}{i_{c_1}} \right|$$

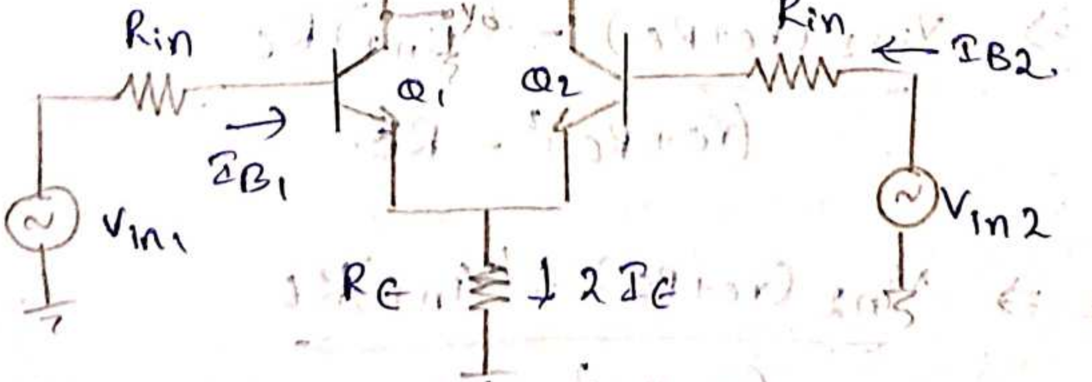
$$R_{o_2} = \left(\frac{v_{o_2}}{i_{c_2}} \right)$$

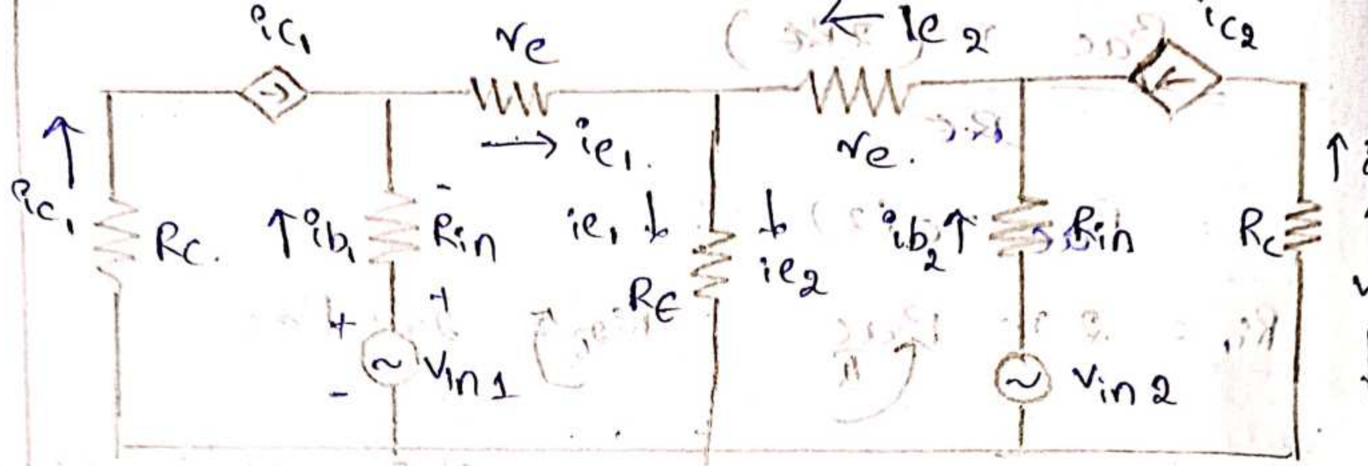
$$R_{o_1} = \left(\frac{-\cancel{i_{c_1}} R_C}{\cancel{i_{c_1}}} \right)$$

$$R_{o_1} = \left| -R_C \right|$$

$$\therefore \boxed{R_{o_1} = R_C} \; ; \; \boxed{R_{o_2} = R_C}$$

Dual input and unbalanced output configuration :-





Apply KVL at loop 1, we get

$$-V_{in1} + R_{in} i_{b1} + r_e i_{e1} + R_E (i_{e1} - i_{e2}) = 0 \rightarrow (1)$$

Apply KVL at loop 2.

$$-V_{in2} + R_{in} i_{b2} + r_e i_{e2} + R_E (i_{e1} - i_{e2}) = 0 \rightarrow (2)$$

Replace i_b with i_c/β.

$$(1) \Rightarrow R_{in}\frac{i_{c1}}{\beta} + r_e i_{e1} + R_E (i_{e1} + i_{e2}) = V_{in1} \rightarrow (3)$$

$$(2) \Rightarrow R_{in}\frac{i_{c2}}{\beta} + r_e i_{e2} + R_E (i_{e1} + i_{e2}) = V_{in2} \rightarrow (4)$$

$R_{in}/\beta \rightarrow$ neglect.

$$(3) \Rightarrow i_{e1}[r_e + R_E] + R_E i_{e2} = V_{in1} \rightarrow (5)$$

$$(4) \Rightarrow i_{e1} R_E + i_{e2}[r_e + R_E] = V_{in2} \rightarrow (6).$$

Solve 5 & 6 Equations using Cramers rule we get

$$i_{e1} \Rightarrow \frac{V_{in1}(r_e + R_E) - (V_{in2}) R_E}{(r_e + R_E)^2 - R_E}$$

$$i_{e2} \Rightarrow \frac{V_{in2}(r_e + R_E) - (V_{in1}) R_E}{(r_e + R_E)^2 - R_E}$$

$V_o = V_{C2}$.

$= -i_{C_2} \cdot R_C$

$= -i_{e_2} R_C$

$$= -R_C \left[\frac{v_{in_2}(r_e + R_E) - v_{in_1} R_E}{(r_e + R_E)^2 - R_E^2} \right]$$

$$= R_C \left[\frac{v_{in1} R_E - v_{in2}(r_e + R_E)}{(r_e + R_E)^2 - R_E^2} \right]$$

$$= R_C \left[\frac{v_{in1} R_E - v_{in2}(r_e + R_E)}{r_e^2 + 2R_E r_e + \cancel{R_E^2} - \cancel{R_E^2}} \right]$$

$$= R_C \left[\frac{v_{in1} R_E - v_{in2}(r_e + R_E)}{r_e (r_e + 2R_E)} \right]$$

$r_e << R_E$, $r_e << 2R_E$

neglect r_e.

$$v_o = R_C \left[\frac{v_{in1} R_E - v_{in2} R_E}{r_e (2R_E)} \right]$$

$$V_o = R_C \left[\frac{(v_{in1} - v_{in2}) \cancel{R_E}}{2 r_e \cancel{R_E}} \right]$$

$$v_o = \frac{R_C}{2r_e} [v_{in1} - v_{in2}]$$

$$v_o = \frac{\cancel{2} R_C}{2 r_e} v_d$$

$$\frac{v_o}{v_d} = \frac{R_C}{2 r_e}$$

$$\boxed{A_d = \frac{R_C}{2 r_e}}$$

Input Resistance (R_{i_1}, R_{i_2}) :

It is defined as Equivalent resistance that can be measure at either of 2 input terminals with other terminal as grounded

$$R_{i_1} = \left|\frac{v_{in1}}{i_{b1}}\right|_{v_{in2}=0}$$

$$R_{i_2} = \left|\frac{v_{in2}}{i_{b2}}\right|_{v_{in1}=0}$$

$$R_{i_1} = \left|\frac{v_{in1}}{\frac{i_{e1}}{\beta_{ac}}}\right|_{v_{in2}=0}$$

$$= \left|\frac{v_{in1} \times \beta_{ac}}{i_{e1}}\right|_{v_{in2}=0}$$

$$= \left|\frac{V_{in1} \times \beta_{ac}}{\frac{V_{in1}(r_e + R_E) + V_{in2} R_E}{(r_e + R_E)^2 - R_E^2}}\right|_{v_{in2}=0}$$

$$= \frac{[(V_{in1})(\beta_{ac})][(r_e + R_E)^2 - R_E^2]}{V_{in1}(r_e + R_E)}$$

$$= \frac{\beta_{ac}(r_e^2 + 2r_e R_E + \cancel{R_E^2} - \cancel{R_E^2})}{r_e + R_E}$$

$$= \frac{\beta_{ac}(r_e^2 + 2r_e R_E)}{r_e + R_E}$$

$$R_{i_1} \Rightarrow \frac{\beta_{ac}\, r_e\,(r_e + 2R_E)}{r_e + R_E}$$

$r_e \ll R_E$, $r_e \ll 2R_E$

$$R_{i_1} = \frac{\beta_{ac}\, r_e\,(2\cancel{R_E})}{\cancel{R_E}}$$

$$R_{i_1} = 2\beta_{ac} r_e .$$

$$\therefore R_{i_1} = R_{i_2} = 2 r_e \beta_{ac}$$

Output Resistance (R_{o1}, R_{o2})

It is defined as equivalent resistance that can be measure at one output terminal.

$$R_{o1} = \left|\frac{V_{o1}}{i_{c1}}\right|$$

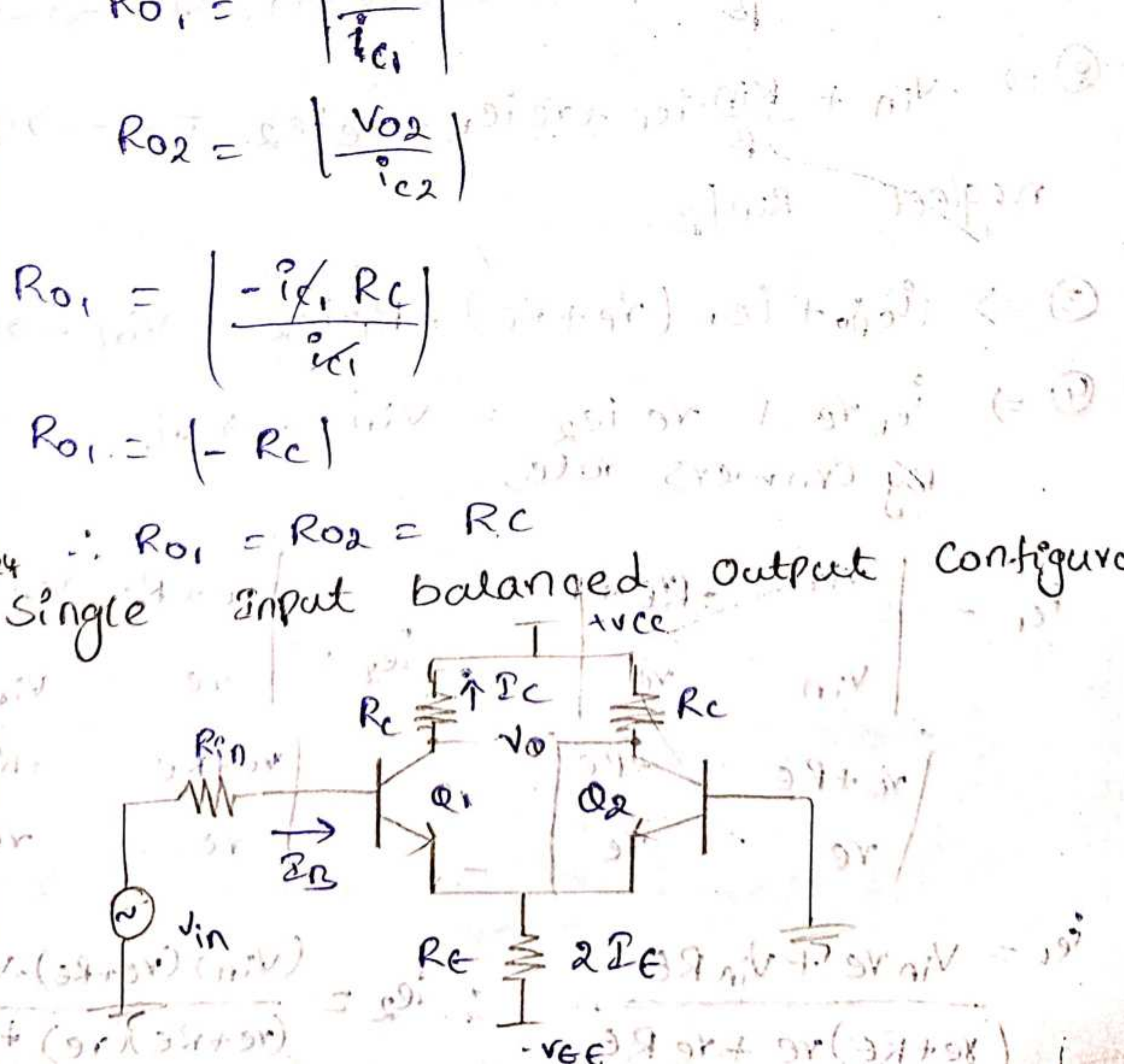
$$R_{o2} = \left|\frac{V_{o2}}{i_{c2}}\right|$$

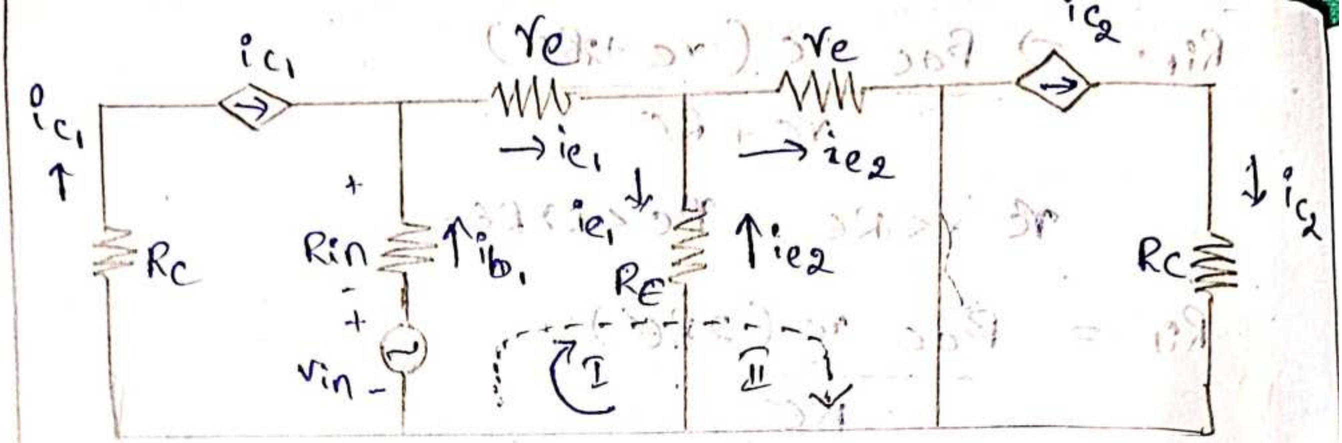
$$R_{o1} = \left|\frac{-\cancel{i_{c1}}\, R_C}{\cancel{i_{c1}}}\right|$$

$$R_{o1} = |-R_C|$$

$$\therefore R_{o1} = R_{o2} = R_C$$

24 Single input balanced output configuration





Apply KVL at loop 1:

$$-v_{in} + R_{in} i_{b_1} + r_e i_{e_1} + R_E (i_{e_1} - i_{e_2}) = 0 \rightarrow ①$$

Apply KVL at loop 2:

$$-v_{in} + R_{in} i_{b_1} + r_e i_{e_1} + r_e i_{e_2} = 0 \rightarrow ②$$

Sub $i_{b_1} = i_{c_1}/\beta$

$$① \Rightarrow -V_{in} + \frac{R_{in}}{\beta} i_{c_1} + r_e i_{e_1} + R_E (i_{e_1} - i_{e_2}) = 0 \rightarrow ③$$

$$② \Rightarrow -V_{in} + \frac{R_{in}}{\beta} i_{c_1} + r_e i_{e_1} + r_e i_{e_2} = 0 \rightarrow ④$$

neglect R_{in}/β.

$$③ \Rightarrow i_{e_1} (r_e + R_E) - R_E i_{e_2} = V_{in} \rightarrow ⑤$$

$$④ \Rightarrow i_{e_1} r_e + r_e i_{e_2} = V_{in} \rightarrow ⑥$$

By Cramers rule,

$$i_{e_1} = \frac{\begin{vmatrix} v_{in} & -R_E \\ V_{in} & r_e \end{vmatrix}}{\begin{vmatrix} r_e + R_E & -R_E \\ r_e & r_e \end{vmatrix}} ; \quad i_{e_2} = \frac{\begin{vmatrix} r_e + R_E & V_{in} \\ r_e & V_{in} \end{vmatrix}}{\begin{vmatrix} r_e + R_E & -R_E \\ r_e & r_e \end{vmatrix}}$$

$$i_{e_1} = \frac{V_{in} r_e + V_{in} R_E}{(r_e + R_E) r_e + r_e R_E} ; \quad i_{e_2} = \frac{(V_{in})(r_e + R_E) - V_{in} r_e}{(r_e + R_E)(r_e) + r_e R_E}$$

output voltage :

$$V_o = V_{C2} - V_{C1}$$
$$= i_{C2}R_C - (-i_{C1}R_C)$$
$$= i_{C1}R_C + i_{C2}R_C$$
$$= R_C[i_{C1} + i_{C2}].$$

$\because i_c \simeq i_e$.

$$V_o = R_C[i_{e1} + i_{e2}]$$

$$= R_C\left[\frac{V_{in}r_e + V_{in}R_E}{(r_e+R_E)r_e + r_eR_E} + \frac{(V_{in})(r_e+R_E) - V_{in}r_e}{(r_e+R_E)r_e + r_eR_E}\right]$$

$$= R_C\left[\frac{V_{in}(r_e+R_E)}{r_e(r_e+R_E+R_E)} + \frac{V_{in}(\cancel{r_e}+R_E-\cancel{r_e})}{r_e(r_e+2R_E)}\right]$$

$$= R_C\left[\frac{V_{in}(r_e+R_E+R_E)}{r_e(r_e+2R_E)}\right]$$

$$= R_C\left[\frac{V_{in}(\cancel{r_e+2R_E})}{r_e(\cancel{r_e+2R_E})}\right]$$

$$= R_C\left(\frac{V_{in}}{r_e}\right)$$

$$\frac{V_o}{V_{in}} = \frac{R_C}{r_e}.$$

$$\boxed{A_d = \frac{R_C}{r_e}}$$

Input Resistance

$$R_{i1} = \frac{V_{in}}{i_{b1}} \qquad i_{b1} = \frac{i_{c1}}{\beta} = \frac{i_{e1}}{\beta_{ac}}$$

$$R_{i1} = \frac{V_{in}.\beta_{ac}}{i_{e1}}. \Rightarrow \frac{V_{in}.\beta_{ac}}{\dfrac{V_{in}r_e + V_{in}R_E}{(r_e+R_E)r_e + r_eR_E}}$$

$$R_{i_1} = \frac{\cancel{V_{in}}\,\beta_{ac}\left[r_e (r_e + 2R_E)\right]}{\cancel{V_{in}}\,(r_e + R_E)}$$

$$= \frac{\beta_{ac}(r_e)\left[r_e + 2R_E\right]}{r_e + R_E}$$

$\because r_e << R_E$; $r_e << 2R_E$.

$$R_{i_1} = \frac{\beta_{ac}\, r_e (2\cancel{R_E})}{\cancel{r_e} + \cancel{R_E}}$$

$$\boxed{R_{i_1} = 2\beta_{ac}\, r_e}$$

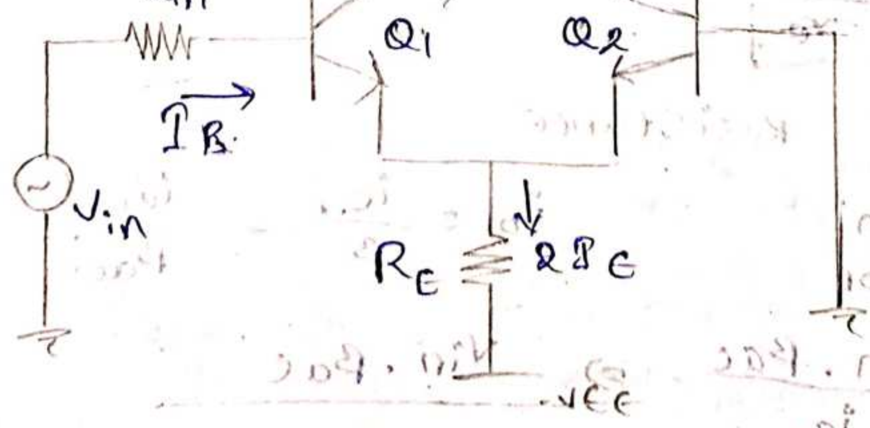
Output Resistance

$$R_{o_1} = \left|\frac{V_{o_1}}{i_{c_1}}\right| \; ; \quad R_{o_2} = \left|\frac{V_{o_2}}{i_{c_2}}\right|$$

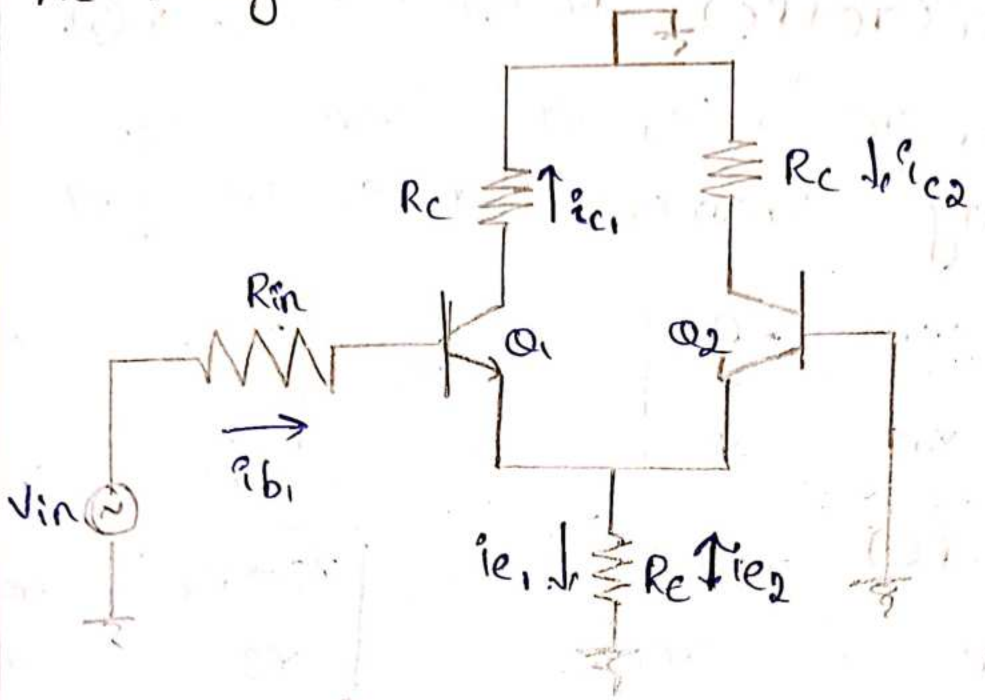
$$= \left|\frac{R_C(-\cancel{i_{c_1}})}{\cancel{i_{c_1}}}\right| \qquad = \frac{\cancel{i_{c_2}}\,R_C}{\cancel{i_{c_2}}}$$

$$\boxed{R_{o_1} = +R_C} \qquad \boxed{R_{o_2} = R_C}$$

Single input unbalance output configuratio



AC Analysis :



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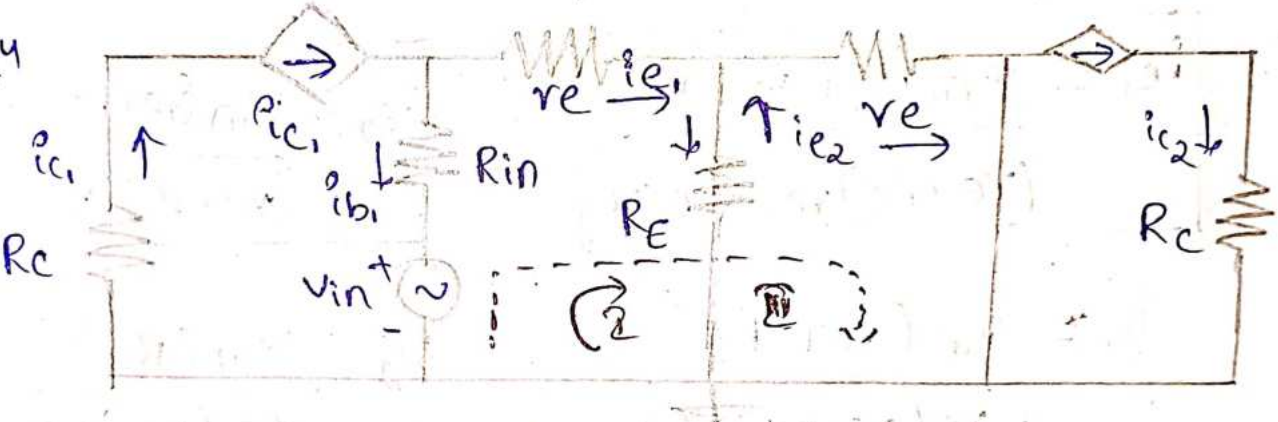


Apply KVL at loop-1

$$-V_{in} + R_{in} i_{b_1} + r_e i_{e_1} + R_E (i_{e_1} - i_{e_2}) = 0 \quad \text{—①}$$

Apply KVL at loop-2

$$-V_{in} + R_{in} i_{b_1} + r_e i_{e_1} + r_e i_{e_2} = 0 \longrightarrow ②$$

$$\because i_{b_1} = \frac{i_{c_1}}{\beta_{ac}}.$$

sub i_{b_1} value in Equ ① & ②

$$① \Rightarrow -V_{in} + \frac{R_{in}}{\beta_{ac}} i_{B_1} + r_e i_{e_1} + R_E (i_{e_1} - i_{e_2}) = 0 \rightarrow ③$$

$$② \Rightarrow -V_{in} + \frac{R_{in}}{\beta_{ac}} i_{c_1} + r_e i_{e_1} + r_e i_{e_2} = 0 \rightarrow ④$$

neglect $\frac{R_{in}}{\beta_{ac}}$. i.e. very small.

③ ⇒ $i_{e1}(r_e + R_E) - R_E\, i_{e2} = v_{in} \rightarrow$ ⑤

④ ⇒ $i_{e1} r_e + i_{e2} r_e = v_{in} \rightarrow$ ⑥

By using cramers rule, we get.

$$i_{e1} = \frac{\begin{vmatrix} v_{in} & -R_E \\ v_{in} & r_e \end{vmatrix}}{\begin{vmatrix} (r_e+R_E) & -R_E \\ r_e & r_e \end{vmatrix}} \quad ; \quad i_{e2} = \frac{\begin{vmatrix} r_e+R_E & v_{in} \\ r_e & v_{in} \end{vmatrix}}{\begin{vmatrix} r_e+R_E & -R_E \\ r_e & r_e \end{vmatrix}}$$

$$\boxed{i_{e1} = \frac{v_{in} r_e + v_{in} R_E}{(r_e+R_E) r_e + R_E \cdot r_e}} \qquad \boxed{i_{e2} = \frac{v_{in}(r_e+R_E) - v_{in} r_e}{(r_e+R_E) r_e + r_e R_E}}$$

$$i_{e1} = \frac{v_{in}(r_e + R_E)}{r_e(r_e+R_E+R_E)} \qquad i_{e2} = \frac{v_{in}(R_E)}{r_e(r_e+2R_E)}$$

$$i_{e1} = \frac{v_{in}(r_e + R_E)}{r_e(r_e+2R_E)}$$

* output voltage :-

$$V_o = V_{c2}$$
$$= i_{c2} R_C$$
$$= i_{e2} R_C.$$
$$= R_C \left[\frac{v_{in}(R_E)}{r_e(r_e+2R_E)}\right]$$

$r_e \ll 2R_E$. so neglect r_e.

$$V_o = R_C \left[\frac{v_{in} \cancel{R_E}}{r_e(2\cancel{R_E})}\right]$$
$$= R_C \left[\frac{v_{in}}{2r_e}\right]$$

$$\frac{v_o}{v_{in}} = \frac{R_C}{2r_e}$$

$$\therefore \frac{v_o}{v_{in}} = A_d$$

$$\boxed{\therefore A_d = \frac{R_C}{2r_e}}$$

Input Resistance:-

$$R_{in} = \frac{V_{in}}{i_{b1}} \qquad i_{b1} = \frac{i_{e1}}{\beta_{ac}}$$

$$= \frac{v_{in} \cdot \beta_{ac}}{i_{e1}}$$

$$= \frac{\cancel{v_{in}}\ \beta_{ac}}{\frac{\cancel{v_{in}}\,(r_e+R_E)}{r_e\,(r_e+2R_E)}}$$

$$= \frac{\beta_{ac}\ r_e\,(r_e+2R_E)}{r_e+R_E}$$

$r_e << R_E$; $r_e << 2R_E$

So neglect r_e.

$$R_{in} = \frac{\beta_{ac}\ r_e\,(2\cancel{R_E})}{\cancel{R_E}}$$

$$\boxed{R_{in} = 2\beta_{ac}\, r_e}$$

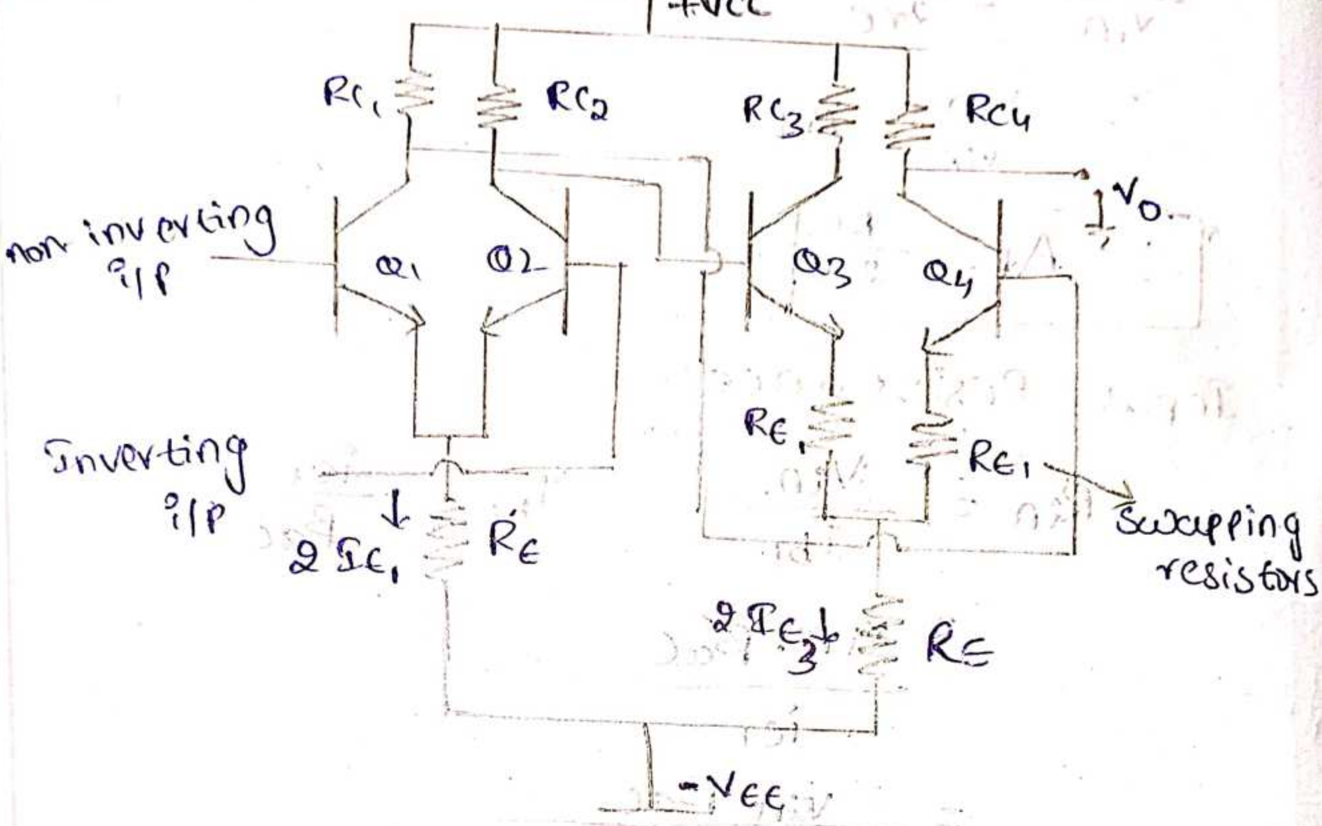
Output Resistance:-

$$R_o = \left|\frac{v_o}{i_{c1}}\right|$$

$$= \frac{\cancel{i_{c1}} \cdot R_C}{\cancel{i_{c1}}}$$

$$\boxed{\therefore R_o = R_C}$$

1/8/24 Cascaded Differential Amplifier...

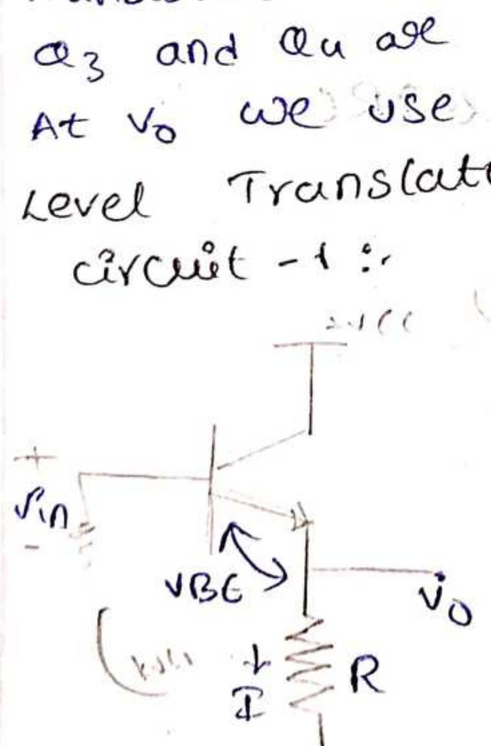


$RC_1 = RC_2$; $RC_3 = RC_4$.

* swapping resistors are used to reduce heat and output distrotions.
* Transistors Q_1 and Q_2 are similar. and Q_3 and Q_4 are also similar.
* At V_o we use level translater.

Level Translator :- To reduce VCC value.

circuit - 1 :-



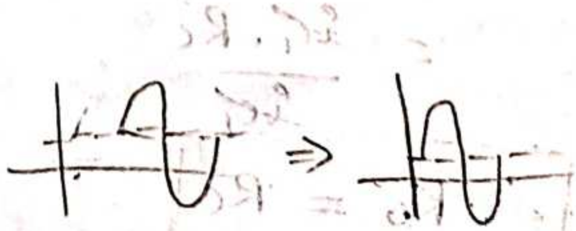
* Simple level shifting N/w

Apply KVL

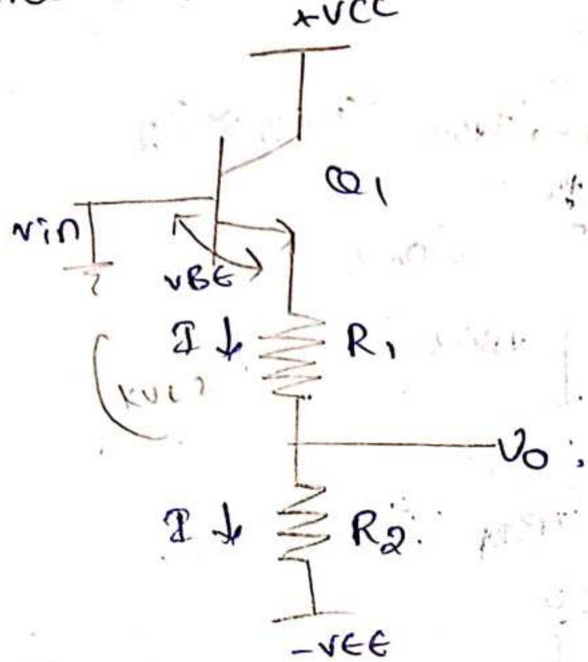
$0 = -V_{in} + V_{BE} + V_o$

$V_o = V_{in} - V_{BE}$

$V_o = V_{in} - 0.7V.$



circuit 2.



* voltage divider.

$$-V_{in} + V_{BE} + I(R_1+R_2) = 0.$$

$$I(R_1+R_2) = V_{in} - V_{BE}.$$

$$I = \frac{V_{in} - V_{BE}}{R_1+R_2}.$$

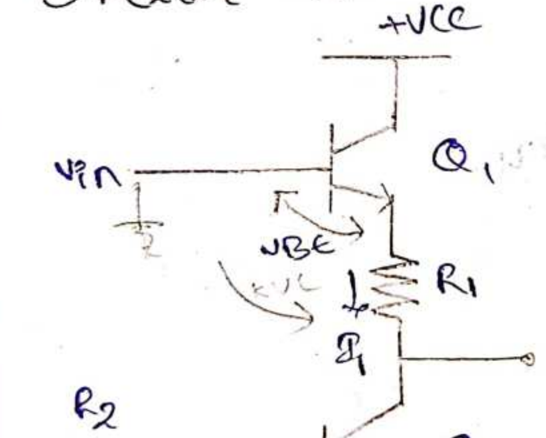
$$V_o = I \cdot R_2.$$

$$V_o = \left[\frac{V_{in}-V_{BE}}{R_1+R_2}\right] R_2.$$

Disadvantages :-

* $R_2 \downarrow = G \downarrow$.
* high output Impedance → It leads to increase loading effect.

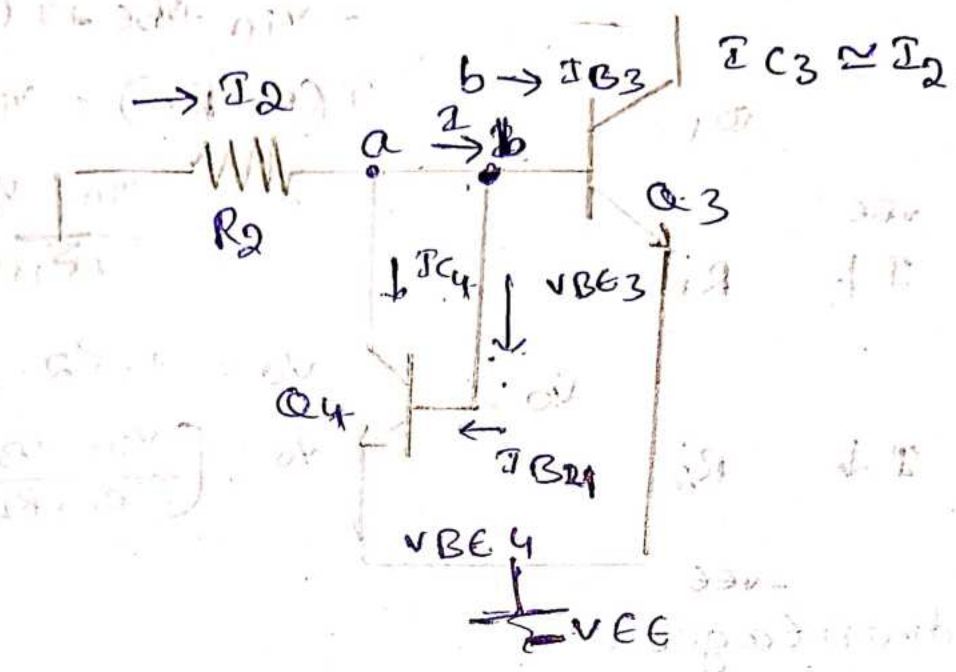
Circuit -3.

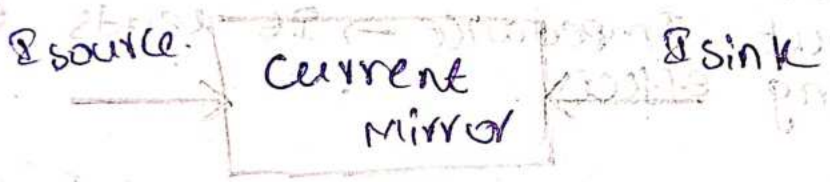


* constant current Bias.

$$V_o = V_{in} - V_{BE} - I_{R_1}.$$

2/8/24 Current Mirror circuit.





$I_{source} = I_{sink}$

Q_3 is similarly Q_4.

$I_{B3} = I_{B4}$,

$I_{C3} = I_{C4}$

$V_{BE3} = V_{BE4}$.

Apply KCL at node a,

$I_2 = I + I_{C4} \rightarrow (1)$

Apply KCL at node b,

$I = I_{B3} + I_{B4}$.

$= 2I_{B3}$ (or) $2I_{B4}$ $[\because I_{B3} = I_{B4}]$

$I_{B3} = \dfrac{I_{C3}}{\beta}$; $I_{B4} = \dfrac{I_{C4}}{\beta}$

$I_2 = 2I_{B3} + I_{C4}$

$= \dfrac{2I_{C3}}{\beta} + I_{C4}$

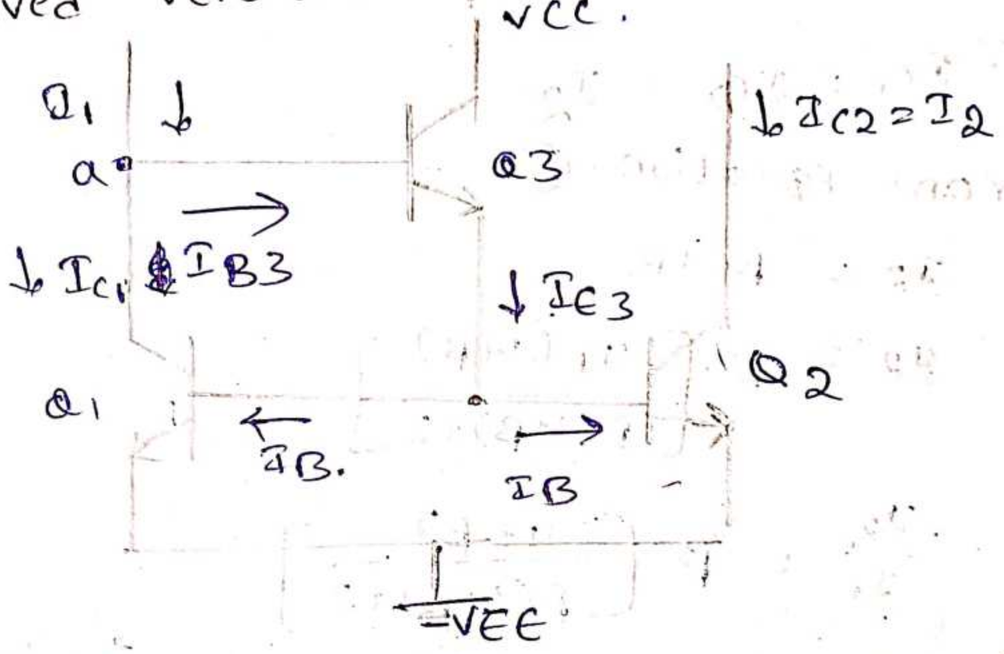
$$I_2 = \frac{2 I_{C3}}{\beta} + I_{C3}$$

$$= I_{C3} + \left(\frac{2}{\beta}\right) I_{C3}$$

$\frac{2}{\beta}$ is so small, neglect

$$\boxed{I_2 = I_{C3}}$$

Improved version of current mirror circuit



Apply KCL at node a

$$I_1 = I_{C1} + I_{B3} \rightarrow (1)$$

$$I_{E3} = I_B + I_B$$
$$= 2 I_B$$

$$\because I_E = I_C + I_B$$
$$= \beta \cdot I_B + I_B$$
$$= (1+\beta) I_B$$

$$I_{E3} = (1+\beta) I_{B3}$$

$$2 I_B = (1+\beta) I_{B3}$$

$$\boxed{I_{B3} = \left(\frac{2 I_B}{1+\beta}\right)}$$

$$\boxed{I_{C1} = \beta \cdot I_B} \rightarrow (2)$$

from equation ①

$$I_1 = \beta \cdot I_B + \frac{2 I_B}{(1+\beta)}$$

$$I_1 = I_B \left(\frac{\beta (1+\beta) + 2}{1+\beta} \right)$$

$$I_B = \frac{I_1 (1+\beta)}{\beta(1+\beta)+2}$$

$\because I_{C_1} = I_{C_2} = I_2$

from equation ②

$$I_2 = \beta \cdot I_B$$

$$I_2 = \beta \left[\frac{I_1 (1+\beta)}{\beta (1+\beta)+2} \right]$$

$$\frac{I_2}{I_1} = \beta \left[\frac{(1+\beta)}{\beta(1+\beta)+2} \right]$$

input and output current is independent on β value.

3/8/24

Problems

1. Determine the output voltage of a Differential amplifier for the input voltages of $V_1 = 300\,\mu V$; $V_2 = 240\,\mu V$. The differential gain of a amplifier is 5000. and the value of CMRR = i) 100 ii, 10^5.

i, CMRR = 100

$$V_d = V_1 - V_2$$

$V_d = 300 - 240$

$= 60\ \mu V$

$V_c = \frac{V_1 + V_2}{2}$

$= \frac{300 + 240}{2}$

$= 270\ \mu V.$

$CMRR = A_d / A_c$

$100 = \frac{5000}{A_C}$

$\therefore A_C = 50$

$V_o = A_d V_d + A_C V_C$

$= 5000(60 \times 10^{-6}) + 50(270 \times 10^{-6})$

$\boxed{V_o = 0.313 V.}$

ii, $CMRR = 10^5$

$V_d = V_1 - V_2$

$= 300 - 240$

$= 60\ \mu V$

$V_C = \frac{V_1 + V_2}{2}$

$= \frac{300 + 240}{2}$

$= \frac{540}{2}$

$= 270\ \mu V.$

$CMRR = A_d / A_C$

$10^{\cancel{5}\,2} = \frac{5000}{A_C}$

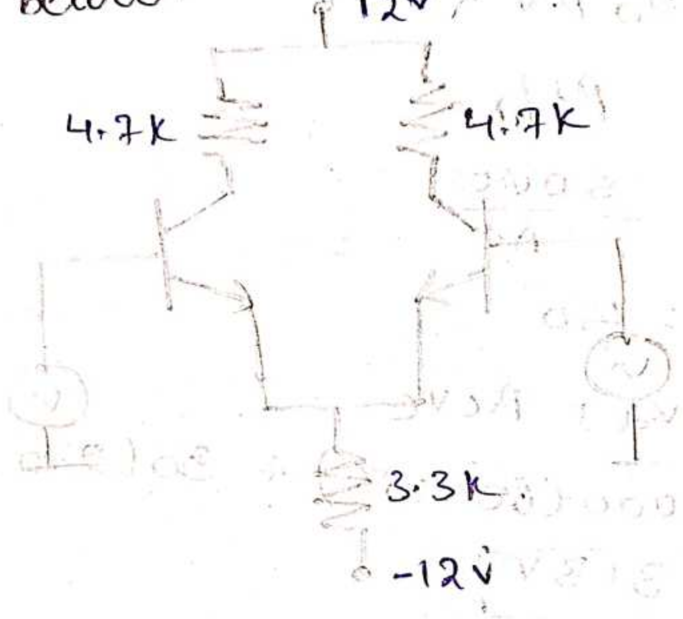
$A_C = \frac{5}{100} = \frac{1}{20} = 0.05.$

$V_o = A_d V_d + A_C V_C$

$= (5000)(60\times10^{-6}) + (0.05)(270\times10^{-6})$

$\boxed{V_o = 0.3V}$

2 Calculate operating point for the circuit shown below.



$R_C = 4.7K\Omega$

$R_E = 3.3K\Omega$

$V_{CC} = 12V$

$-V_{EE} = -12V$

$\therefore V_{EE} = 12V$

(I_{CQ}, V_{CEQ})

$I_E = \dfrac{V_{EE} - V_{BE}}{2R_E}$

$= \dfrac{12 - 0.7}{2\times3.3\times10^3}$

$I_E = 1.712\times10^{-3} A$

$\therefore I_C = 1.712 mA.$

$V_{CEQ} = V_{CC} + V_{BE} - I_C R_C$

$= 12 + 0.7 - (1.712\times10^{-3})(4.7\times10^{3})$

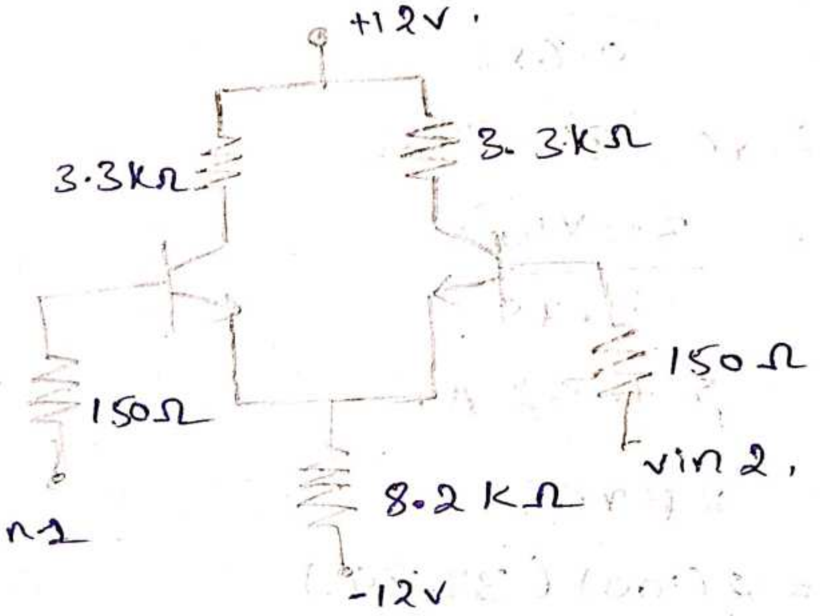
$= 12 + 0.7 - (1.712\times4.7)$

$V_{CEQ} = 4.65\,V$

$\therefore (I_{CQ}, V_{CEQ}) = (1.712\,mA, 4.65\,V)$

3. For differential amplifiers shown below. calculate i, operating point
ii, voltage gain.
iii, input/output Impedance.

Assume $\beta = 100$.



$R_C = 3.3\,k\Omega$

$R_E = 8.2\,k\Omega$

$R_{in} = 150\,\Omega$

$V_{CC} = 12V$

$V_{CE} = 12V$; $\beta = 100$.

(I_{CQ}, V_{CEQ})

$$I_E = \frac{V_{EE} - V_{BE}}{2R_C}$$

$$= \frac{12 - 0.7}{2 \times 8.2 \times 10^3}$$

$I_E = 0.688\,mA$.

$V_{CEQ} = V_{CC} + V_{BE} - I_C R_C$

$= 12 + 0.7 - (0.688 \times 10^{-3} \times 3.3 \times 10^3)$

$= 12 + 0.7 - (0.688 \times 3.3)$

$V_{CEQ} = 10.4296.$

$(I_{CQ}, V_{CEQ}) = (0.688mA, 10.42V)$

$$A_d = \frac{R_C}{r_e}$$

$$r_e = \frac{2.6mV}{I_E}$$

$$= \frac{26}{0.688}$$

$\therefore r_e = 37.79 \Omega$

$$A_d = \frac{3.3 \times 10^3}{37.79}$$

$= 87.32 A.$

$R_{i1} = 2 \beta r_e$

$= 2(100)(37.79)$

$= 7.558 k\Omega$

$R_O = R_C = 3.3k\Omega$

5/8/24

4. An operational amplifier has a differential gain of 80 dB and CMRR of 95 dB. If $V_1 = 2\mu V$, $V_2 = 1.6\mu V$ then calculate the differential and common mode output values

$A_d = 80 dB$

$CMRR = 95 dB.$

$V_1 = 2\mu V$

$V_2 = 1.6 \mu V.$

$A_d = 20 \log A_d$

$80\,dB = 20 \log A_d$

$A_d = 10^4$.

Differential output $= A_d V_d$.

$= 10^4 (2 - 1.6) \times 10^{-6}$

$= \boxed{4\ mV}$

$CMRR = A_d / A_C$,

$5.623 \times 10^4 = \dfrac{10^4}{A_C}$.

$A_C = \dfrac{10^4}{5.623 \times 10^4}$

$\boxed{A_C = 0.1778}$

CMRR in dB.

$95 = 20 \log CMRR$

$\dfrac{95}{20} = \log CMRR$

$CMRR = 10^{(95/20)}$

$\boxed{\therefore CMRR = 5.632 \times 10^4}$

Common mode output,

$= A_C \left(\dfrac{V_1 + V_2}{2} \right)$

$= 0.1778 \left(\dfrac{2 + 1.6}{2} \right) \times 10^{-6}$

$= (0.1778) \left(\dfrac{3.6}{2} \right) \times 10^{-6}$

$= \boxed{0.32\ \mu V}$

5. The common mode input to a certain differential amplifier having differential gain of 125 is $4 \sin 200\pi t$ V. Determine the common mode output if CMRR is 60 dB.

$A_d = 125$

$V_c = 4 \sin 200\pi t$ V.

$CMRR = 60\,dB$.

$A_C V_C = ?$

$60\,dB = 20 \log CMRR$

$CMRR = 10^3 = 1000.$

$\because CMRR = Ad/Ac \Rightarrow$

$$Ac = \frac{125}{1000} \quad \frac{Ad}{CMRR}.$$

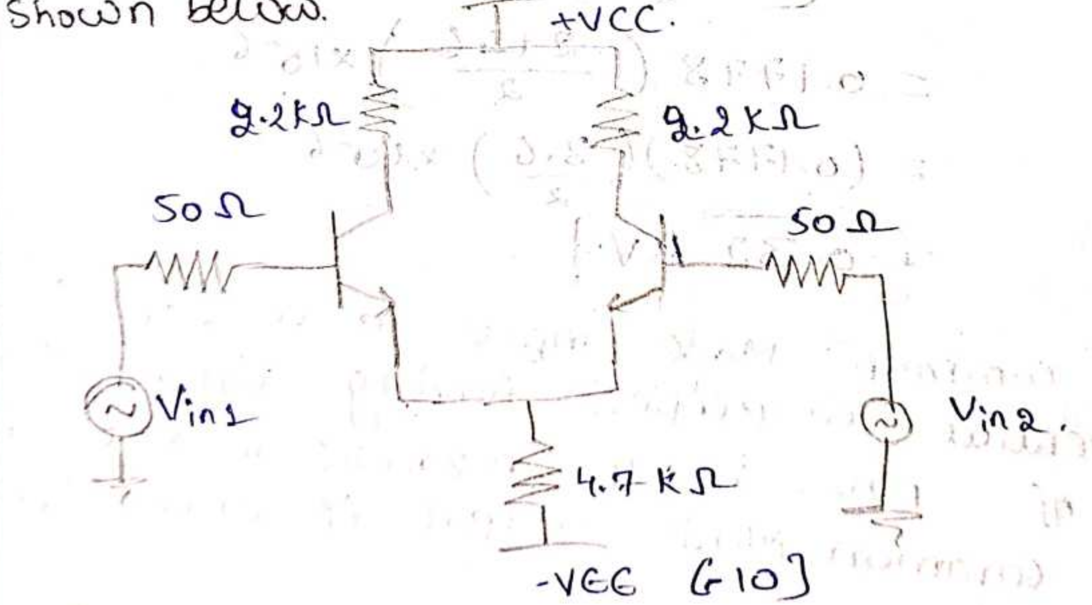
$$\Rightarrow \frac{Ad}{CMRR} = 0.125$$

$$\therefore \boxed{Ac = 0.125}$$

$$Ac \cdot Vc = 0.125 (4 \sin 20\pi t)$$

$$\therefore AcVc = \boxed{0.5 \sin 200\pi t.}$$

i, Determine the Q point for each transistor.
ii, Determine the voltage gain
iii, Determine input and output resistances
iv, Determine output voltage if $V_{in1} = 50$ mv. and $V_{in2} = 20$ mv. For the differential amplifier shown below.



[10]

$$(I_{CEQ}, V_{CEQ}) = (0.987\,mA, 8.57V)$$

$$I_E = \frac{V_{EE} - V_{BE}}{2R_E}.$$

$$I_E = \frac{10 - 0.7}{2(4.7 \times 10^3)} = 0.987\,mA.$$

$$V_{CEQ} = V_{CC} + V_{BE} - I_C R_C$$

$$= 10 + 0.7 - (0.987 \times 2.2)$$

$$= 8.57V.$$

ii, $A_d = \dfrac{R_C}{r_e}$

$$r_e = \frac{26\,mV}{I_E}$$

$$= \frac{26 \times \cancel{10^{-3}}}{0.987 \times \cancel{10^{-3}}}$$

$$= \frac{26}{0.98}$$

$$r_e = 26.34\,\Omega$$

$$A_d = \frac{2.2 \times 10^3}{26.34}$$

$$= 83.52.$$

iii, $R_i = 2\beta r_e$ $\quad \because \beta = 100$

$$= 2 \times 100 \times 26.34$$

$$= 5.268\,k\Omega$$

$$R_o = 2.2\,k\Omega$$

iv, $V_o = I_C R_C$

$$= 0.987 \times 2.2$$

$$V_o = 2.17V$$

8/8/24

UNIT-2

CHARACTERISTICS OF OP-AMP

Advantages of Integrated circuit Technology :-

① Small size:-
The size of an IC is 1000 times smaller than discrete circuit.

② Low cost :-
1000's of silicon chips consisting number of components are produced simultaneously, is known as Mask Protection. Due to this IC's are available in low cost.

③ weight :-
Even though number of components are fabricated on single silicon chip weight is very less compared to descrete circuit.

④ Low supply voltages :-
IC's work at lower voltages.

⑤ Power consumption :-
The power consumption of IC is very low.

⑥ Highly Reliable :-
Due to the absence of soluring techniques and inter connections, the IC's are highly realiable.
Due to low Power consumption, the

raise of temperature is very low which prolong the life of an IC.

(7) Speed of Response :-

speed of response is high.

Disadvantages of IC's Technology :-

Classification of IC's :-

- IC
  - Linear
    - Monolithic Technology
      - Bipolar [using BJT]
        - using P-N Junction isolation Technique.
        - using Dielectric isolation Technique
      - Unipolar [using FET]
        - MOSFET
        - JFET.
    - Thick & Thin film Technology
    - Hybrid Technology
  - Digital.

9/8/24 Monolithic Technology.

The term "monolithic technology" refers to the process of fabricating all the components of the circuit including transistors, resistors, capacitor and interconnections on a single semiconductor substrate, typically silicon. where individual component are assembled on a printed circuit board.

Thin and Thick film Technology :-

Thin film Technology :- Thin film technology involves depositing a very thin layer of conductive, resistive, or insulating material onto a subtrate, typically using techniques like vaccum deposition, etc. The film thickness is usually in the range of a few nanometers to micrometers. It is more expensive.

Thick film Technology :- Thick film technology involve screen printing or stenciling pastes containing conductive, resistive or dielectric materials onto a substrate. The layers are typically much thicker than those in thin films, ranging from a few micrometers to several tens of micrometers. less in price.

In linear IC

Thin film: used when high accuracy is needed.

Thick film: used when durability and lower costs are more important.

Hybrid Technology :- It refers to the combination of different types of components such as integrated circuit, and desatele circuit components to create a single function unit.

Types of Package :-
1. Metal Can Package
2. Dual - in - line Package.
3. Flat Package.

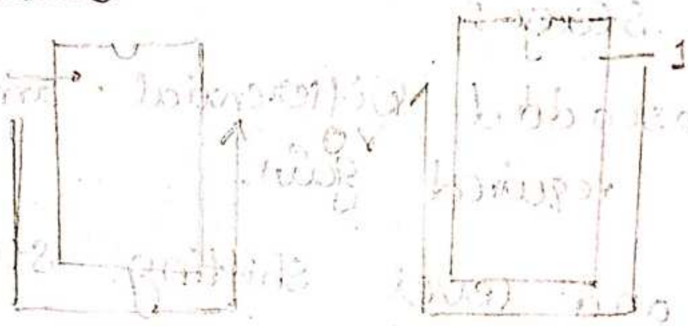
Type		Criteria.
Metal Can Package.		① Heat dissipation is important. ② used in high power applications like power-amplifier Voltage regulator.
Dual - in - line Package.		① for Experimenal or bread boarding purpose, we use.
Flat Package.		① More reliability is required. ② light in weight ③ suitable for airborne applications.

Level of Integration	No. of components used.
1. Small Scale Integration [SSI]	100
2. Medium Scale Integration [MSI]	100 - 1000
3. Large Scale Integration [LSI]	1000 - 10,000
4. Very Large Scale Integration [VLSI]	10,000 - 1,00,000
5. Ultra Large Scale Integration [ULSI]	above million

Features

* Factors affecting in selection of IC Package.
1. Relative Cost [DIP]
2. Reliability [flat Package]
3. weight of Package [less - flat Pack, o.w - Metal Can]
4. Ease of IC fabrication [DIP]
5. Power to be dissipated [Metal Can Pack]
6. Need of External heat sink [DIP, flat Packs].

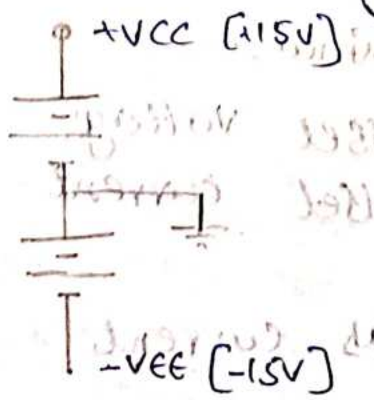
* Pin Identification :-



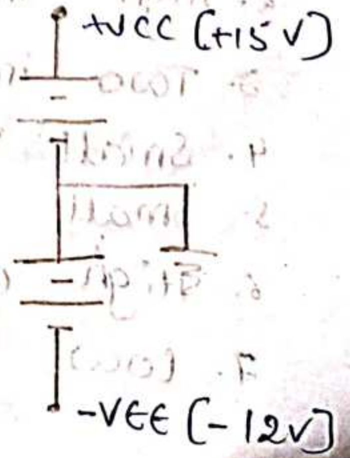
* Temperature Ranges :-
1. Military Temperature range.
   -50° to 125°C [-55° to +85°C]
2. Industrial Temperature range.
   -20° to 85°C [-40°C to +85°C]
3. Commercial Temperature range
   0°C to 70°C [0° to 75°C] - life span - ↓

* Power Supply

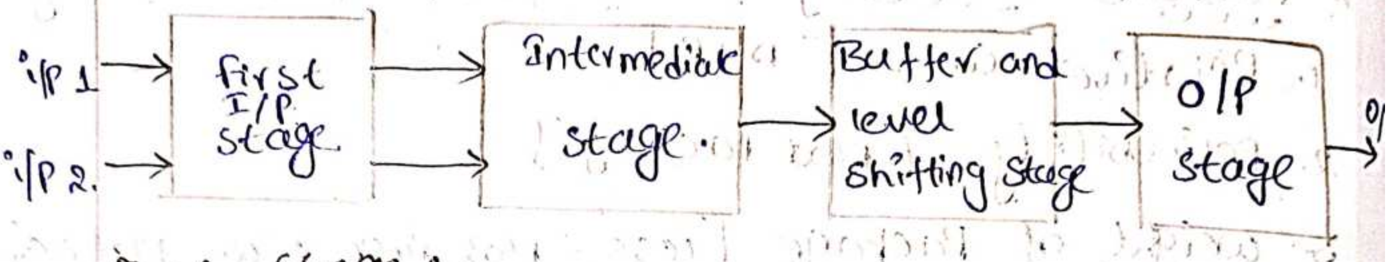
Balanced Power Supply



UnBalanced Power Supply



10/8/24 Block Diagram of Operational Amplifier



First stage :-

D.I.B.O Differential Amplifier.

High input impedance [∵ reduce load Effect]

High gain.

second Stage :-

cascaded Differential amplifier to attain required gain.

Buffer and level shifting stage :-

DC level to maintain ground.

output stage :-

low O/P impedance, voltage.

To maintain the above requirements, we use Darlington Amplifier.

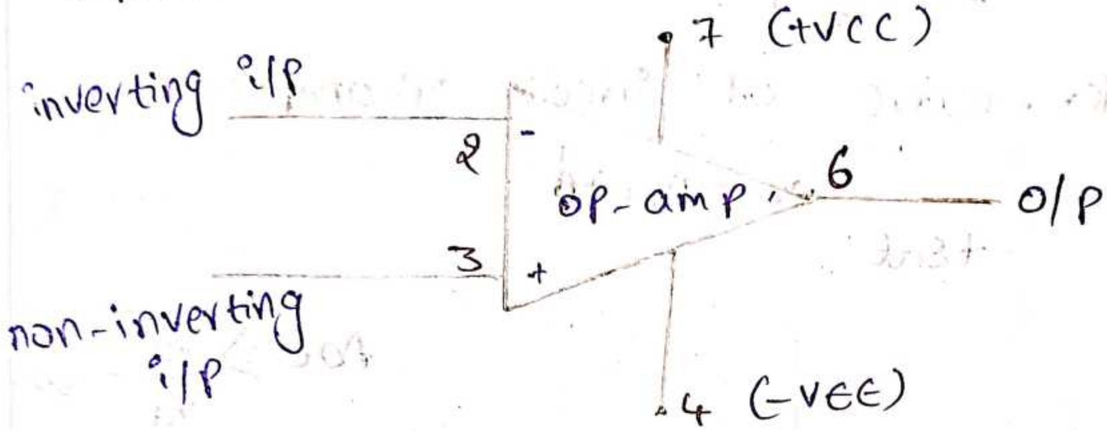
Requirement of input stage of OP-amp:-

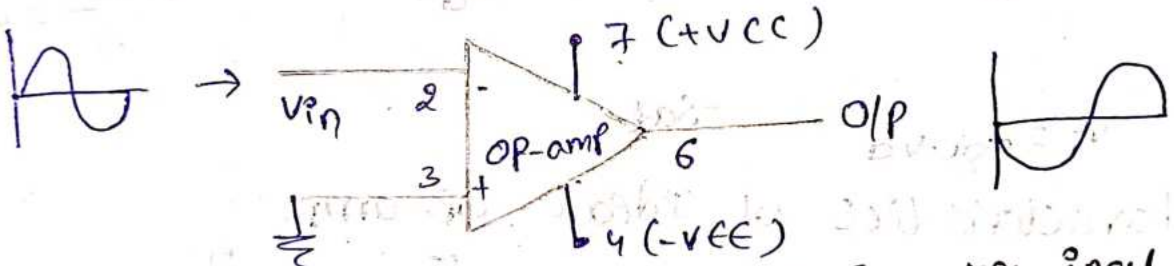
1. High voltage gain.
2. High input Impedance.
3. Two input Terminals.
4. Small input offset voltage.
5. small input offset current.
6. High CMRR
7. Low input bias current

Requirements of output stage:-

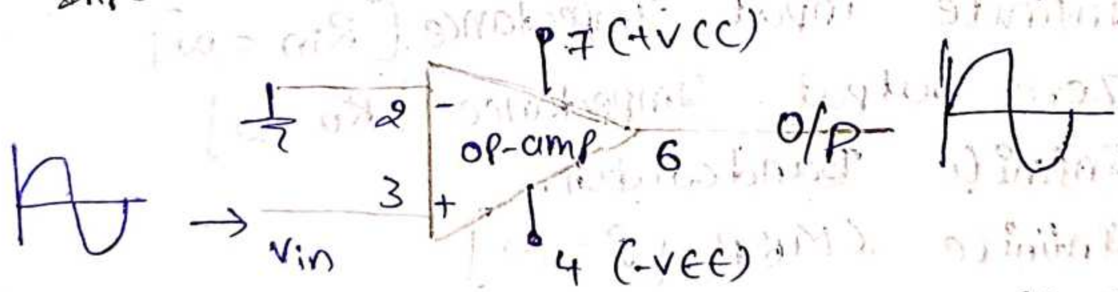
1. Large output voltage swing capability
2. Large output current swing capability.
3. Low output impedance.
4. Low Power dissipation.
5. Short circuit protection.

Operational - Amplifier symbol.



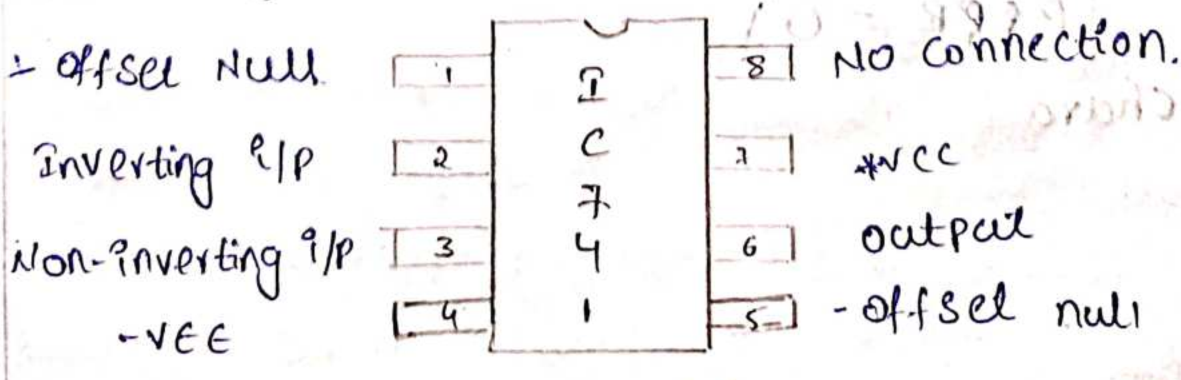


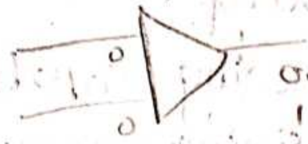
Input applied to Inverting i/p terminal.



Input applied to non-inverting i/p terminal

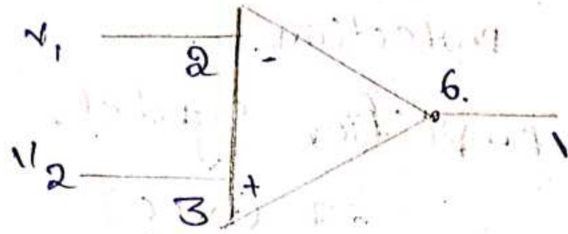
* Pin Diagram of operational - amplifier.



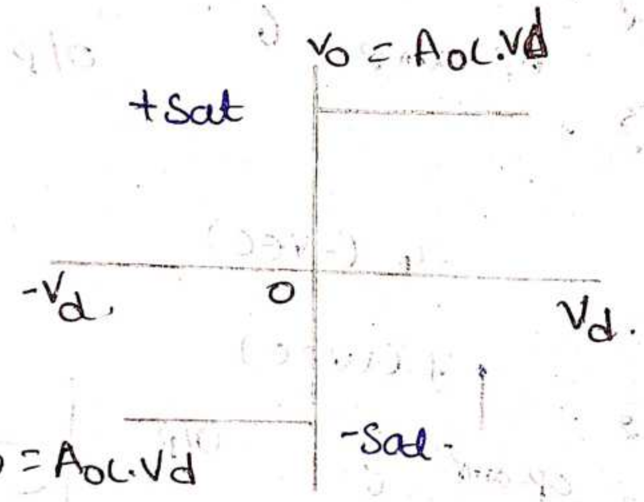


we connect potentiometer to pin 1 & 5 to

Ideal OP-amp :-



Transfer - curve of ideal OP-amp :-



$$A_{OL} \Rightarrow \frac{V_o}{V_d} = \infty.$$

$$\Rightarrow \frac{V_o}{0} = \infty.$$

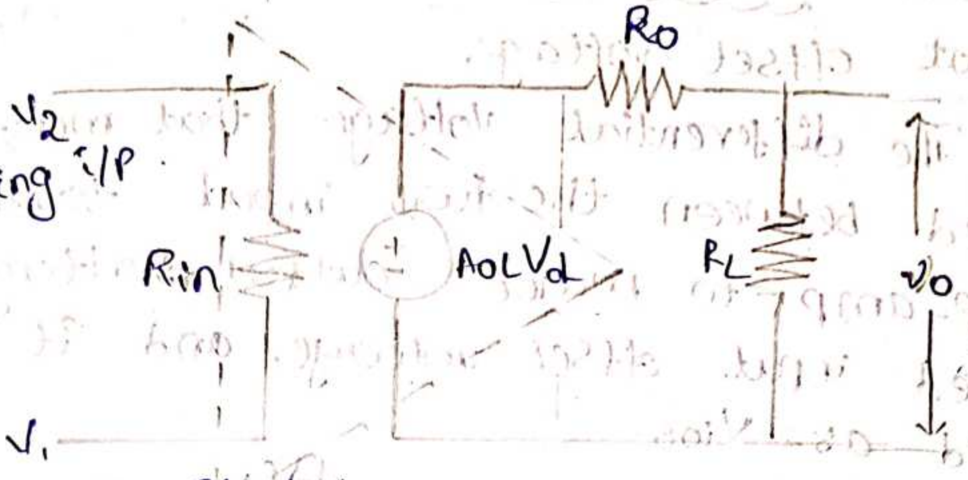
$V_o = A_{OL} \cdot V_d$

Characteristics of Ideal OP-amp.

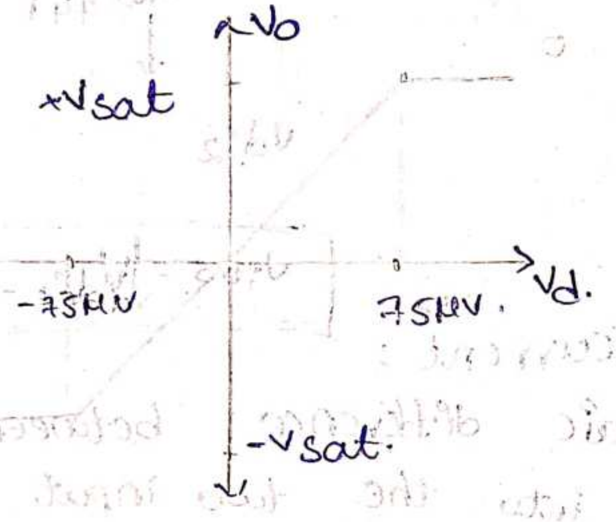
1. Infinite Voltage gain. $[A_{OL} = \infty]$
2. Infinite input Impedance $[R_{in} = \infty]$
3. Zero output Impedance $[R_o = 0]$
4. Infinite Bandwidth.
5. Infinite CMRR $[\rho = \infty]$
6. Infinite slew rate. $[S = \infty]$
7. No. affect of Temperature
8. Power supply rejection ratio is Equal to ze $[PSSR = 0]$

Chara

12/8/24 Practical OP-amp.



Transfer Curve of Practical op-amp.



$\uparrow A_{OL} = \frac{V_o}{V_d}$.

$A_{OL} = 2 \times 10^5$.

$V_o = \pm V_{sat}$
$= \pm 15 V$.

$V_d = \frac{V_o}{A_{OL}} = \frac{\pm 15}{2 \times 10^5}$
$= \pm 75 \mu V$.

Characteristics of Practical op-amp:

1. High voltage gain
2. High input impedance.
3. Low output impedance.
4. High Bandwidth.
5. High CMRR.
6. High slew rate.

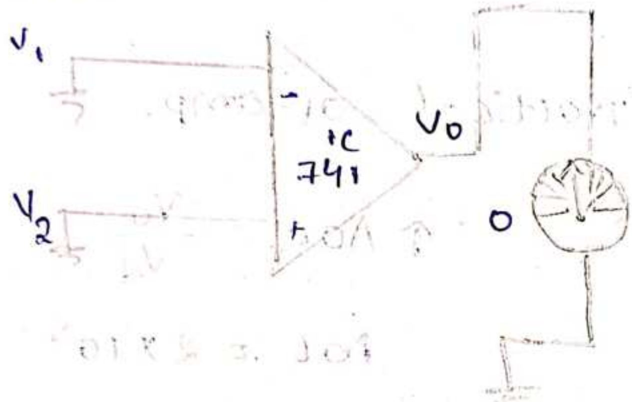
Features of IC - 741.

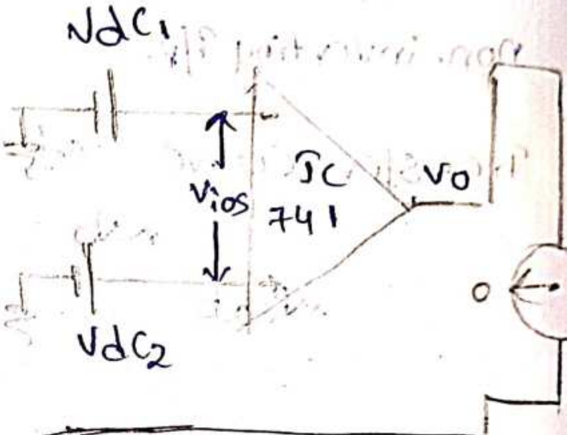
1. No frequency compensation required.
2. Short circuit protection required.
3. Offset voltage null capability.
4. Large common mode and differential voltage range.

* Op-amp Parameters :-

1. Input offset Voltage.

The differential voltage that must be applied between the two input terminal of an op-amp. to make output voltage '0'. is called input offset voltage. and it can be denoted as Vios.





$$\boxed{V_{ios} = |V_{dc_1} - V_{dc_2}|}$$

2. Input offset current :

The algebraic difference between the currents flowing into the two input terminals of op-amp is called input offset current and it can be denoted as (I_{ios}).

* Mathematically it can be represented as

$$\boxed{I_{ios} = |I_{b_1} - I_{b_2}|}$$

In Ideal case, I_{ios} value $= 0$

$I_{b_1} \rightarrow$ current Entering into non-inverting terminal

$I_{b_2} \rightarrow$ current Entering into inverting terminal

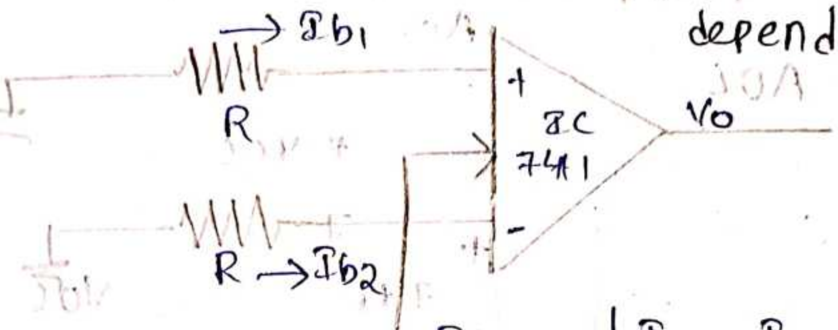
* I_{ios} value of IC-741 is "200nA."

3. Input Bias current :-

The average value of two currents flowing into the op-amp input terminals is called input bias current and it can be represented as "I_b".

$$\boxed{I_b = \frac{I_{b1} + I_{b2}}{2}}$$

Ideally, I_b value is 0. IC-741 has maximum value of 500 nA. I_{ios} & I_b are Temperature dependent Parameter



$$I_{ios} = |I_{b1} - I_{b2}|$$

$$I_b = \frac{I_{b1} + I_{b2}}{2}$$

14/8/24

4. Differential input Resistance :-

It is Equivalent resistance measured at either the inverting or non-inverting input terminals with other input terminal grounded. It is denoted as "R_i"

5. Input capacitance :-

It is Equivalent capacitance measured at either inverting or non-inverting input terminals with other input terminal as grounded. It is denoted as "C_i"

Range of capacitange used in IC741 is "1 - 4 PF"

6. open Loop voltage Gain :-

It is the ratio of output voltage to differential input voltage. It is also called large signal voltage gain. It is denoted as A_{OL}

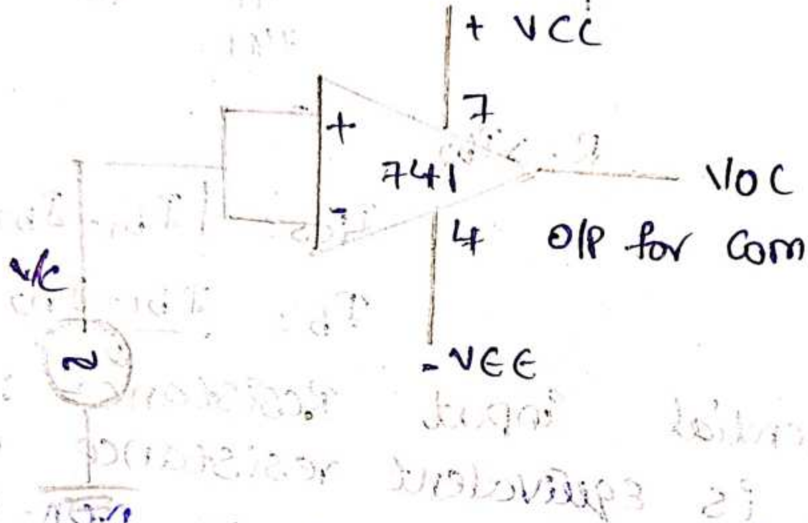
$$A_{OL} = \frac{V_o}{V_d}.$$

7. Common Mode Regection Ratio :-

It is the ratio of differential voltage gain to the common mode voltage gain. It is denoted as "CMRR."

$$CMRR = \frac{Ad}{Ac}$$

$Ad = AOL$.



The common mode input Vc is applied to both input terminals of OP-amp. then output Voc is measured. Therefore the common mode gain $Ac = \frac{V_{oc}}{V_c}$.

The value of CMRR in IC741 is 90dB.

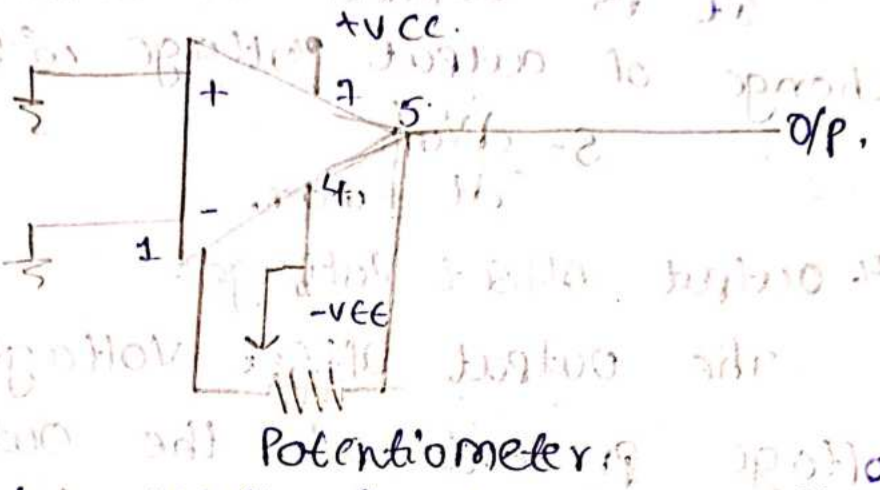
8. Output Voltage swing:-

The OP-amp output voltage swing is limited. It is decided by supply voltage. It never exceeds the limits. +Vcc & -VEE. Above this values, OP-amp get saturated.

9. Output Resistance :-

It is equivalent resistance measured between the output terminals of OP-amp and ground.

10. Offset voltage adjustment range.



The range for which input offset voltage can be adjusted using potentiometer so, as to reduce output to '0' is called offset voltage adjustment range.

11. Power supply Rejection Ratio

It is defined as the ratio of change in input offset voltage due to the change in supplied voltage keeping other supply voltage as constant. It is also called power supply sensitivity.

$$PSRR = \left.\frac{\Delta V_{ios}}{\Delta V_{CC}}\right|_{V_{EE} = \text{constant}}$$

(or)

$$PSRR = \left.\frac{\Delta V_{ios}}{\Delta V_{EE}}\right|_{V_{CC} = \text{constant}}$$

It is also called as supply voltage rejection ratio. "SVRR".

12. Power consumption :

It is amount of power to be considered by OP-amp with 0 input voltage for its proper functioning. It is denoted as "P_C".

13. Slew Rate :-

It is defined as maximum rate of change of output voltage with time.

$$S = \left.\frac{dV_o}{dt}\right|_{max}$$

14. Output offset Voltage :-

The output offset voltage is the DC voltage present at the output terminals when both inputs terminals are grounded.
It is denoted as (V_{oos})

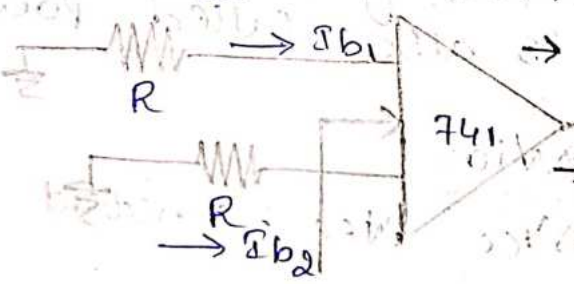
16/8/24

Input Bias current :

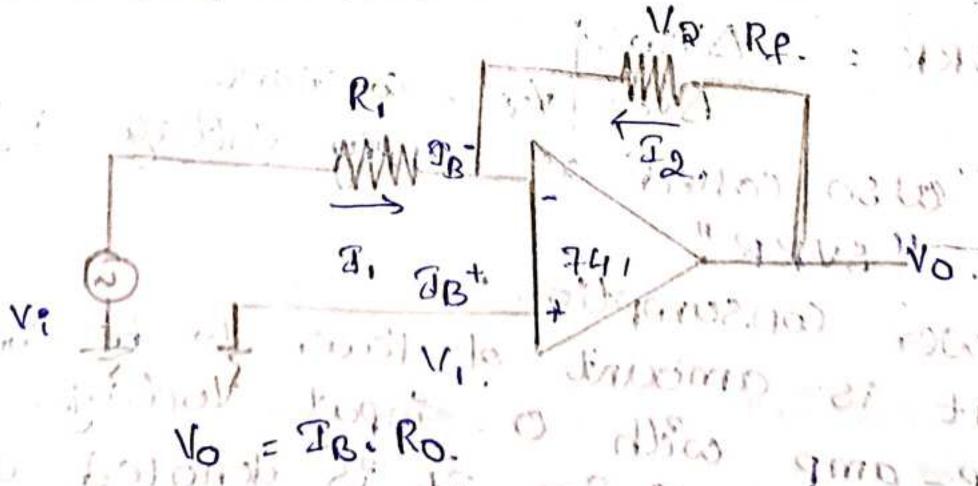
The average voltage of 2 currents flowing into the op-amp input terminals is called input Bias current

DC Characteristics
→ i/p Bias current
→ i/p offset current
→ i/p offset Voltage
→ Thermal Drift

$$I_b = \frac{I_{b1} + I_{b2}}{2}.$$



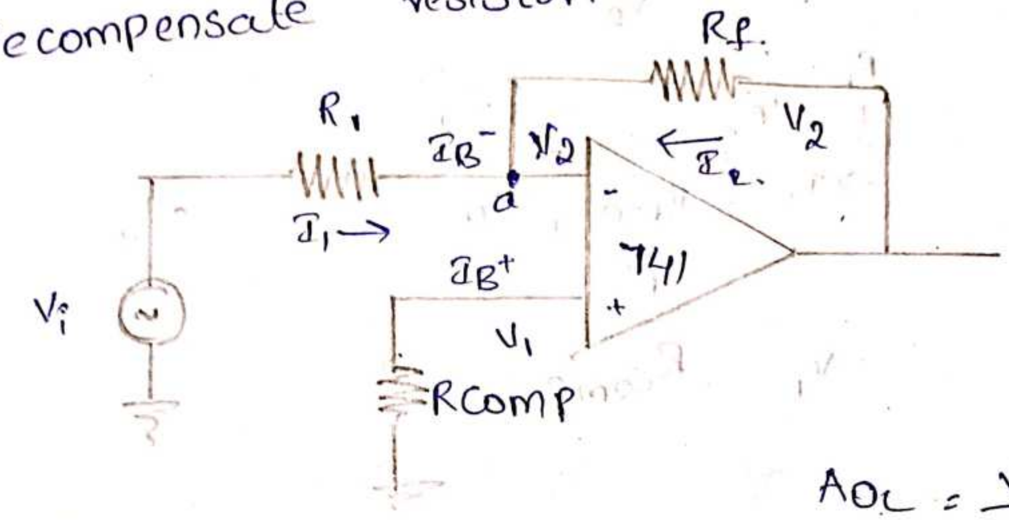
input bias current :-



$$V_o = I_B \cdot R_o.$$
$$= 500\,nA \times 1M\Omega$$
$$= 500 \times 10^{-9} \times 10^{6}$$
$$= 500 \times 10^{-3}$$

$\therefore V_o = 500\,mV.$

To compensate this V_o value, we are placing Recompensate resistor.



$V_d \Rightarrow 0 \rightarrow$ Virtual ground concept.

$$A_{OL} = \frac{V_o}{V_d}.$$

$$\infty = \frac{V_o}{V_d}.$$

$$V_d = 0.$$

$$V_d = V_1 - V_2.$$

$$0 = V_1 - V_2.$$

$$\therefore V_1 = V_2$$

Apply KCL at node a

$$I_B^- = I_1 + I_2.$$

$$I_B^- = \frac{V_2}{R_1} + \frac{V_2}{R_f}. \quad [V_1 = V_2]$$

$$I_B^- = V_2\left[\frac{1}{R_1} + \frac{1}{R_f}\right] \longrightarrow (1).$$

$$I_B^+ = \frac{V_1}{R_{comp}} \quad [\because V_1 = V_2]$$

$$I_B^+ = \frac{V_2}{R_{comp}} \longrightarrow (2).$$

Assume $I_B^- = I_B^+$.

$$\cancel{V_2}\left[\frac{1}{R_1} + \frac{1}{R_f}\right] = \frac{\cancel{V_2}}{R_{comp}}$$

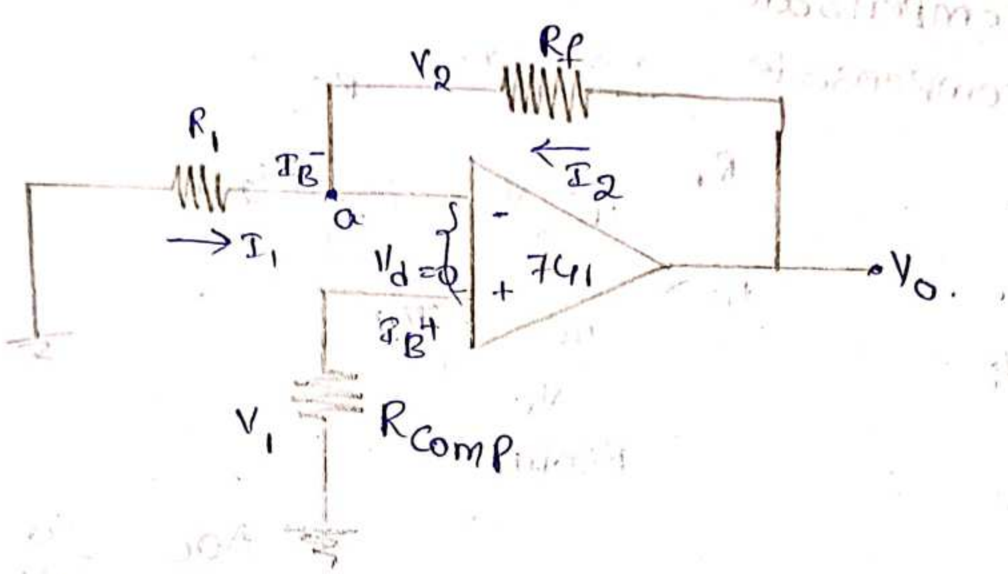
$$\frac{1}{R_1} + \frac{1}{R_f} = \frac{1}{R_{comp}}$$

$$\frac{R_1 + R_f}{R_1 R_f} = \frac{1}{R_{comp}}$$

$$\boxed{R_{comp} = \frac{R_1 R_f}{R_1 + R_f}} \qquad R_1 \parallel R_f$$

Parallel combination of R_1 and R_f makes output voltage (V_o) as zero

Input offset current :-



$\because V_d = 0.$

$V_1 = V_2.$

$V_1 = I_B^+ . R_{comp}.$

$I_1 = \frac{V_2}{R_1} = \frac{V_1}{R_1}.$

$I_1 = \frac{I_B^+ . R_{comp}}{R_1} \rightarrow (1).$

Apply KCL at node a.

$I_B^- = \frac{I_B^+ . R_{comp}}{R_1} + I_2$

$I_2 = I_B^- - \frac{I_B^+ . R_{comp}}{R_1}$

$V_o = V_1 - V_2 \Rightarrow I_B^+ . R_{comp} - I_2 R_f.$

$V_o = I_B^+ . R_{comp} - R_f \left[\frac{I_B^- . R_1 - I_B^+ . R_{comp}}{R_1} \right]$

$= I_B^+ . R_{comp} - \frac{R_f . I_B^- \cancel{R_1}}{\cancel{R_1}} + \frac{I_B^+ R_{comp} . R_f}{R_1}$

$= I_B^+ . R_{comp} + \frac{I_B^+ . R_{comp} . R_f}{R_1} - R_f . I_B^-$

$= I_B^+ . R_{comp} \left[1 + \frac{R_f}{R_1} \right] - R_f . I_B^-$

$= I_B^+ R_{comp} \left[\frac{R_1 + R_f}{R_1}\right] - R_f \cdot I_B^-$

$= I_B^+ \left[\frac{\cancel{R_1} R_f}{R_1 + R_f}\right]\left[\frac{\cancel{R_1 + R_f}}{\cancel{R_1}}\right] - R_f \cdot I_B^-$

$= R_f I_B^+ - R_f I_B^-$

$V_o = R_f [I_B^+ - I_B^-]$

$$\boxed{V_o = R_f I_{os}}$$

$V_o = R_f I_{ios}$

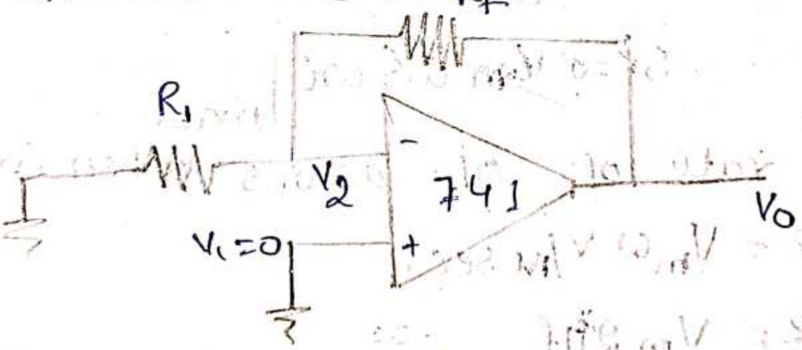
$= 1 M\Omega \times 200 nA$

$= 1 \times 10^6 \times 200 \times 10^{-9}$

$= 200 \times 10^{-3}$

$V_o = 200 mV$

Input - offset Voltage:



By voltage dividing rule

$V_2 = \frac{V_o \cdot R_1}{R_1 + R_f}$

$V_o = \frac{V_2 (R_1 + R_f)}{R_1} \rightarrow ①$

$V_{ios} = (V_1 - V_2)$

$V_{ios} = V_2$

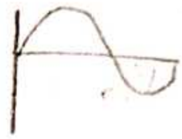
$V_o = \frac{V_{ios} (R_1 + R_f)}{R_1}$

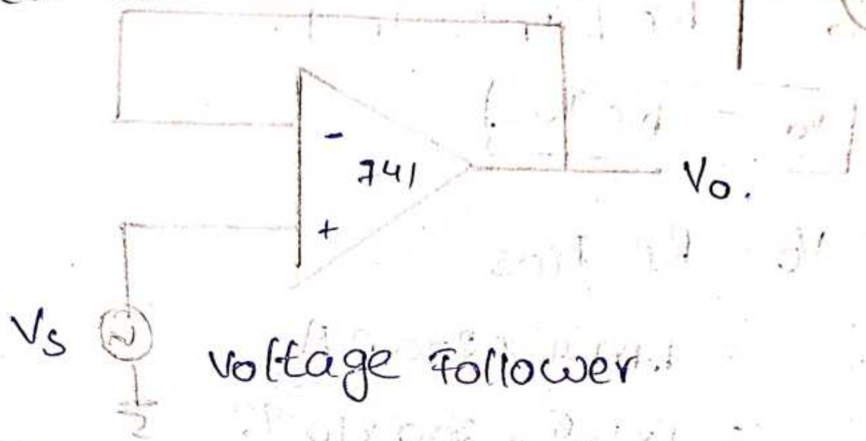
Total o/p offset Voltage is

$$V_{OT} = V_{ios}\left[1 + \frac{R_f}{R_1}\right] + R_f \cdot I_{ios}.$$

AC characteristics :-

① Slew Rate.





voltage Follower.

$V_s = V_m \sin \omega t = V_o.$

Slew Rate, $SR = \frac{dv_o}{dt}$

$$= \frac{d}{dt}\left[V_m \sin\omega t\right]$$

$$SR = \omega V_m \cos \omega t \Big|_{max}$$

Maximum rate of o/p across when $\cos\omega t = 1$.

$$SR = V_m \omega \ V/\mu sec.$$

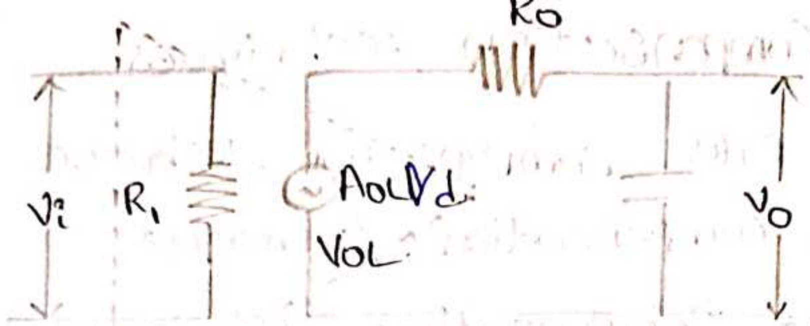
$$SR = V_m 2\pi f$$

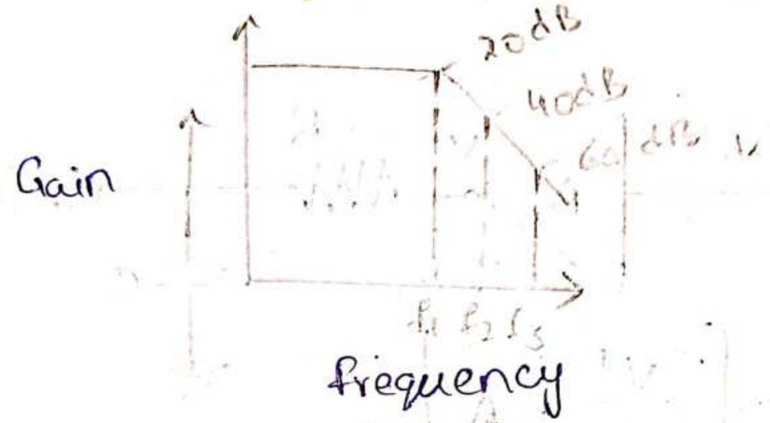
$$SR = V_m (6.28) f.$$

$$f_{max} = \frac{SR}{(6.28) V_m}$$

② Frequency Response :-

frequency response is defined as gain of Op-Amp. response to different frequen--cies.





Voltage divide rule.

$$V_o = \frac{A_{OL} V_d (1/j\omega C)}{R_o + (1/j\omega C)}$$

$$\frac{V_o}{V_d} = \frac{A_{OL}\,(1/j\omega C)}{\frac{R_o j\omega C + 1}{j\omega C}} = \frac{\frac{A_{OL}}{j\omega C}}{\frac{R_o j\omega C + 1}{\cancel{j\omega C}}}$$

$$= \frac{A_{OL}}{R_o j\omega C + 1}$$

$$= \frac{A_{OL}}{1 + R_o C j (2\pi f)}$$

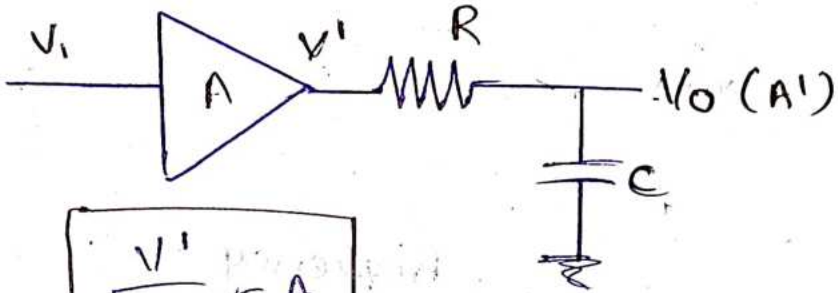
$$|A| = \frac{A_{OL}}{1 + j(f/f_1)} \qquad \because f_1 = \frac{1}{2\pi R_o C}$$

$$= \frac{A_{OL}}{\sqrt{1 + (f/f_1)^2}}$$

19/8/24 Frequency Compensation Techniques.

1. Dominant Pole Compensation Technique.
2. Pole - Zero Compensation Technique.

1. Dominant Pole Compensation Technique:-



$$\boxed{\frac{V'}{V_1} = A}$$

By voltage divide rule,

$$V_0 = \frac{V' \cdot (1/j\omega C)}{R + 1/j\omega C}$$

$$= \frac{\frac{V'}{j\omega C}}{\frac{Rj\omega C + 1}{\cancel{j\omega C}}}$$

$$= \frac{V'}{1 + Rj C 2\pi f} \qquad \because f_a = \frac{1}{2\pi RC}$$

$$V_0 = \frac{V'}{1 + j(f/f_a)}$$

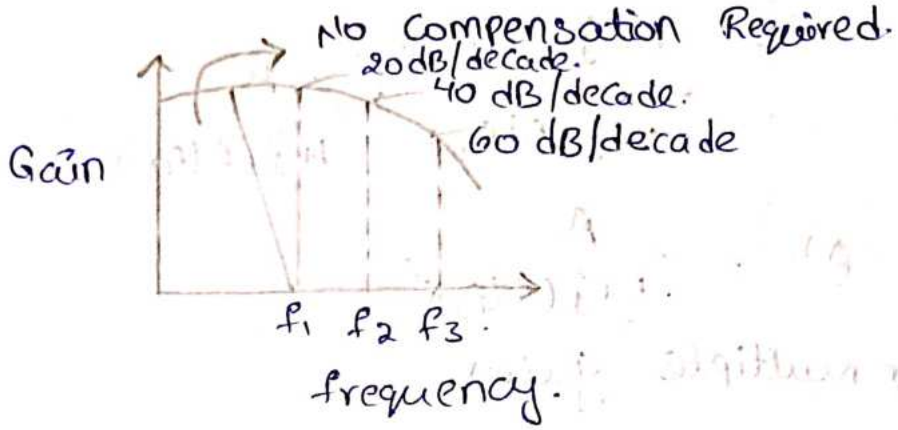
$$\boxed{\frac{V_0}{V'} = \frac{1}{1 + j(f/f_a)}}$$

$$A' = \frac{V_0}{V'} \times \frac{V'}{V_1}$$

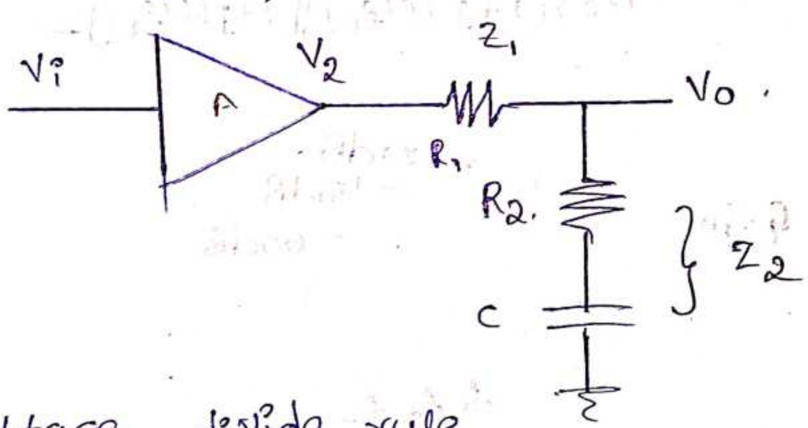
$$= \left(\frac{1}{1 + j(f/f_a)}\right) \cdot A$$

$$A' = \frac{A}{1 + j(f/f_a)}$$

$$A' = \frac{A'}{1+j(f/f_a)\left[1+j(f/f_1)\right]\left[1+j(f/f_2)\right]\cdots}$$



2. Pole - zero compensation Technique.



By voltage divide rule.

$$V_o = \frac{V_2 \, Z_2}{Z_1 + Z_2}$$

$$= V_2 \frac{R_2 + \frac{1}{j\omega C}}{R_1 + R_2 + \frac{1}{j\omega C}}$$

$$= V_2 \frac{1 + j\omega C R_2 / \cancel{j\omega C}}{\frac{R_1 j\omega C + R_2 j\omega C + 1}{\cancel{j\omega C}}}$$

$$= V_2 \frac{(1 + j\omega C R_2)}{R_1 j\omega C + R_2 j\omega C + 1}$$

$$\frac{V_o}{V_2} = \frac{1 + j2\pi f C R_2}{1 + R_1 j2\pi f C (R_1 + R_2)} \qquad \therefore f_a = \frac{1}{2\pi R_2 C}$$

$$f_o = \frac{1}{2\pi (R_1 + R_2) C}$$

$$\frac{V_o}{V_2} = \frac{1 + j(f/f_a)}{1 + j(f/f_o)}.$$

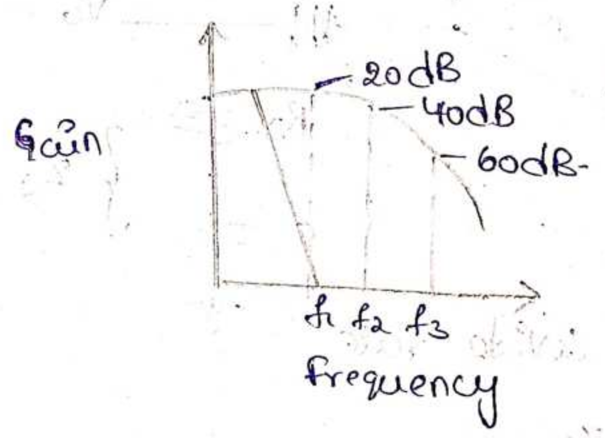
$$A' = \frac{V_0}{V_2} \times \frac{V_2}{V_1}$$

$$= \frac{1 + j(f/fa)}{1 + j(f/f_0)} \times \frac{A}{1 + j(f/fa)}$$

$$A' = \frac{A}{1 + j(f/f_0)}$$

for multiple gains,

$$A' = \frac{A}{1 + j(f/f_0)[1 + j(f/f_1)][1 + j(f/f_2)] \cdots}$$

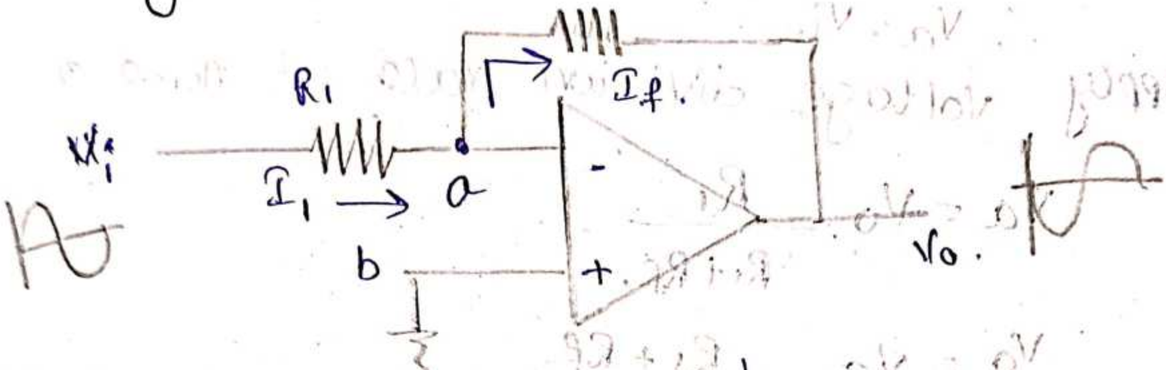


2. Pole - zero compen

Unit - 3

Linear Application of op - amp

Inverting Amplifier.



According to virtual ground.

$V_a = V_b$.

Here $V_b = 0$ ∴ $V_a = 0$

Apply KCL at node a.

$$I_1 = I_f.$$

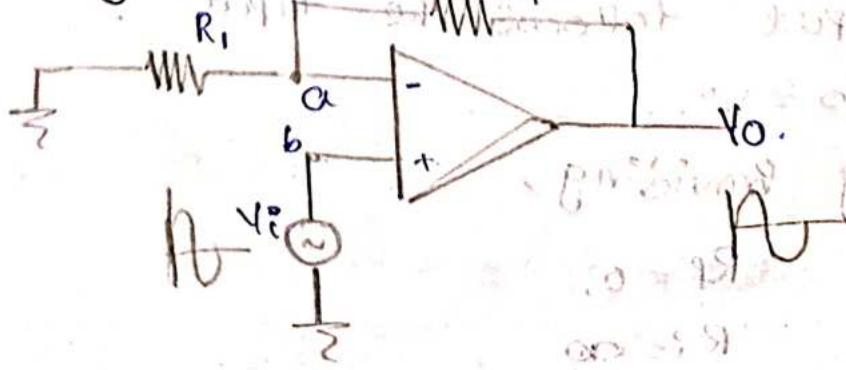
$$\frac{V_i - V_a}{R_1} = \frac{V_a - V_o}{R_f}.$$

$$\frac{V_i}{R_1} = \frac{-V_o}{R_f} \quad [\because V_a = 0]$$

$$\therefore V_o = -\frac{R_f \cdot V_i}{R_1}$$

$$\boxed{V_o = -\frac{R_f}{R_1} V_i.} \qquad \boxed{\frac{V_o}{V_i} = \frac{-R_f}{R_1}}$$

Non - Inverting Amplifier :.



According to virtual ground,

$V_a = V_b$.

Here, $V_b = V_i$

$\therefore V_a = V_i$.

Apply voltage division rule at node a.

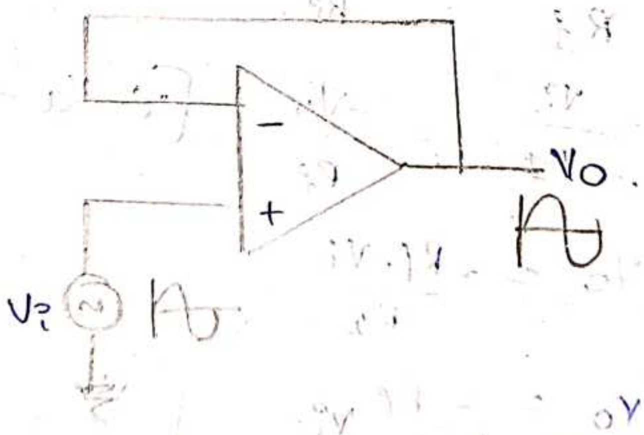
$$V_a = V_o \cdot \frac{R_1}{R_1 + R_f}$$

$$V_o = V_a \cdot \frac{R_1 + R_f}{R_1 \cancel{+R_f}}$$

$$V_o = V_i \; \frac{R_1 + R_f}{R_1}$$

$$\therefore \boxed{V_o = V_i \left(1 + \frac{R_f}{R_1}\right)} \qquad \boxed{\frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}}$$

Voltage follower:



It is the special case of non-inverting amplifier.

output follows the input.

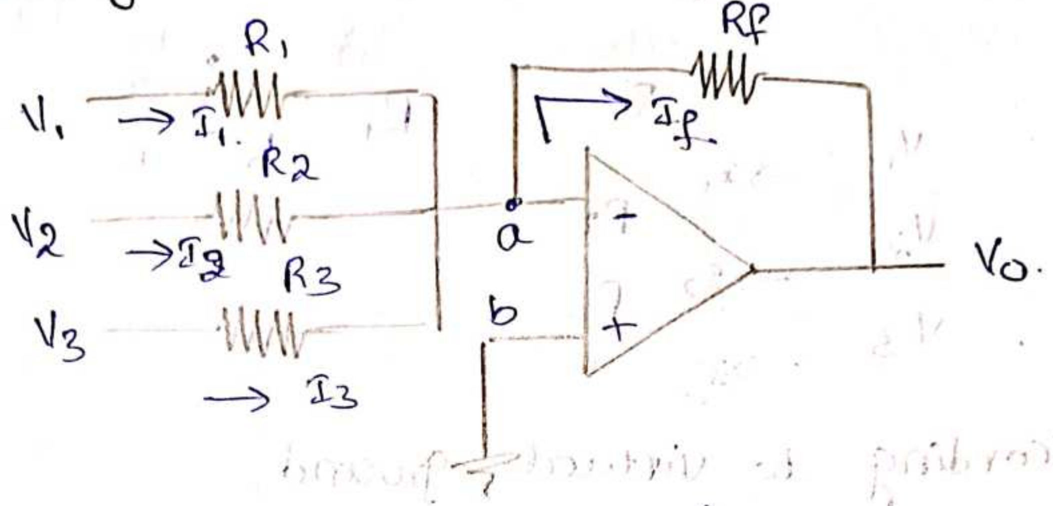
$V_o = V_i$.

By Providing,

$R_f = 0$,

$R_i = \infty$

Inverting summing amplifier :-



according to virtual ground.

$V_a = V_b$.

Here, V_b is directly connected to ground, so

$V_b = 0$. $[\because V_a = V_b]$.

$\therefore V_a = 0$

Apply KCL at node a.

$$I_1 + I_2 + I_3 = I_f.$$

$$\frac{V_1 - V_a}{R_1} + \frac{V_2 - V_a}{R_2} + \frac{V_3 - V_a}{R_3} = \frac{V_a - V_o}{R_f}$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_o}{R_f} \quad [\because V_a = 0]$$

If $R_1 = R_2 = R_3 = R_f$ then.

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_o}{R_f} \text{ becomes}$$

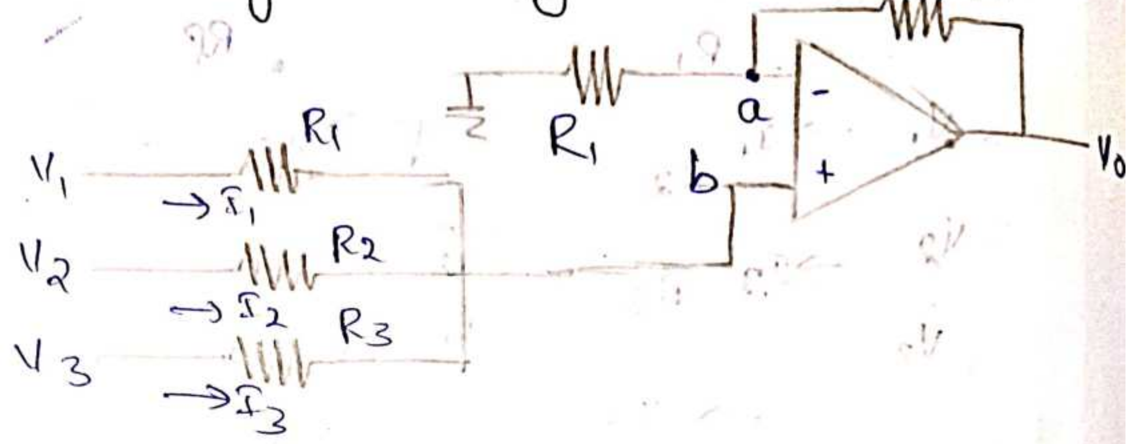
$$\frac{V_1}{R_f} + \frac{V_2}{R_f} + \frac{V_3}{R_f} = \frac{-V_o}{R_f}.$$

$$V_o = -R_f\left(\frac{V_1}{R_f} + \frac{V_2}{R_f} + \frac{V_3}{R_f}\right)$$

$$V_o = \frac{-\cancel{R_f}}{R_f}(V_1 + V_2 + V_3)$$

$$\boxed{\therefore V_o = -(V_1 + V_2 + V_3).}$$

Non-Inverting summing Amplifier:



According to virtual ground,

$V_a = V_b$.

Apply voltage division rule at node a.

$$V_a = V_o \cdot \frac{R_1}{R_1 + R_f}$$

$$V_o = V_a \frac{R_1 + R_f}{R_1}$$

$$V_o = V_a \left(1 + \frac{R_f}{R_1}\right) \longrightarrow ①$$

Apply KCL at node b,

$$I_1 + I_2 + I_3 = 0.$$

$$\frac{V_1 - V_b}{R_1} + \frac{V_2 - V_b}{R_2} + \frac{V_3 - V_b}{R_3} = 0$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \left[-\frac{V_b}{R_1} - \frac{V_b}{R_2} - \frac{V_b}{R_3}\right] = 0$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{V_b}{R_1} + \frac{V_b}{R_2} + \frac{V_b}{R_3}$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = V_b\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]$$

$$V_b = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}.$$

Take, $V_o = V_a\left(1 + \frac{R_f}{R_1}\right)$ from equ ①.

Since $V_a = V_b$. sub V_b value in equ ①

$$V_o = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \left[1 + \frac{R_f}{R_1}\right].$$

If $R_1 = R_2 = R_3 = \frac{R_f}{2}$ then.

$$V_o = \frac{\frac{V_1}{R_f/2} + \frac{V_2}{R_f/2} + \frac{V_3}{R_f/2}}{\frac{1}{R_f/2} + \frac{1}{R_f/2} + \frac{1}{R_f/2}} \left[1 + \frac{R_f}{R_f/2}\right]$$

$$= \frac{\frac{2V_1}{R_f} + \frac{2V_2}{R_f} + \frac{2 \cdot V_3}{R_f}}{\frac{2}{R_f} + \frac{2}{R_f} + \frac{2}{R_f}} \left[1 + \frac{2R_f}{R_f}\right]$$

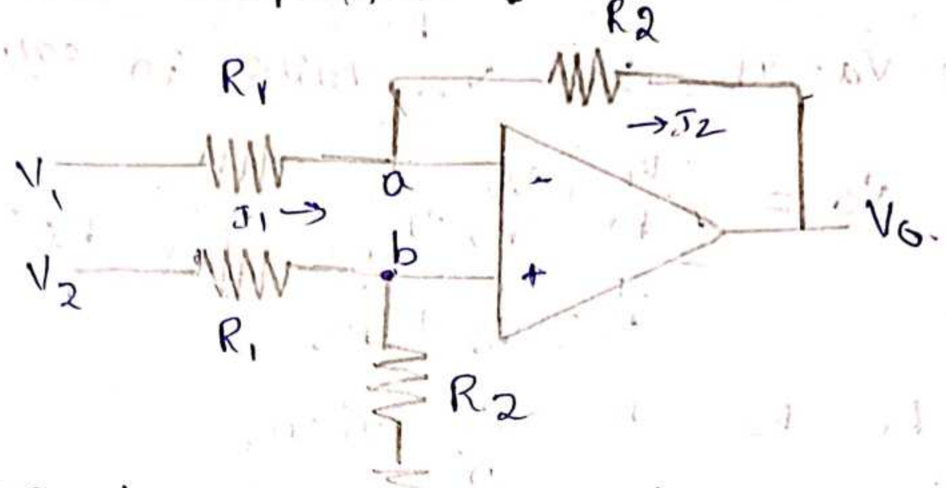
$$= \frac{\frac{2}{R_f}(V_1 + V_2 + V_3)}{\frac{(2+2+2)}{R_f}} [1+2]$$

$$= \frac{\frac{\cancel{2}}{R_f}(V_1 + V_2 + V_3)}{\frac{\cancel{6}\,\cancel{3}}{R_f}} [\cancel{3}]$$

$$= \frac{\frac{V_1 + V_2 + V_3}{R_f}}{\frac{1}{R_f}}$$

$$\boxed{V_o = V_1 + V_2 + V_3}$$

Difference Amplifier :-



According to virtual ground,

$V_a = V_b$.

By voltage division rule, at b

$$V_b = V_2 \frac{R_2}{R_1+R_2}. \qquad I_1 = I_2.$$

$$V_a = V_2 \frac{R_2}{R_1+R_2}$$

Apply KCL at node a,

$$\frac{V_1 - V_a}{R_1} = \frac{V_a - V_o}{R_2} \qquad [\because V_a = V_b).$$

$$\frac{V_1 - V_b}{R_1} = \frac{V_b - V_o}{R_2}.$$

$$\frac{V_1}{R_1} - \frac{V_b}{R_1} = \frac{V_b}{R_2} - \frac{V_o}{R_2}$$

$$\frac{V_1}{R_1} + \frac{V_o}{R_2} = \frac{V_b}{R_2} + \frac{V_b}{R_1}.$$

$$\frac{V_1}{R_1} + \frac{V_o}{R_2} = V_b\left[\frac{1}{R_1} + \frac{1}{R_2}\right].$$

$$\frac{V_1}{R_1} + \frac{V_o}{R_2} = V_2\left(\frac{R_2}{R_1+R_2}\right)\left(\frac{1}{R_1+R_2}\right)$$

$$\frac{V_1}{R_1} + \frac{V_0}{R_2} = V_2 \cdot \frac{R_2}{\cancel{R_1+R_2}} \left(\frac{\cancel{R_1+R_2}}{R_1 R_2}\right)$$

$$= V_2 \left(\cancel{R_2}\right)\left(\frac{1}{R_1 \cancel{R_2}}\right)$$

$$\frac{V_1}{R_1} + \frac{V_0}{R_2} = \frac{V_2}{R_1}.$$

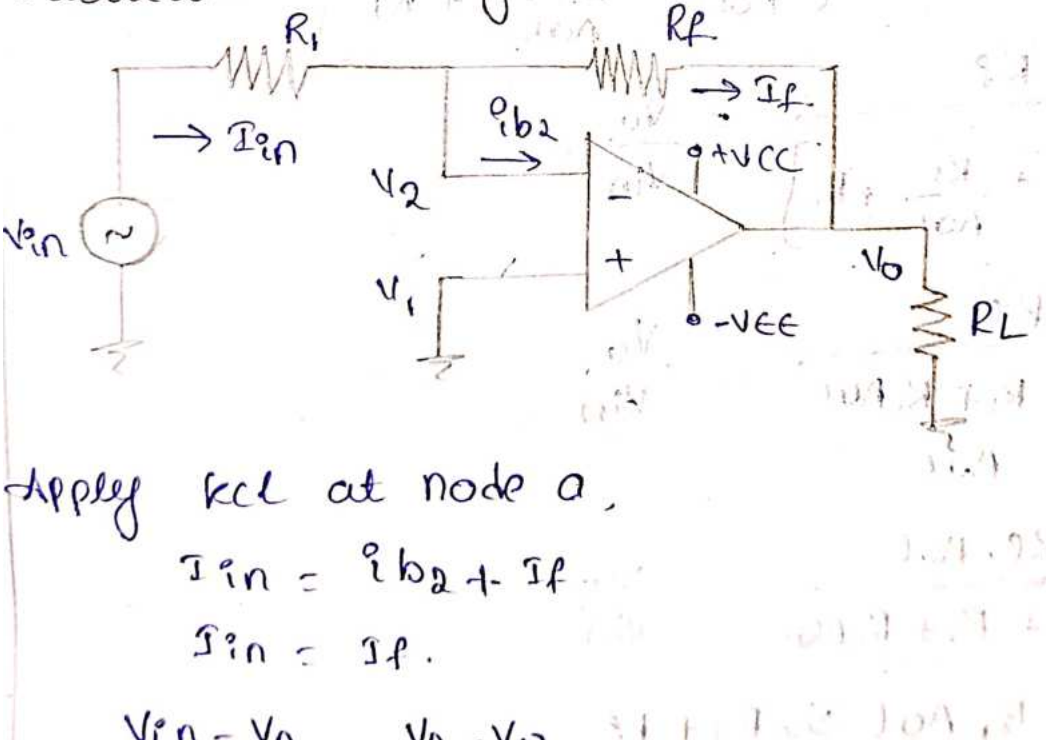
$$\frac{V_0}{R_2} = \frac{V_2}{R_1} - \frac{V_1}{R_1}$$

$$\frac{V_0}{R_2} = (V_2 - V_1)\frac{1}{R_1}$$

$$\boxed{V_0 = \frac{R_2}{R_1}(V_2 - V_1)} \qquad R_2 = R_1$$

$$V_0 = V_2 - V_1$$

Practical Inverting Amplifier :-



Apply kcl at node a.

$$I_{in} = i_{b2} + I_f$$

$$I_{in} = I_f.$$

$$\frac{V_{in} - V_A}{R_1} = \frac{V_A - V_0}{R_f} \longrightarrow (1)$$

$$A_{OL} = \frac{V_0}{V_d}.$$

$$V_0 = A_{OL} \cdot V_d.$$

$$= A_{OL}[V_1 - V_2] \qquad V_1 = 0 \text{ (Ground)}$$

$$V_0 = A_{OL}[-V_2]$$

$$V_o = -A_{OL}(V_2)$$

$$V_o = -A_{OL}(V_A).$$

$$V_A = \frac{-V_o}{A_{OL}}$$

Sub V_A value in equation ①.

$$\frac{V_{in} + \frac{V_o}{A_{OL}}}{R_1} = \frac{\frac{-V_o}{A_d} - V_o}{R_f}$$

$$R_f\left(V_{in} + \frac{V_o}{A_{OL}}\right) = \left(-\frac{V_o}{A_{OL}} - V_o\right)R_1$$

$$V_{in}R_f + \frac{V_o}{A_{OL}}R_f = \frac{-V_o}{A_{OL}}R_1 - V_oR_1.$$

$$V_{in}R_f = -V_o\left[\frac{R_f}{A_{OL}} + \frac{R_1}{A_{OL}} + R_1\right]$$

$$\frac{-R_f}{\left[\frac{R_f}{A_{OL}} + \frac{R_1}{A_{OL}} + R_1\right]} = \frac{V_o}{V_{in}}$$

$$\frac{-R_f}{\frac{R_f + R_1 + R_1A_{OL}}{A_{OL}}} = \frac{V_o}{V_{in}}$$

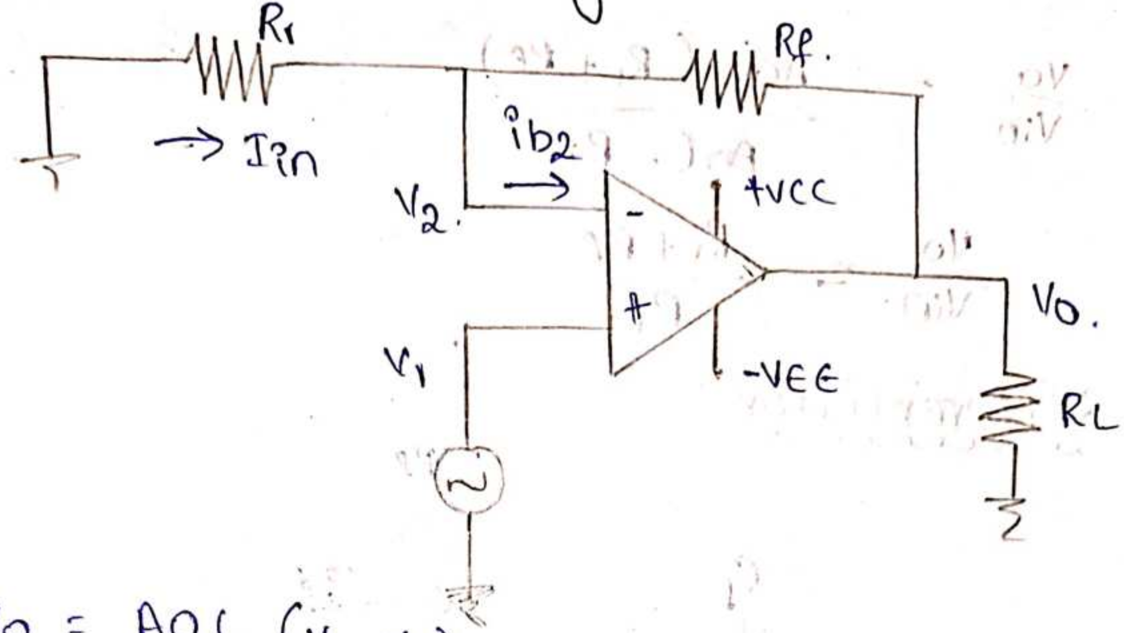
$$\frac{-R_f \cdot A_{OL}}{R_f + R_1 + R_1A_{OL}} = \frac{V_o}{V_{in}}.$$

$$R_1A_{OL} \gg R_1 + R_f$$

$$\frac{V_o}{V_{in}} = \frac{-R_f\cancel{A_{OL}}}{R_1 A_{OL}}.$$

$$\frac{V_o}{V_{in}} = \frac{-R_f}{R_1}.$$

Practical Non-inverting amplifier:-



$$V_o = A_{OL}(V_1 - V_2)$$

$$V_o = A_{OL}(V_{in} - V_2).$$

By voltage division rule,

$$V_2 = \frac{V_o \cdot R_1}{R_1 + R_f}.$$

$$V_o = A_{OL}\left(V_{in} - \frac{V_o R_1}{R_1 + R_f}\right)$$

$$V_o = A_{OL} V_{in} - A_{OL}\left(\frac{V_o \cdot R_1}{R_1 + R_f}\right)$$

$$V_o + V_o \cdot \frac{A_{OL} \cdot R_1}{R_1 + R_f} = A_{OL} \cdot V_{in}.$$

$$V_o\left(1 + \frac{A_{OL} \cdot R_1}{R_1 + R_f}\right) = A_{OL} \cdot V_{in}.$$

$$\frac{V_o}{V_{in}} = \frac{A_{OL}}{1 + \frac{A_{OL} \cdot R_1}{R_1 + R_f}} \qquad \therefore \frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_1}.$$

$$\frac{V_o}{V_{in}} = \frac{A_{OL}}{\frac{(R_1 + R_f) + A_{OL} \cdot R_1}{R_1 + R_f}}$$

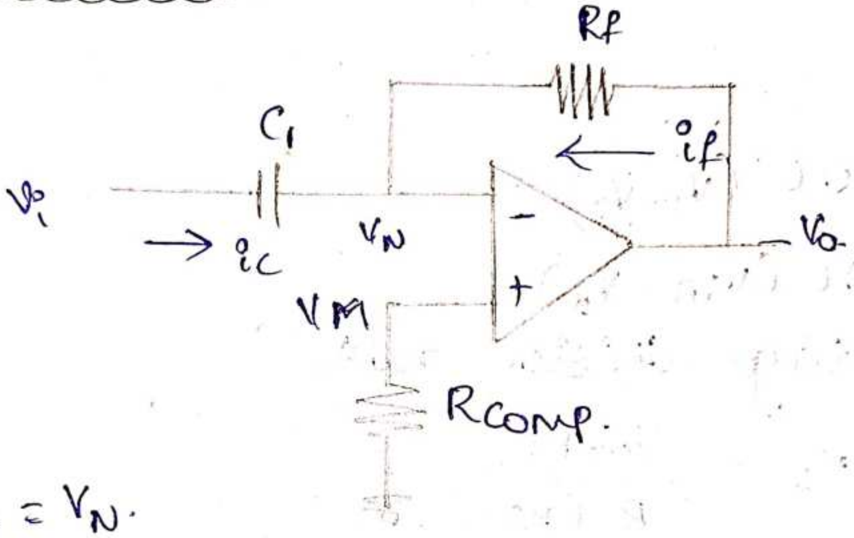
$$\frac{V_o}{V_{in}} = \frac{A_{OL} \cdot (R_1 + R_f)}{R_1 + R_f + A_{OL} \cdot R_1}$$

$$R_1 + R_f << A_{OL} \cdot R_1$$

$$\frac{V_o}{V_{in}} = \frac{A_{OL}(R_1 + R_f)}{A_{OL} \cdot R_1}$$

$$\frac{V_o}{V_{in}} = \frac{R_1 + R_f}{R_1}.$$

* **Differentiator :-**



$V_M = V_N$.

$V_M = 0$

$V_N = 0$.

$$i = C \cdot \frac{dv}{dt}.$$

$$i_C = C_1 \cdot \frac{d}{dt}(v_i - V_N).$$

$$i_C = C_1 \frac{dv_i}{dt}$$

Apply KCL at node N.

$$i_C + i_f = 0.$$

$$C_1 \cdot \frac{dv_i}{dt} + \frac{V_o}{R_f} = 0.$$

$$\frac{V_o}{R_f} = -C_1 \cdot \frac{dv_i}{dt}.$$

$$\boxed{V_o = -C_1 \cdot R_f \cdot \frac{dv_i}{dt}}$$

Phasor Equivalent:

$$V_o(S) = -R_f\, C_1\, S \cdot V_i(S)$$

$$\frac{V_o(S)}{V_i(S)} = \left|\frac{V_o}{V_1}\right|$$

$$\left|\frac{V_o}{V_1}\right| = \left|-R_f C_1 S\right|$$

$$= \left|-j\omega R_f C_1\right|$$

$$= \omega R_f C_1$$

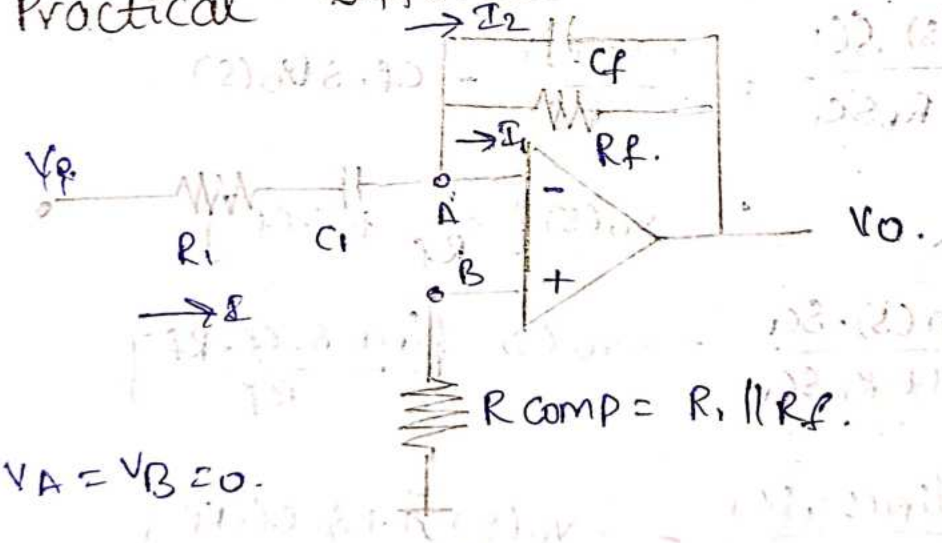
$$= 2\pi f R_f C_1.$$

$$|A| = \frac{f}{f_a}.$$

$$\left[\because\ f_a = \frac{1}{2\pi R_f C_1}\right]$$

20/8/24

Practical Differentiator.



$$V_A = V_B = 0.$$

Apply KCL at node A.

$$I = I_1 + I_2.$$

Calculation for I,

$$I = \frac{V_{in} - V_A}{R_1 + 1/j\omega C_1}$$

$$I = \frac{V_{in}}{R_1 + 1/SC_1}$$

$$I = \frac{V_{in}(S)}{R_1(SC_1) + 1/SC_1}$$

$$I = \frac{V_{in}(S) \cdot (SC_1)}{1 + R_1 SC_1} \rightarrow ①$$

Calculation for I_1.

$$I_1 = \frac{V_A - V_o}{R_f}$$

$$I_1 = \frac{-V_o(S)}{R_f} \rightarrow ②$$

Calculation for I_2

$$I_2 = C_f \cdot \frac{d(V_A - V_o)}{dt}$$

$$= -C_f \cdot \frac{dV_o}{dt}$$

$$I_2 = -C_f S V_o(S) \rightarrow ③$$

Sub equ ①, ②, ③ in (A)

$$\frac{V_{in}(S) SC_1}{1 + R_1 SC_1} = \frac{-V_o(S)}{R_f} - C_f \cdot S V_o(S)$$

$$= -V_o(S)\left[\frac{1}{R_f} + S \cdot C_f\right]$$

$$\frac{V_{in}(S) \cdot SC_1}{1 + R_1 SC_1} = -V_o(S)\left[\frac{1 + S \cdot C_f \cdot R_f}{R_f}\right]$$

$$\frac{R_f V_{in}(S) \cdot SC_1}{1 + R_1 SC_1} = -V_o(S)\left[1 + S \cdot C_f \cdot R_f\right]$$

$$\frac{R_f \cdot SC_1}{1 + R_1 SC_1} = \frac{-V_o(S)}{V_{in}(S)}\left[1 + S \cdot C_f \cdot R_f\right]$$

$$\frac{R_f \cdot SC_1}{(1 + R_1 SC_1)(1 + R_f C_f \cdot S)} = \frac{-V_o(S)}{V_{in}(S)}$$

$$\because R_1 C_1 = R_f C_f$$

$$\frac{R_f \cdot SC_1}{(1 + R_1 SC_1)(1 + R_1 SC_1)} = \frac{-V_o(S)}{V_{in}(S)}$$

$$\Rightarrow \frac{SC_1 \cdot R_f}{(1+SR_1C_1)^2} = \frac{-V_o(S)}{V_{in}(S)}.$$

$$\frac{V_o(S)}{V_{in}(S)} = \frac{-SC_1 \cdot R_f}{(1+S \cdot R_1C_1)^2}.$$

S with $j\omega$.

$$\frac{V_o(j\omega)}{V_{in}(j\omega)} = \frac{-j\omega C_1 R_f}{(1+j\omega R_1 C_1)^2}.$$

$$\frac{V_o(j\omega)}{V_{in}(j\omega)} = \frac{-j2\pi f C_1 R_f}{(1+j2\pi f C_1 R_1)^2}$$

$$= \frac{-j(f/f_a)}{1+j(f/f_b)^2}$$

where $f_a = \frac{1}{2\pi C_1 R_f}$.

$$f_b = \frac{1}{2\pi C_1 R_1}$$

$$\frac{V_o(j\omega)}{V_{in}(j\omega)} = \frac{-j(f/f_a)}{[1+j(f/f_b)^2]}$$

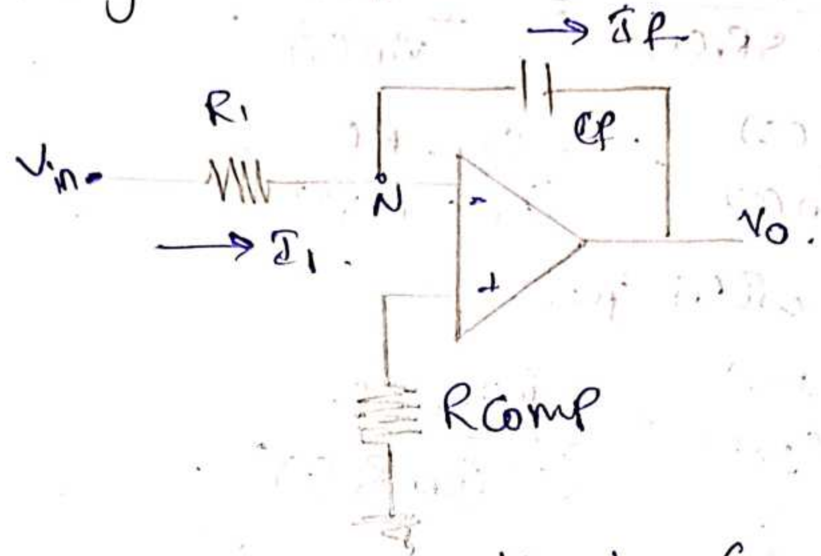
Apply Mod. at above step.

$$\frac{V_o(j\omega)}{V_{in}(j\omega)} = \left|\frac{-j(f/f_a)}{1+j(f/f_b)^2}\right|$$

$$\left|\frac{V_o(j\omega)}{V_{in}(j\omega)}\right| = \frac{\sqrt{(f/f_a)^2}}{\sqrt{(1+(f/f_b)^2)^2}}$$

$$A. = \frac{f/f_a}{1+(f/f_b)^2}$$

Integrator



Apply kcl at node N. $(I_1 + I_f) = 0$

$$I_1 + I_f = 0$$

$$\frac{V_{in}}{R_1} + C_f \cdot \frac{dv_o}{dt} = 0$$

$$C_f \cdot \frac{dv_o}{dt} = \frac{-V_{in}}{R_1}$$

$$\frac{dv_o}{dt} = \frac{-V_{in}}{R_1 C_f} \rightarrow ①$$

Integrate both side.

$$\int_0^t \frac{dv_o}{dt} = \frac{-1}{R_1 C_f} \int_0^t v_{in}(t) \cdot dt$$

$$V_o(t) - V_o(0) = \frac{-1}{R_1 C_f} \int_0^t v_{in}(t) \cdot dt$$

$$V_o(t) = -\frac{1}{R_1 C_f} \int_0^t v_{in}(t) \cdot dt$$

Frequence response. [laplace]

$$V_o(s) = \frac{-1}{s R_1 C_f} V_{in}(s)$$

$$\frac{V_o(s)}{V_{in}(s)} = \frac{-1}{s R_1 C_f}$$

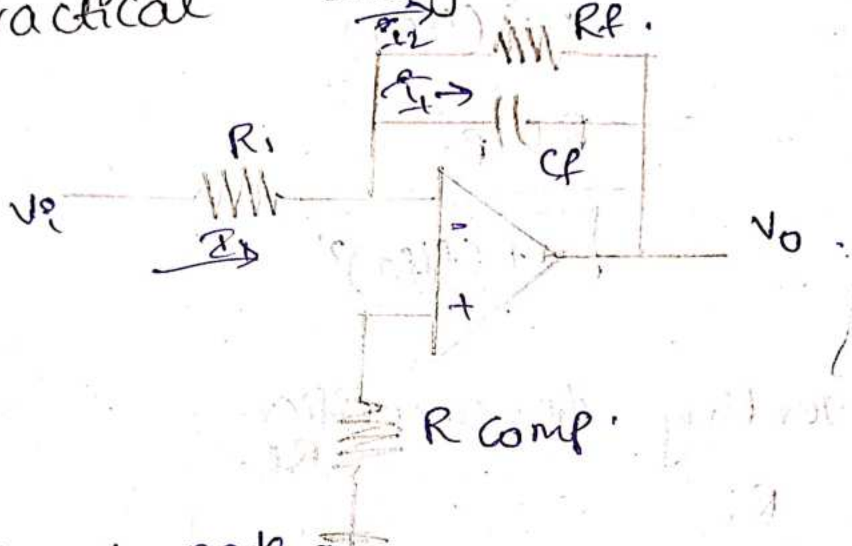
$$A = \frac{-1}{j\omega R_1 C_f}$$

$$= \frac{-1}{j2\pi f R_1 C_f}$$

$$= \frac{-1}{j(f/fa)} \quad \because fa = \frac{1}{2\pi C_f R_f}$$

$$|A| = \frac{1}{f/fa}$$

Practical Integrator.



Apply KCL at node a.

$$I + I_1 + I_2 = 0$$

$$\frac{V_i(S)}{R_1} + S \cdot C_f V_o(S) + \frac{V_o(S)}{R_f} = 0$$

$$\frac{V_{in}(S)}{R_1} = -V_o(S)\left[S \cdot C_f + \frac{1}{R_f}\right]$$

$$\frac{V_{in}(S)}{R_1} = -V_o(S)\frac{[S R_f C_f + 1]}{R_f}$$

$$\frac{-V_o(S)}{V_{in}(S)} = \frac{R_f}{R_1}\left[\frac{1}{1 + S \cdot R_f C_f}\right]$$

$$\frac{V_o(S)}{V_{in}(S)} = \frac{-R_f}{R_1}\left[\frac{1}{1+S \cdot R_f \cdot C_f}\right]$$

$$\frac{V_o(S)}{V_{in}(S)} = \frac{-R_f/R_1}{1 + S \cdot R_f \cdot C_f}$$

freq response.

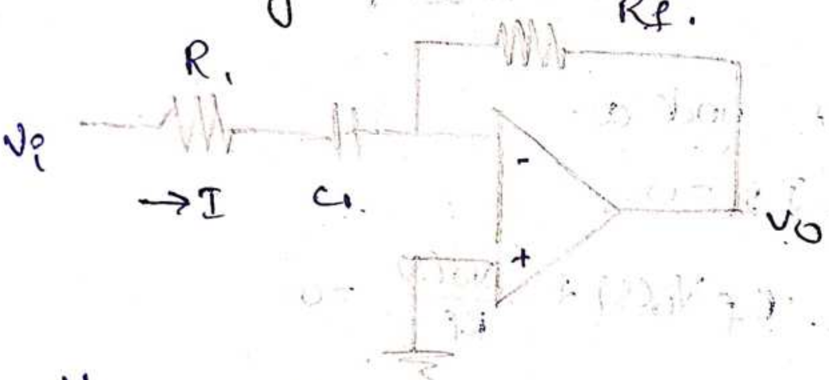
$$A = \frac{-R_f/R_1}{1+j\omega C_f \cdot R_f}$$

$$= \frac{-R_f/R_1}{1+j2\pi f C_f \cdot R_f}$$

$$|A| = \frac{-R_f/R_1}{1+j(f/f_a)}$$

$$= \frac{R_f/R_1}{\sqrt{1+(f/f_a)^2}}$$

28/8/24 Inverting AC - ampeifier.



$$\frac{V_o}{V_i} = \frac{-R_f}{R_1}$$

$$= \frac{-R_f}{R_1 + X_{C_1}}$$

$$= \frac{-R_f}{R_1 + \frac{1}{j\omega C_1}}$$

$$= \frac{-R_f\, j\omega C_1}{j\omega C_1 R_1 + 1}$$

let $R_f = R_1$.

$$\frac{V_o}{V_i} = \frac{-R_f\, j\omega C_1}{1+j\omega C_1 R_f}$$

Both numerator and denominator are divide with $j\omega C_1 R_f$.

$$\frac{V_o}{V_i} = \frac{-1}{1+\frac{1}{j\omega C_1 R_f}}$$

$$= \frac{-1}{1+\frac{1}{j2\pi f C_1 R_f}}$$

$$= \frac{-1}{1+\frac{1}{j(f/f_a)}} \quad \because f_a = \frac{1}{2\pi C_1 R_f}.$$

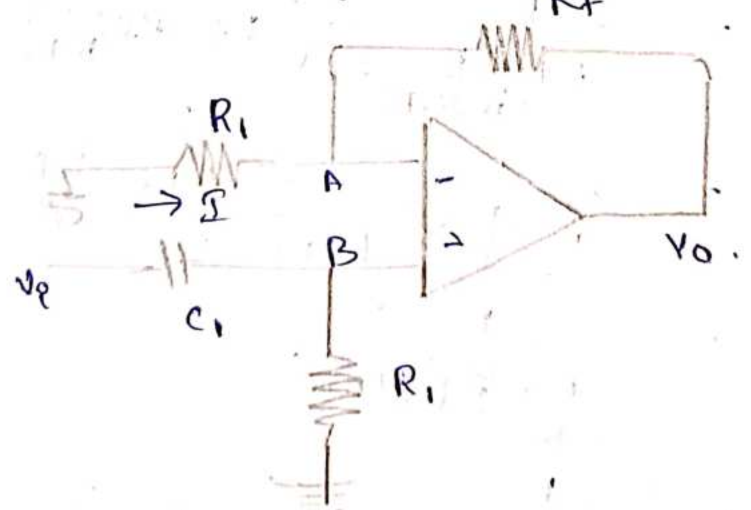
$$\frac{V_o}{V_i} = \frac{1}{1+\frac{1}{j}(f_a/f)}$$

$$= \frac{1}{1-j(f_a/f)}.$$

$$\left|\frac{V_o}{V_i}\right| = \left|\frac{1}{1-j(f_a/f)}\right|$$

$$A = \frac{1}{\sqrt{1+(f_a/f)^2}}.$$

Non-Inverting AC-amplifier.



By virtual ground, $V_A = V_B$.

By Voltage division rule.

$$V_A = \frac{V_o R_1}{R_1+R_f}.$$

$$V_O = V_A\left(\frac{R_1 + R_f}{R_1}\right)$$

$$V_O = V_A\left(1 + \frac{R_f}{R_1}\right)$$

W.K.T $V_A = V_B$

$$V_O = V_B\left(1 + R_f/R_1\right).$$

Node B. voltage division rule.

$$V_B = \frac{V_i R_1}{R_1 + \frac{1}{j\omega C_1}}$$

$$V_B = \frac{V_i\, j\omega C_1 R_1}{1 + \frac{1}{j\omega C_1 R_1}}$$

sub V_B value in V_O.

$$V_O = \left(\frac{V_i\,(j\omega C_1 R_1)}{1 + j\omega C_1 R_1}\right)\left(1 + \frac{R_f}{R_1}\right)$$

$$\frac{V_O}{V_i} = \frac{j\omega C_1 R_1}{1 + j\omega C_1 R_1}\left(1 + \frac{R_f}{R_1}\right).$$

$$= \frac{1}{1 + \frac{1}{j\omega C_1 R_1}}\left(1 + R_f/R_1\right)$$

$$= \frac{1}{1 + \frac{1}{j}\frac{1}{2\pi f C_1 R_1}}\left(1 + \frac{R_f}{R_1}\right)$$

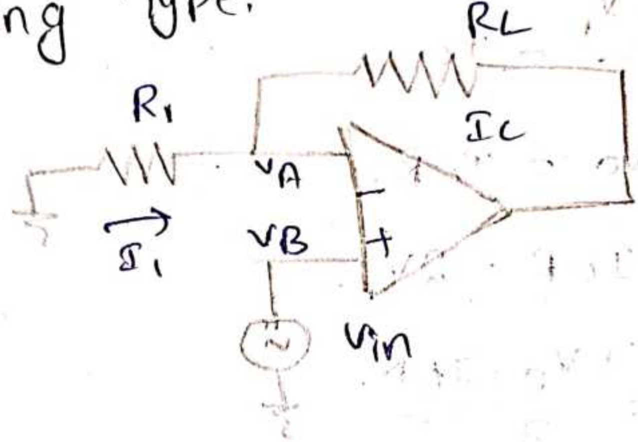
$$= \frac{1}{1 - j\,(f_a/f)} \qquad \frac{1}{1 + \frac{1}{j}\frac{1}{(f/f_a)}}$$

$$\left(\frac{V_O}{V_i}\right) = \frac{1}{\sqrt{1 + (f_a/f)^2}}\left[1 + \frac{R_f}{R_1}\right]$$

Voltage to Current & Current to Voltage Convertor

Voltage to current :-

1. floating Type.



$V_A = V_B = V_{in}$

$V_A = R_L I_L$

$$I_L = \frac{V_A}{R_L}$$

$$I_L = \frac{V_{in}}{R_L}$$

$V_A = R_1 . I_1$

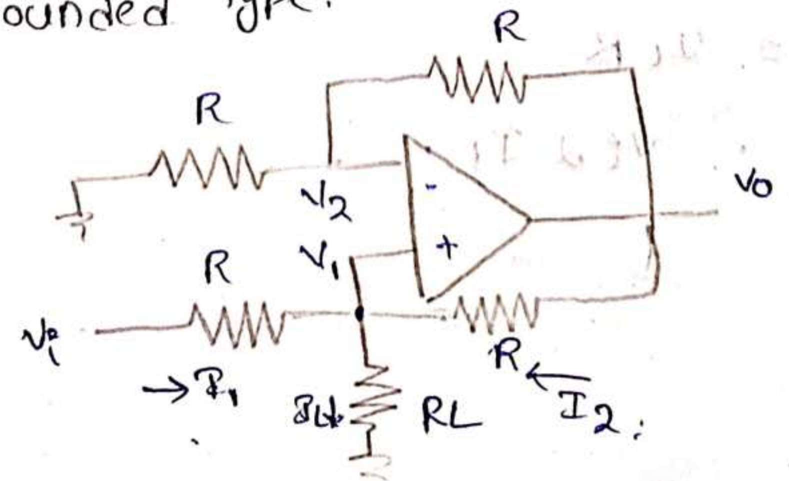
$$I_1 = \frac{V_A}{R_1}$$

$$I_1 = \frac{V_{in}}{R_1}$$

$$I_1 = I_L = \frac{V_{in}}{R_L} \quad [\because R_1 = R_L]$$

$$\boxed{\therefore \; I_L \propto V_{in}}$$

2. Grounded Type.



Apply KCL at V_1

$I_1 + I_2 = I_L$.

$$\frac{V_i - V_1}{R} + \frac{V_0 - V_1}{R} = I_L.$$

$$\frac{V_i - V_1 + V_0 - V_1}{R} = I_L$$

$V_i - 2V_1 + V_0 = I_L R$.

$V_i + V_0 - I_L R = 2V_1$.

$$V_1 = \frac{V_i + V_0 - I_L R}{2}.$$

$$\frac{V_0}{V_i} = \left(1 + R_f/R_1\right)$$ $\because$ Non-interting +

$V_0 = 2V_i$

from ckt,

$$\frac{V_0}{V_i} = \left(1 + R/R\right)$$

$= (1+1)$

$= 2 \quad [\because R_L = R]$

$V_0 = 2V_1$

$$V_0 = \cancel{2}\left(\frac{V_i + V_0 - I_L R}{\cancel{2}}\right)$$

$\cancel{V_0} = V_i + \cancel{V_0} - I_L R$

$V_i = I_L R$

$\therefore V_i \propto I_L$.

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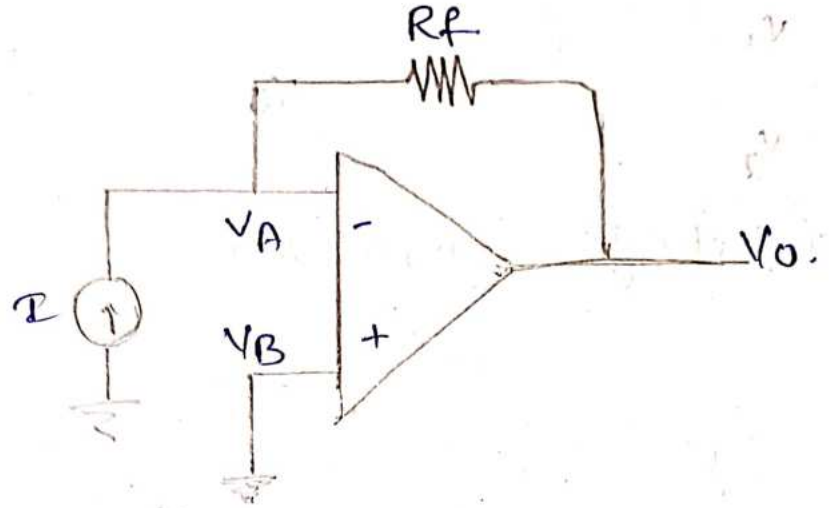
Current to Voltage :



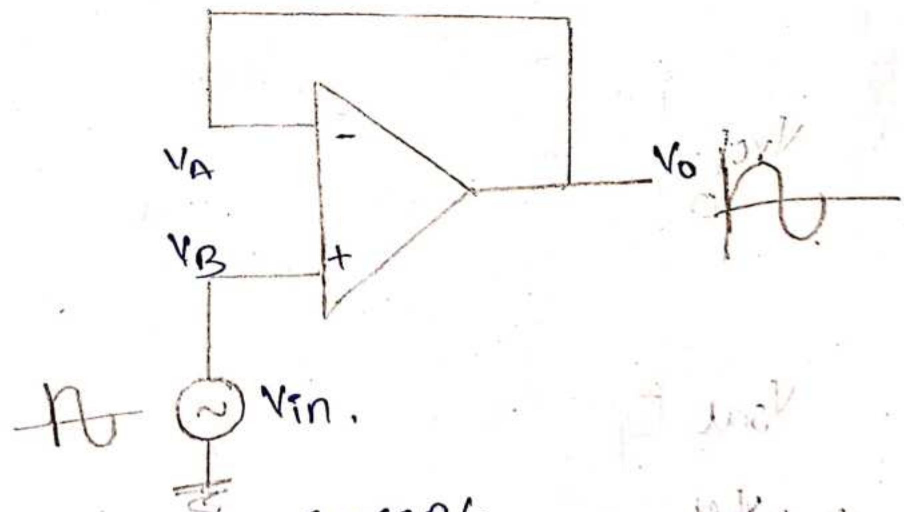
$V_A = V_B = 0$.

$$I = \frac{V_A - V_o}{R_f}$$

$$I = \frac{-V_o}{R_f}$$

$I \propto V_o$.

Buffer :



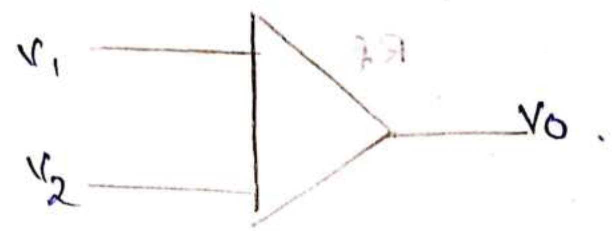
By virtual ground concept,

$V_A = V_B = V_{in}$.

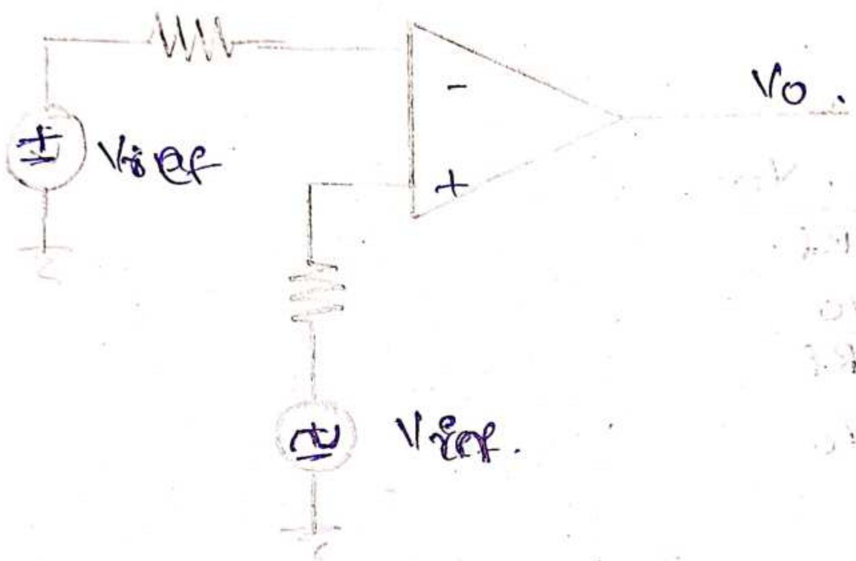
$V_A = V_o$

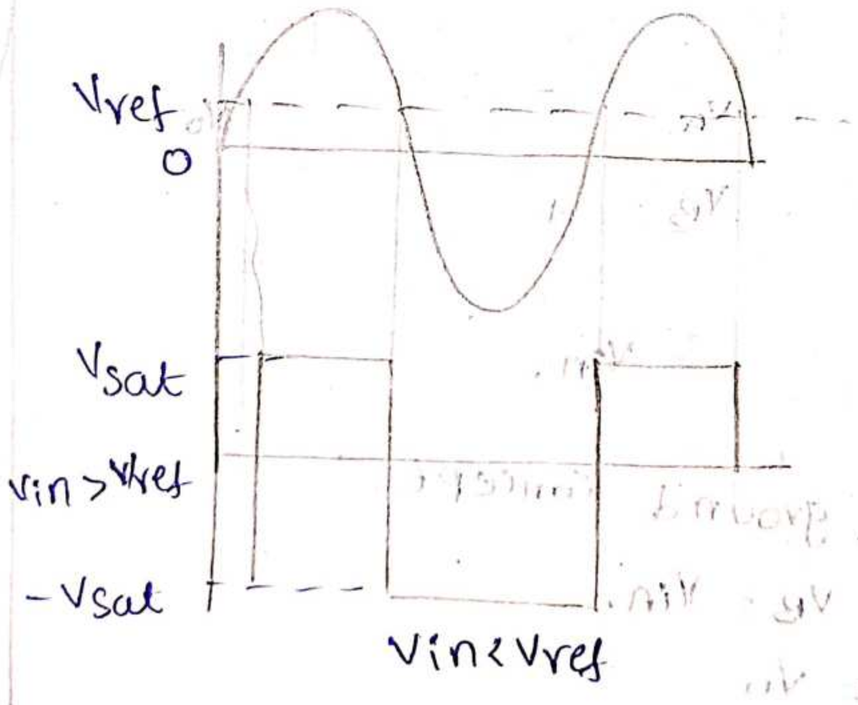
$$\boxed{V_{in} = V_o}$$

Comparators :-

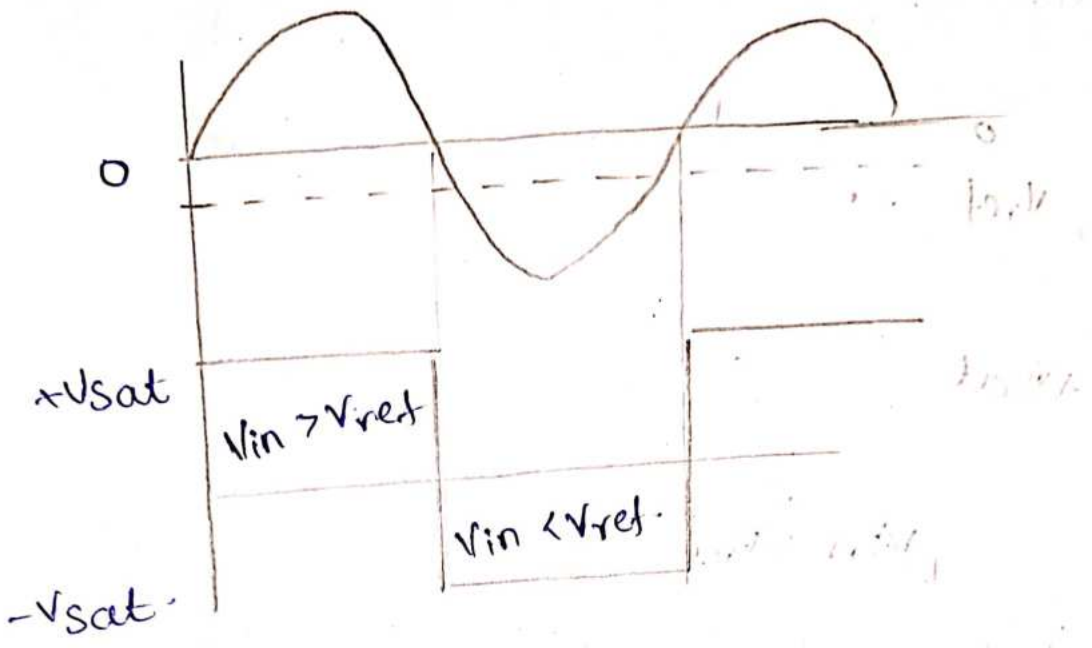


Non- Inverting Comparators.

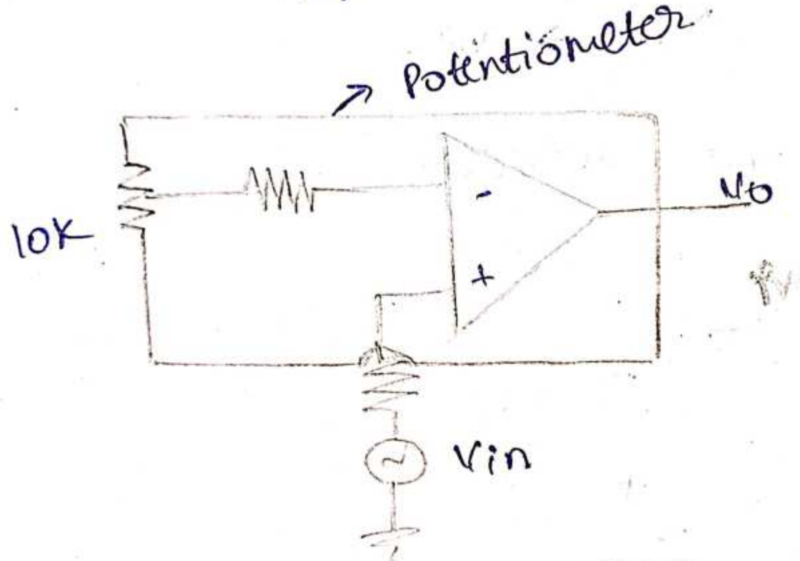




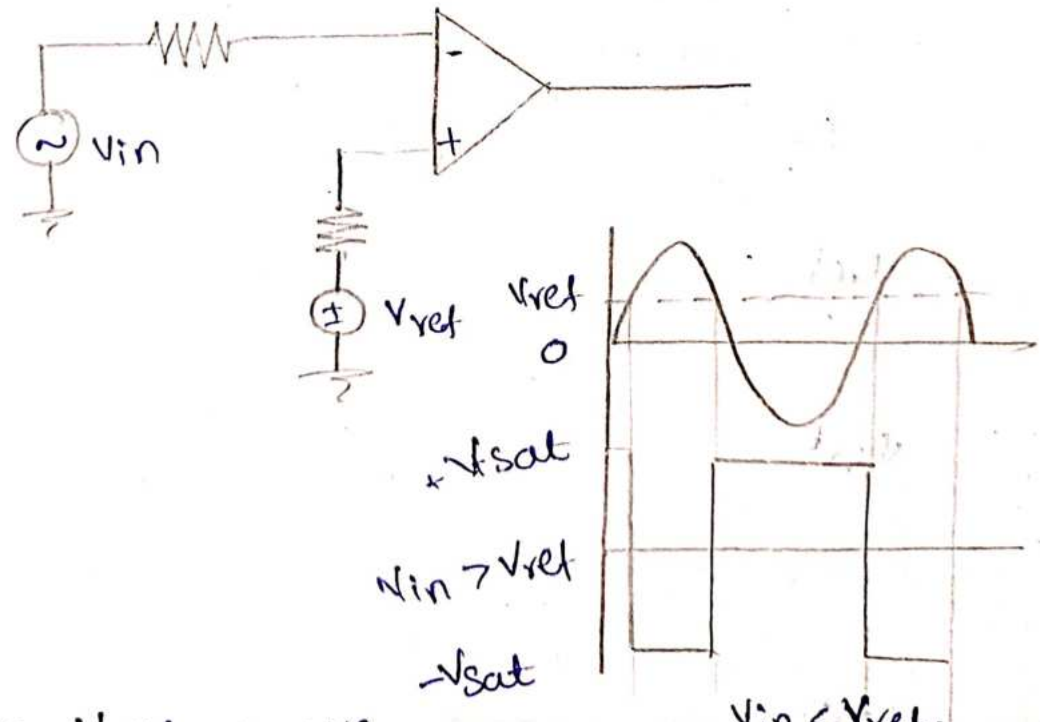
$V_{ref} \rightarrow +ve.$



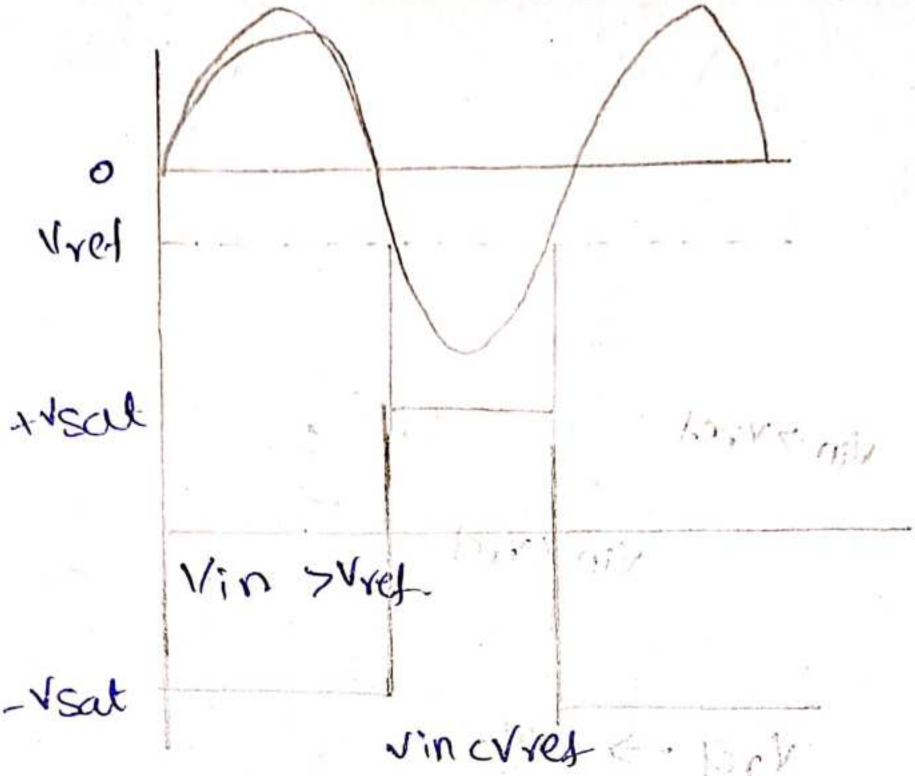
$V_{ref} \rightarrow -ve.$



Inverting COMPORATOR

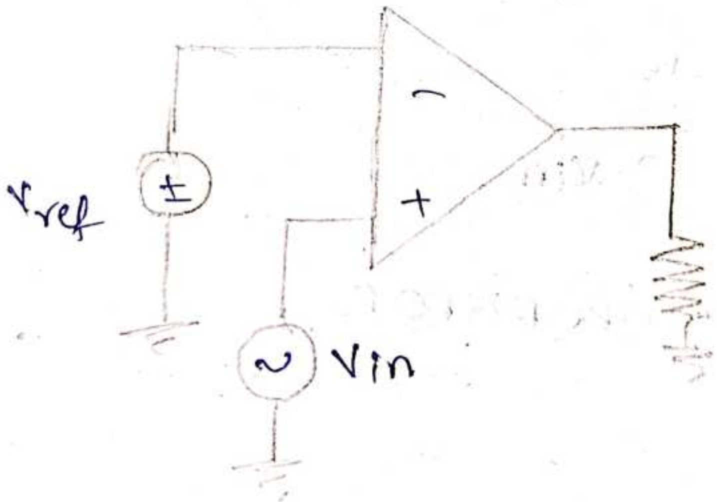


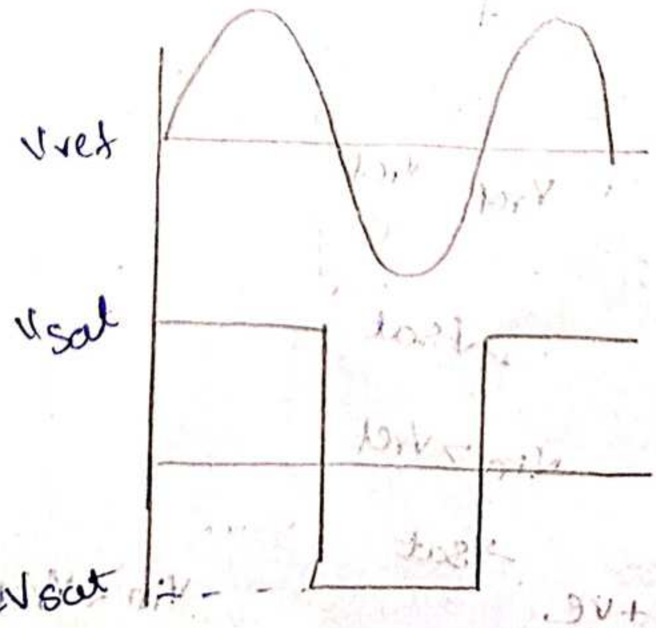
* $V_{ref} \rightarrow +ve.$



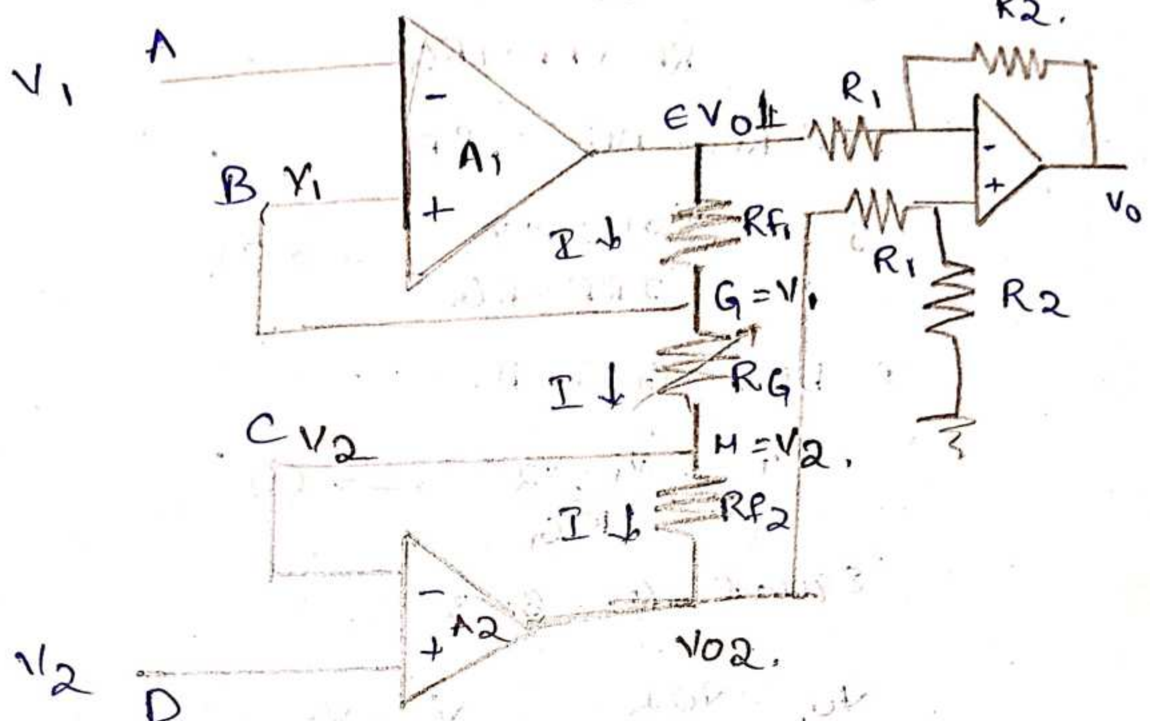
Vref → -ve.

Zero crossing detector.





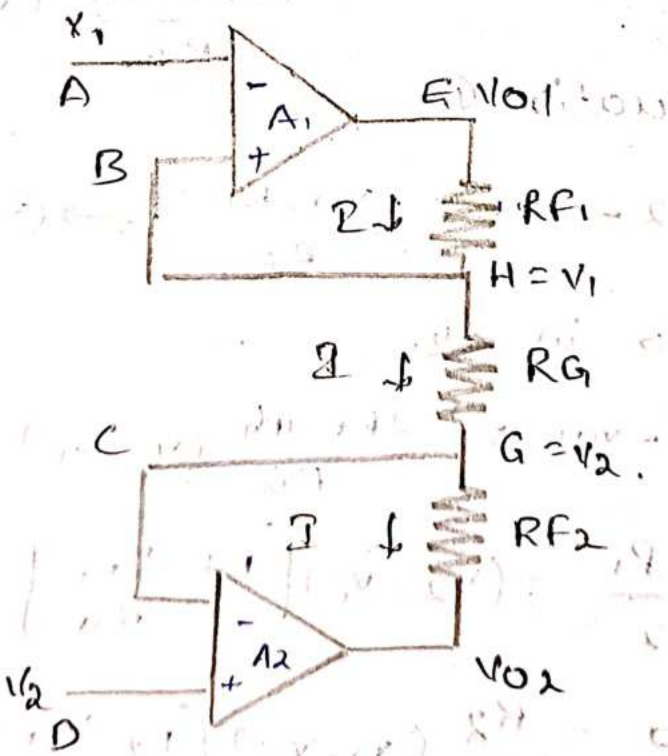
Instrumentation Amplifier.



3rd op-amp output

$$V_o = (V_{o2} - V_{o1})(R_2/R_1) \longrightarrow ①$$

Modified / simplified version of Instruction Amplifier.



$V_0 = (V_{02} - V_{01})\ (R_2/R_1) \rightarrow ①$.

I b/w nodes E & F.

$$I = \frac{V_{01} - V_{02}}{R_{F1} + R_{F2} + R_G}$$

$\because R_{F1} = R_{F2} = R_F$.

$$I = \frac{V_{01} - V_{02}}{2R_F + R_G} \rightarrow ②$$

I b/w G & H.

$$I = \frac{V_1 - V_2}{R_G} \rightarrow ③$$

Equate ② & ③

$$\frac{V_{01} - V_{02}}{2R_F + R_G} = \frac{V_1 - V_2}{R_G}$$

$$\frac{\cancel{-}(V_{02} - V_{01})}{2R_F + G} = \frac{\cancel{-}(V_2 - V_1)}{R_G}$$

$$V_{02} - V_{01} = \frac{(2R_F + R_G)}{R_G}(V_2 - V_1) \rightarrow ④$$

from Equation ①,

$$V_{02} - V_{01} = \frac{V_0 . R_1}{R_2} \rightarrow ⑤$$

5 in ④

$$V_{02} - V_{01} = \frac{2R_F + R_G}{R_G}(V_2 - V_1)$$

$$\frac{V_0 . R_1}{R_2} = (V_2 - V_1)\left[1 + \frac{2R_F}{R_G}\right]$$

$$V_0 = \frac{R_2}{R_1}(V_2 - V_1)\left[1 + \frac{2R_F}{R_G}\right]$$

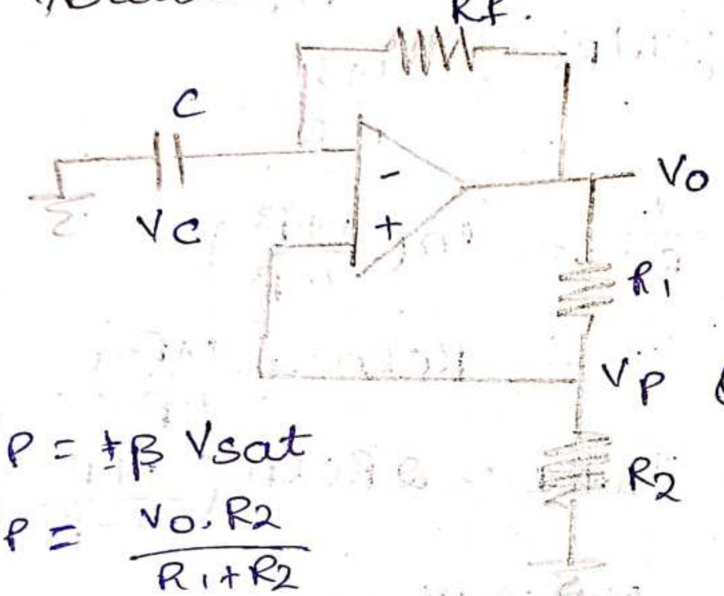
$$V_0 = \frac{R_2}{R_1} V_d \left(1 + 2R_F / R_G\right)$$

$$\frac{V_o}{V_d} = \frac{R_2}{R_1}\left(1 + 2R_f/R_G\right)$$

$$A = \frac{R_2}{R_1}\left(1 + 2R_f/R_G\right).$$

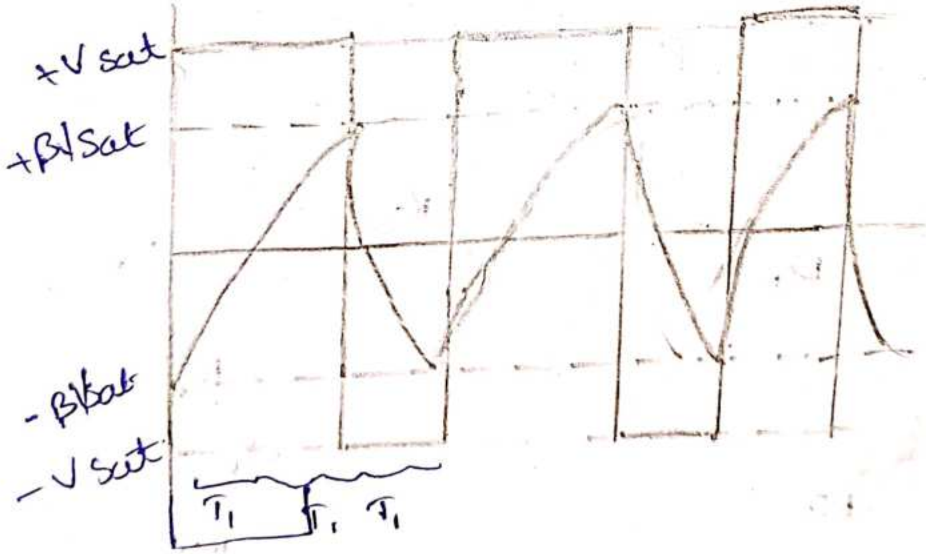
Multivibrator

1. Astable Multivibrator.



$V_P = \pm\beta V_{sat}$

$$V_P = \frac{V_o \cdot R_2}{R_1 + R_2}$$



A.t $t = T_1$.

$V_o = +\beta V_{sat}$.

$V_f = +V_{sat}$

$V_i = -\beta V_{sat}$.

$$\beta V_{sat} = V_{sat} - (V_{sat} + \beta V_{sat})e^{-t/RC}.$$

$$(V_{sat} + \beta V_{sat})\, e^{-t/RC} = V_{sat} - \beta V_{sat}.$$

$$\cancel{V_{sat}}\,(1+\beta)\, e^{-t/RC} = \cancel{V_{sat}}\,(1-\beta)$$

$$(1+\beta)\, e^{-t/RC} = \cancel{V_{sat}}\ 1-\beta$$

$$e^{-T_1/RC} = \frac{1-\beta}{1+\beta}.$$

$$e^{T_1/RC} = \frac{1+\beta}{1-\beta}.$$

$$\frac{T_1}{RC} = \ln\left(\frac{1+\beta}{1-\beta}\right)$$

$$T_1 = RC\ln\left(\frac{1+\beta}{1-\beta}\right).$$

$$T_{total} = 2RC\ln\left(\frac{1+\beta}{1-\beta}\right).$$

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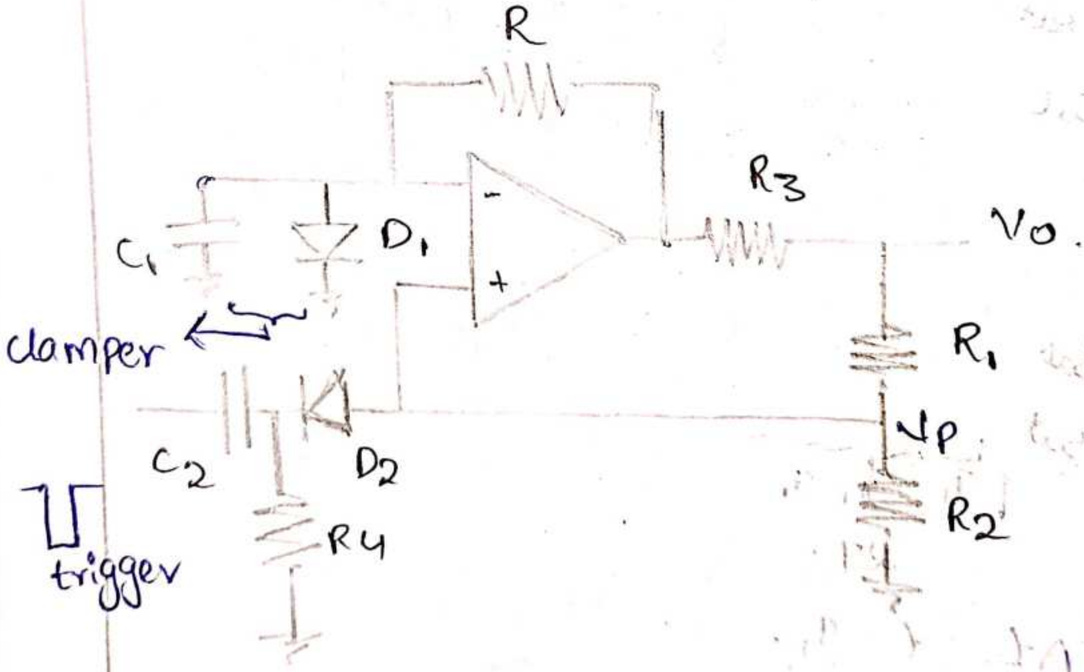
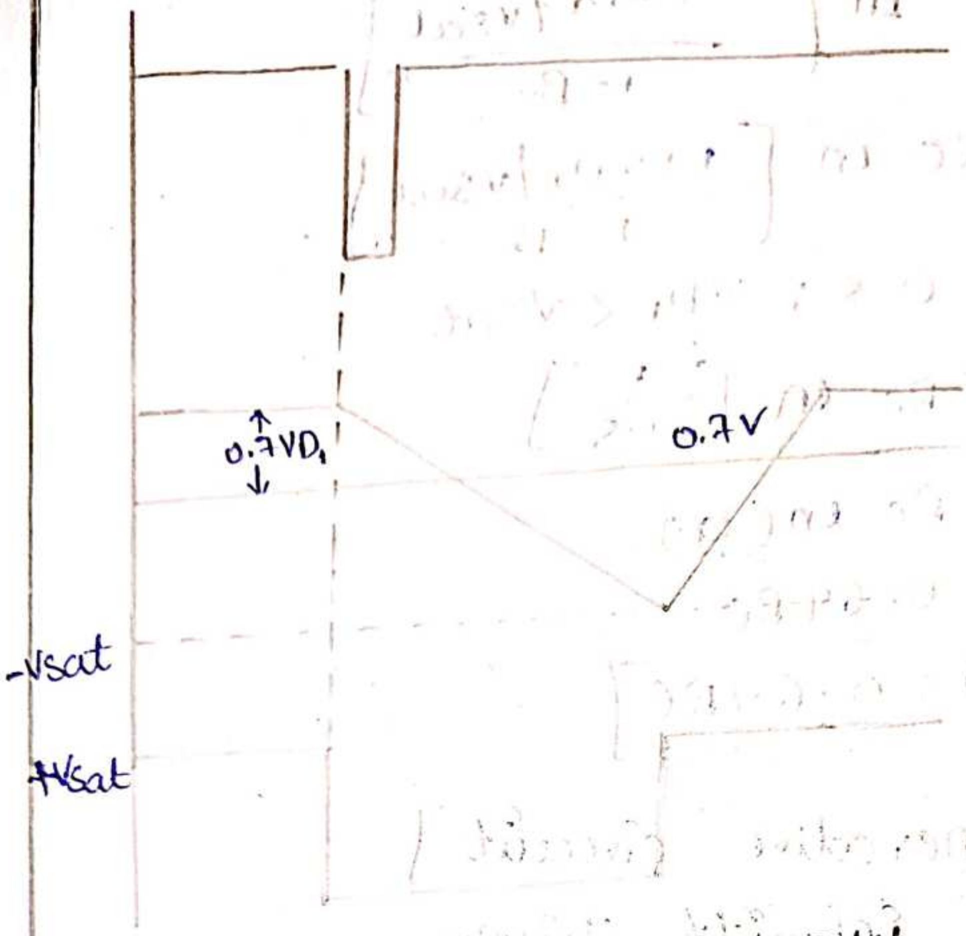
* Monostable. Multivibrator :-





Expression for pulse width.

$$V_o = V_f - (V_f - V_i) e^{-t/RC}$$

$$V_f = -V_{sat}$$

$$V_i = V_{D1}$$

$$V_o = -\beta V_{sat}$$

$$-\beta V_{sat} = -V_{sat} - (-V_{sat} - V_{D1}) e^{-T/RC}$$

$$V_{sat}(1-\beta) = (V_{sat} + V_{D1}) e^{-T/RC}$$

$$\cancel{V_{sat}}(1-\beta) = \cancel{V_{sat}}\left(1 + \frac{V_{D1}}{V_{sat}}\right) e^{-T/RC}$$

$$1-\beta = \left[1 + \frac{V_{D1}}{V_{sat}}\right] e^{-T/RC}$$

$$\frac{1-\beta}{1+\frac{V_{D1}}{V_{sat}}} = e^{-T/RC}$$

$$e^{+T/RC} = \frac{1+\frac{V_{D1}}{V_{sat}}}{1-\beta}$$

$$\frac{T}{RC} = \ln\left[\frac{1 + V_{D1}/V_{sat}}{1-\beta}\right]$$

$$T = RC \ln\left[\frac{1 + V_{D1}/V_{sat}}{1-\beta}\right].$$

$\beta = 0.5$; $V_{D1} < V_{sat}$.

$$T = RC \ln\left[\frac{1}{0.5}\right]$$

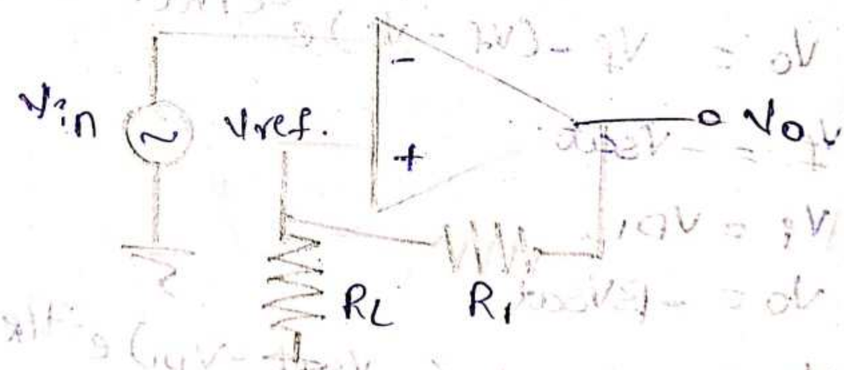
$$= RC \ln(2)$$

$$= 0.69\,RC.$$

$$\boxed{\therefore T = 0.69RC}$$

13/9/24 [Regenerative circuit]

Schmitt Trigger



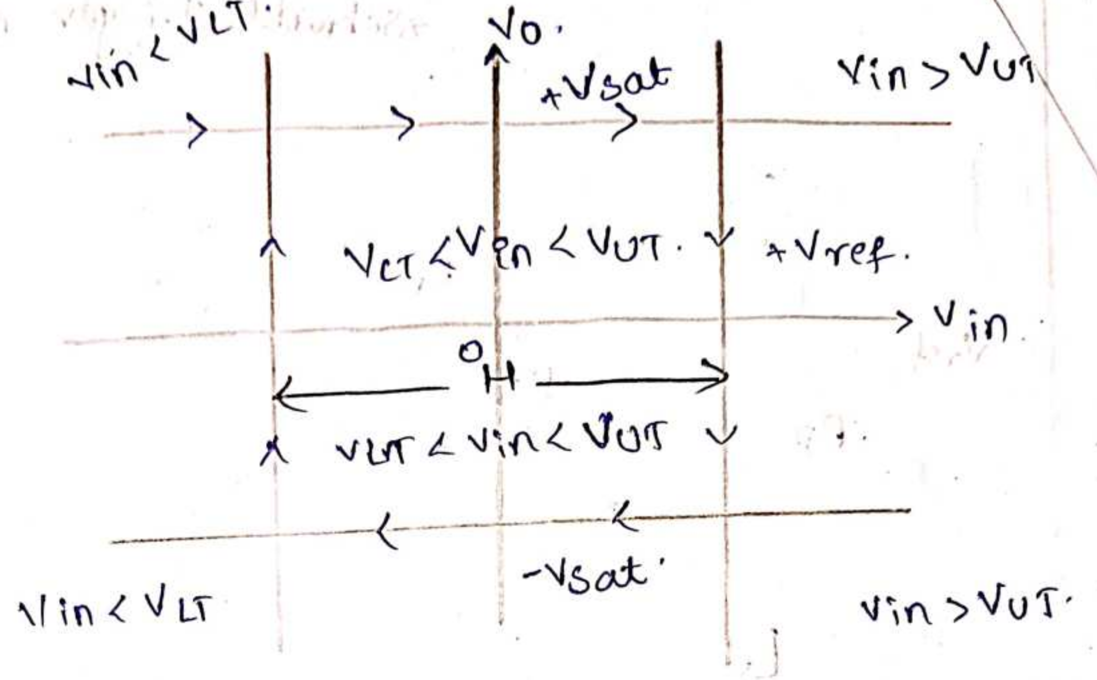
$$V_{ref} = \frac{V_o \cdot R_L}{R_1 + R_L}$$

$$= \frac{\pm V_{sat} \cdot R_L}{R_1 + R_L}$$

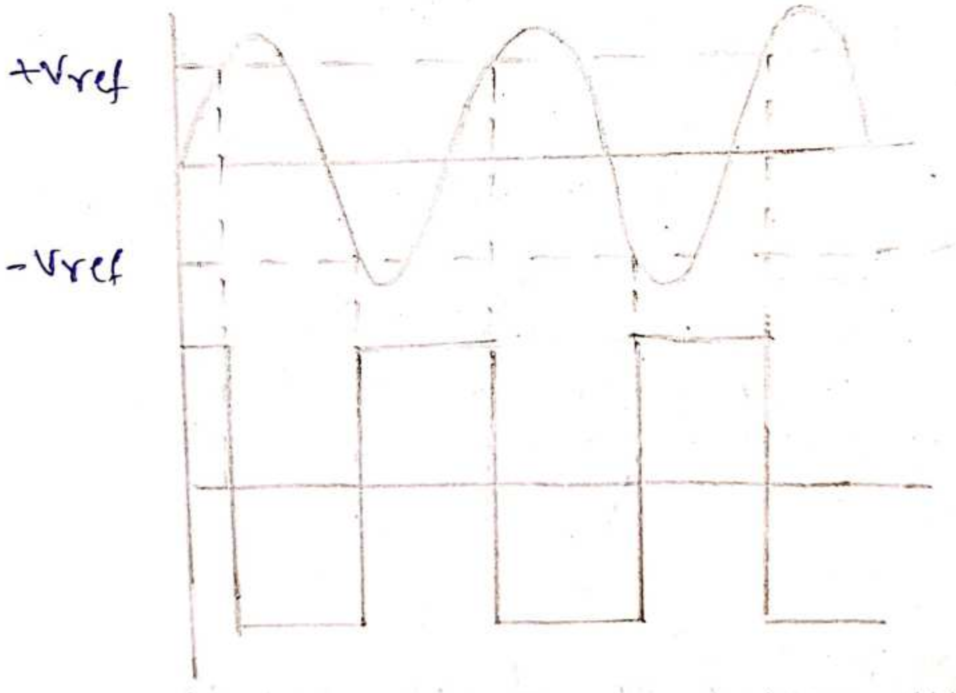
$$UTP = \frac{+V_{sat} \cdot R_L}{R_1 + R_L}$$

$$LTP = \frac{-V_{sat} \cdot R_L}{R_1 + R_L}.$$

H·



Hystersis curve.

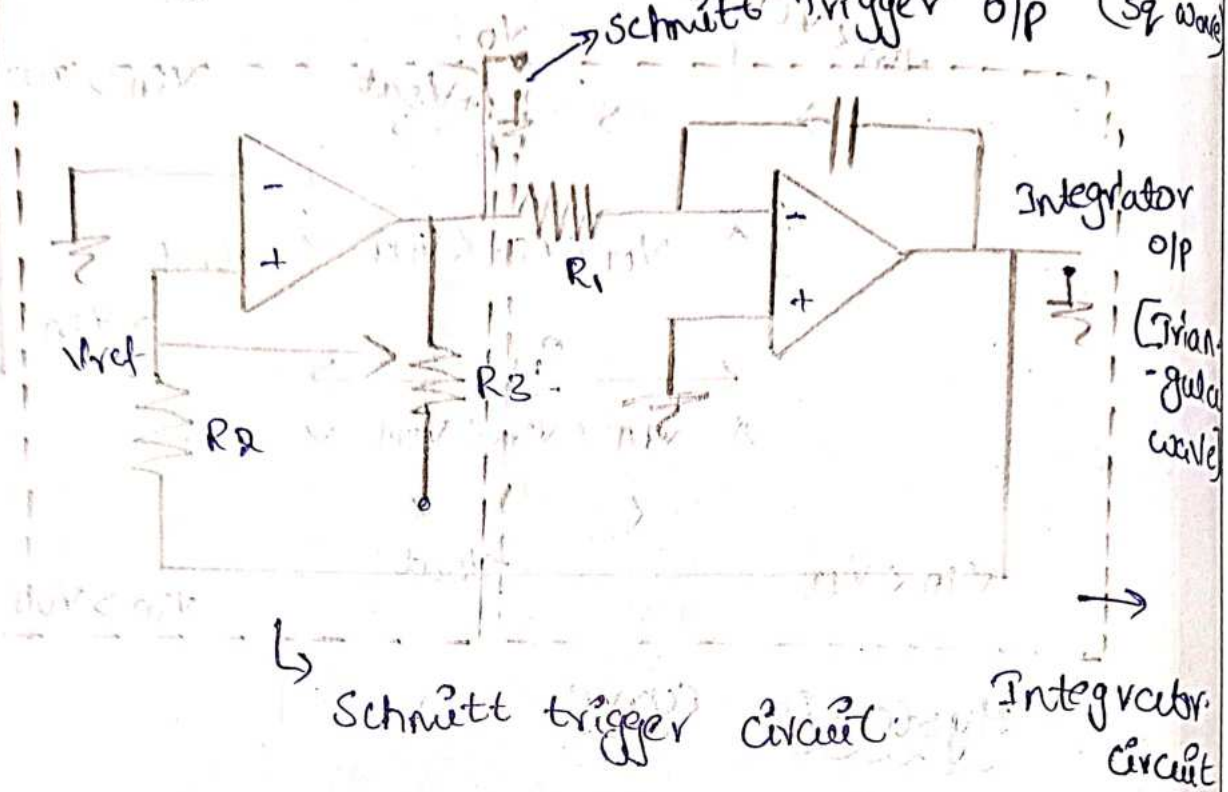


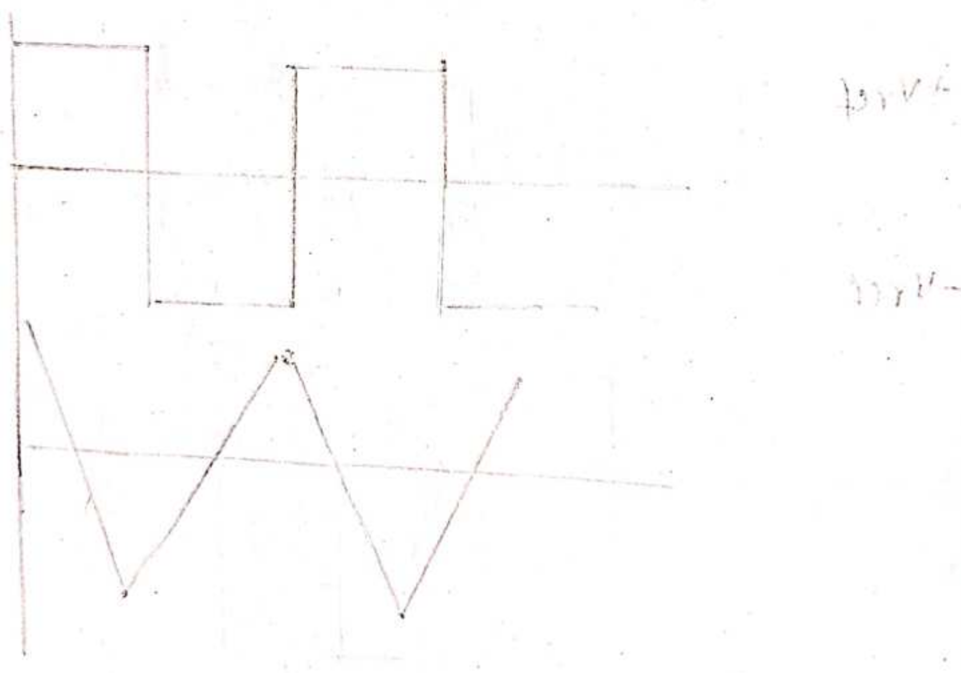
$V_{in} < V_{LT} \rightarrow$ state change.

$V_{in} > V_{UT} \rightarrow$ State change.

$V_{LT} < V_{in} < V_{UT} \rightarrow$ No change

Triangular and Square Wave Generators.





19/9/24 Fundamentals of Log Amplifier.

Basic equation of Diode.

$$I = I_0 . e^{V/nV_T} - 1.$$

Forward bias, Diode - ON.

$$e^{V/nV_T} >> 1.$$

$$I_f = I_0 . e^{V/nV_T}$$

$I_f \rightarrow$ Diode current

$V \rightarrow$ Diode voltage.

$I_0 \rightarrow$ reverse saturation current.

$\eta = 1$ for Si

2 for Ge.

$V_T = KT$.

$k \rightarrow$ boltsmen constant.

$T \rightarrow$ Temperature.

$I_f = I_0 \cdot e^{V/\eta V_T}$

ln apply on both sides

$\ln(I_f) = (\ln) I_0 \cdot e^{V/\eta V_T}$

$\ln I_f = \ln I_0 + \ln e^{V/\eta V_T}$

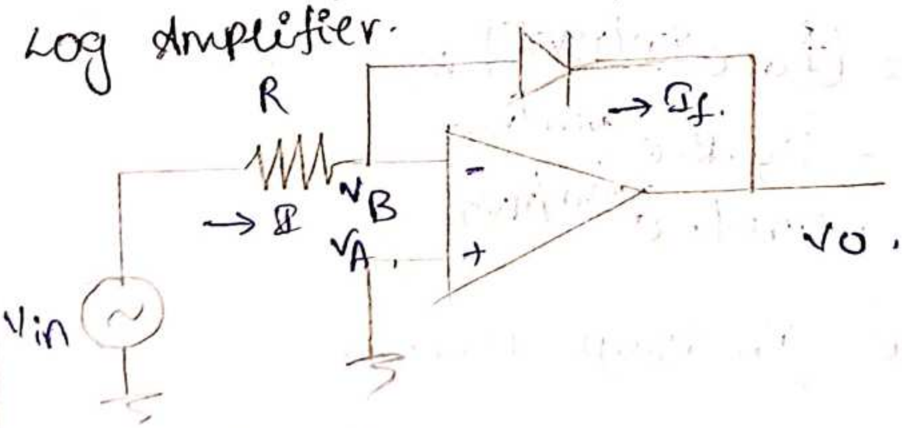
$\ln I_f - \ln I_0 = \frac{V}{\eta V_T}$.

$\frac{V}{\eta V_T} = \ln\left[\frac{I_f}{I_0}\right]$

$V = \ln\left[\frac{I_f}{I_0}\right] \eta V_T$.

$\boxed{\therefore V = \eta V_T \ln\left[\frac{I_f}{I_0}\right]}$

Log Amplifier.



$I = I_f$.

$\frac{V_{in} - V_B}{R} = I_f$.

$\boxed{\frac{V_{in}}{R} = I_f}$

By virtual ground

$V_A = V_B$

$V_B = 0 \because V_A = 0$.

voltage across Diode,

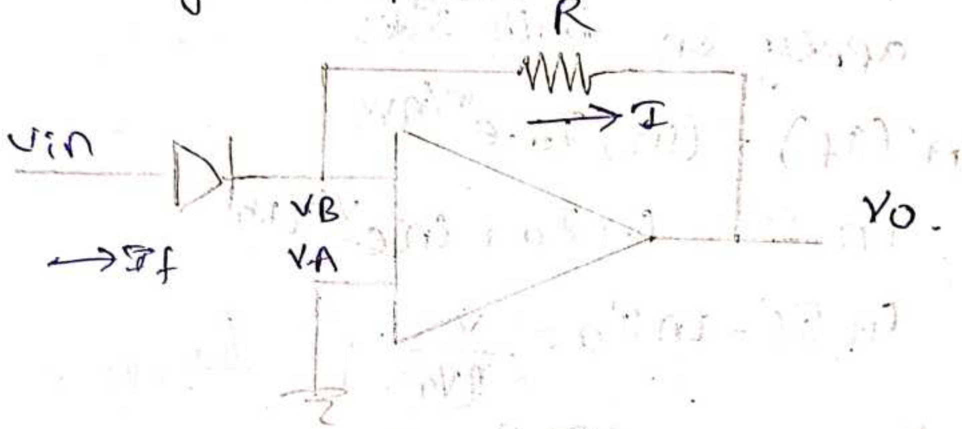
$$V_B - V_O = \eta V_T \ln\left(\frac{I_f}{I_O}\right)$$

$$-V_O = \eta V_T \ln\left(\frac{V_{in}}{R I_O}\right)$$

$$-V_O = \eta V_T \ln\left(\frac{V_{in}}{V_{ref}}\right)$$

$$\boxed{V_O = -\eta V_T \ln\left(\frac{V_{in}}{V_{ref}}\right)}$$

Anti-Log Amplifier



$$V_A = 0 = V_B$$

$$I_f = I.$$

$$I_O \cdot e^{V_{in}/\eta V_T} = \frac{V_B - V_O}{R}$$

$$I_O \cdot e^{V_{in}/\eta V_T} = \frac{-V_O}{R}$$

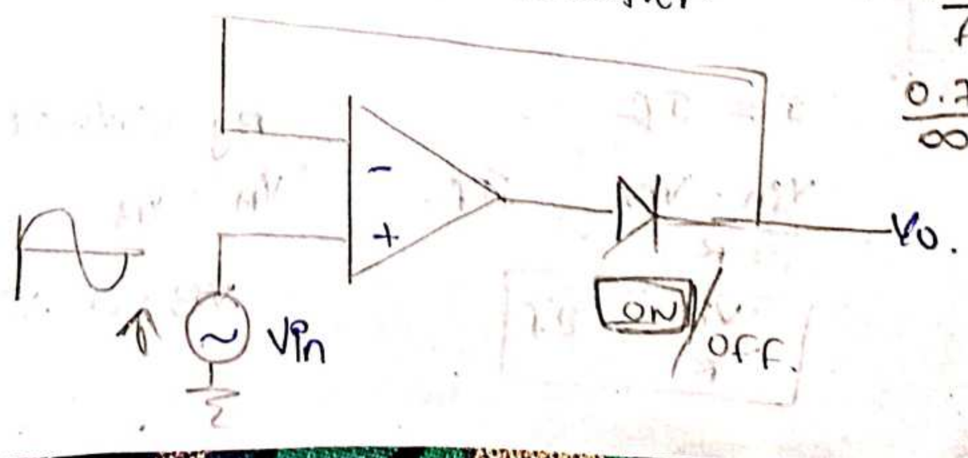
$$-V_O = \left[I_O \cdot e^{V_{in}/\eta V_T}\right] R$$

$$V_O = -I_O \cdot R \cdot e^{V_{in}/\eta V_T}$$

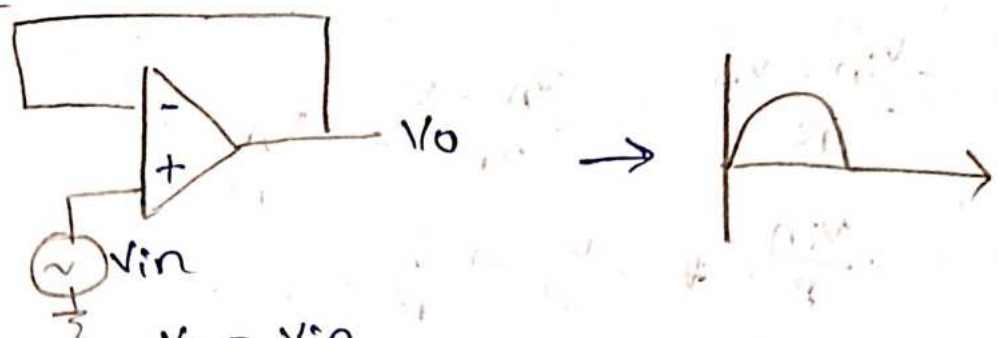
$$\boxed{V_O = -V_{ref} \cdot e^{V_{in}/\eta V_T}}$$

Half-wave Precision Rectifier. $0.7 \rightarrow \frac{V_O}{A_{OL}}$

$\frac{0.7}{\infty} \uparrow$



Case - i, $V_i > 0V$.

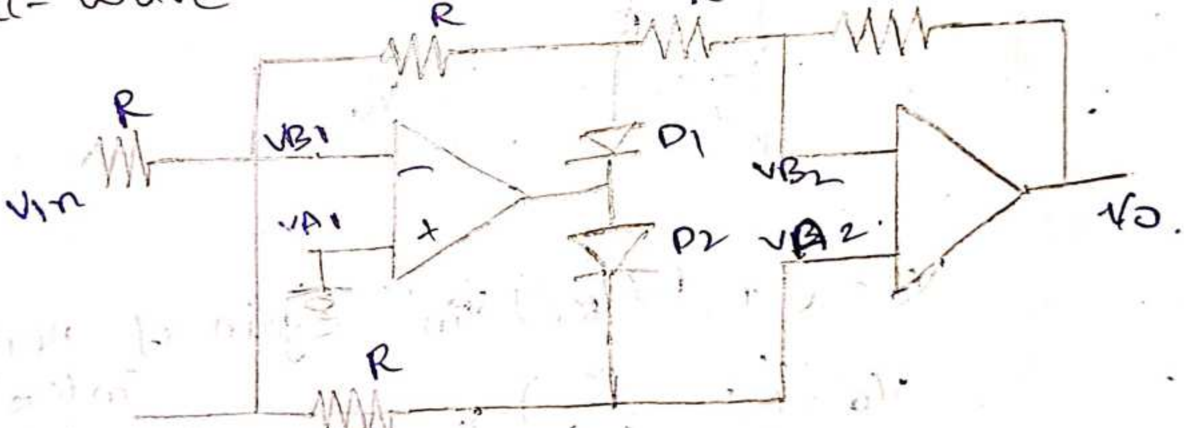


$V_o = V_{in}$

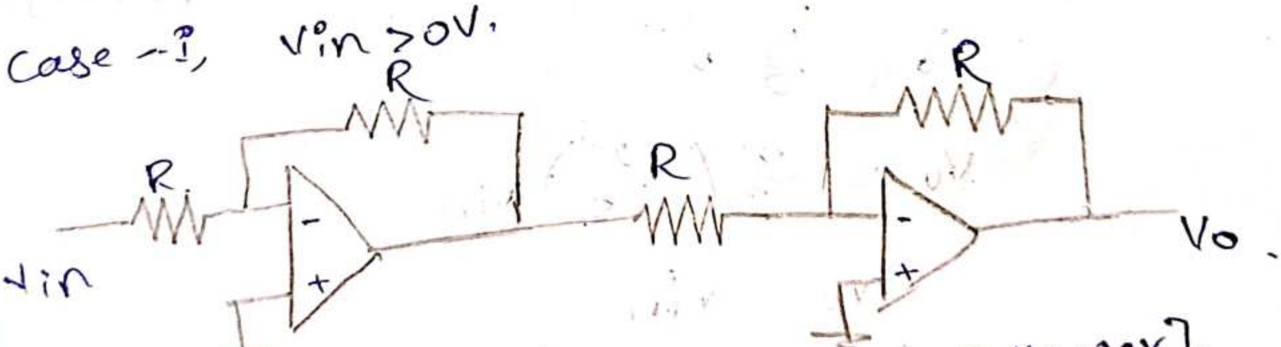
Case - ii, $V_i < 0V$.

$V_o = 0$.

Full- wave Rectifier. precision.

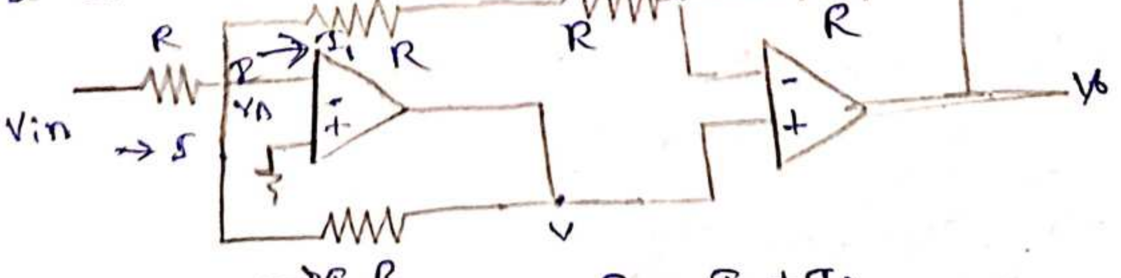


Case - i, $V_{in} > 0V$.



$V_o = V_{in}$. [voltage follower].

case - ii, $V_{in} < 0V$.



$I = I_1 + I_2$.

$I = I_1 + I_2.$

$$\frac{V_{in} - V_A}{R} = \frac{V_A - V}{2R} + \frac{V_A - V}{R}$$

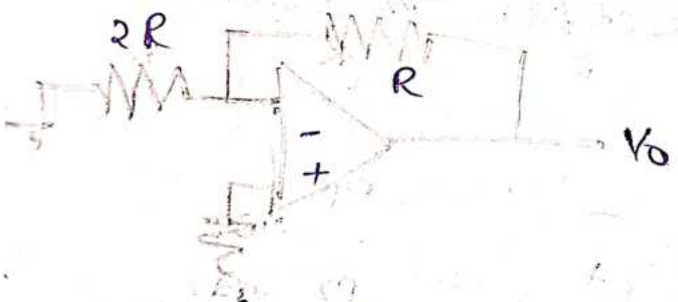
$$\frac{V_{in}}{R} + \frac{V}{2R} + \frac{V}{R} = 0$$

$$\frac{2v_{in} + v + 2v}{2R} = 0$$

$$2v_{in} + 3v = 0$$

$$3v = -2v_{in}$$

$$v = -\frac{2}{3} v_{in}.$$



$V_o = (1 + R_f/R_1) V_{in}$ [gain of non-inverting amplifier].

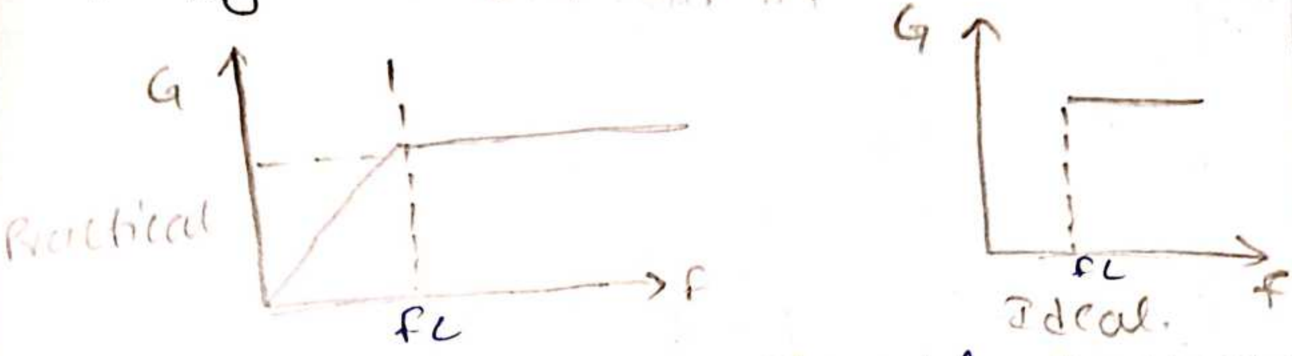
$$V_o = \left(1 + \frac{R}{2R}\right) v$$

$$V_o = \frac{3}{2} v$$

$$V_o = \frac{3}{2}\left(-\frac{2}{3}\right) v_{in}$$

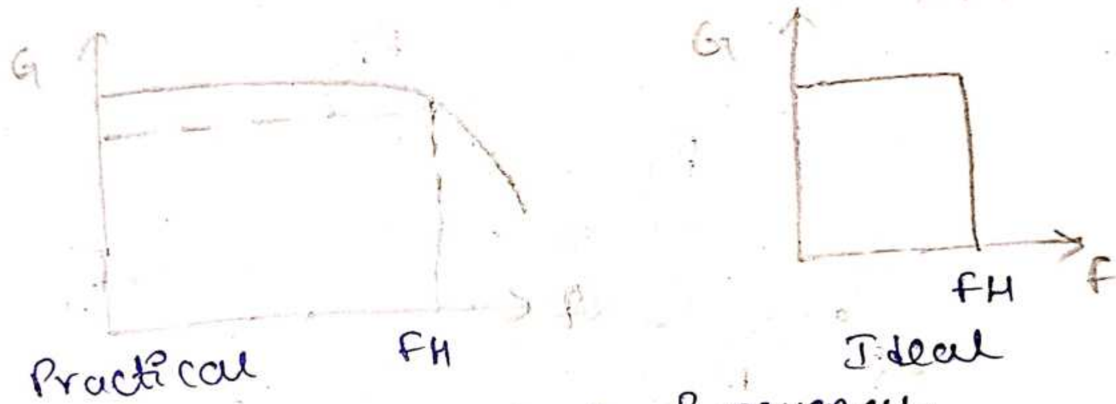
$$\therefore V_o = -V_{in}$$

1. High Pass filter



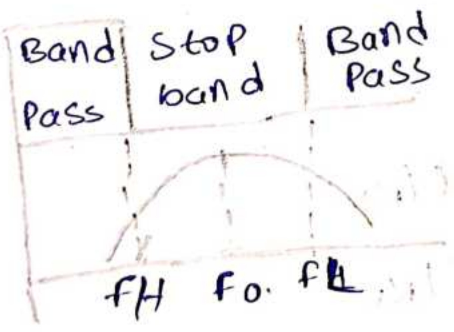
HPF which allows High Frequency.

2. Low Pass filter



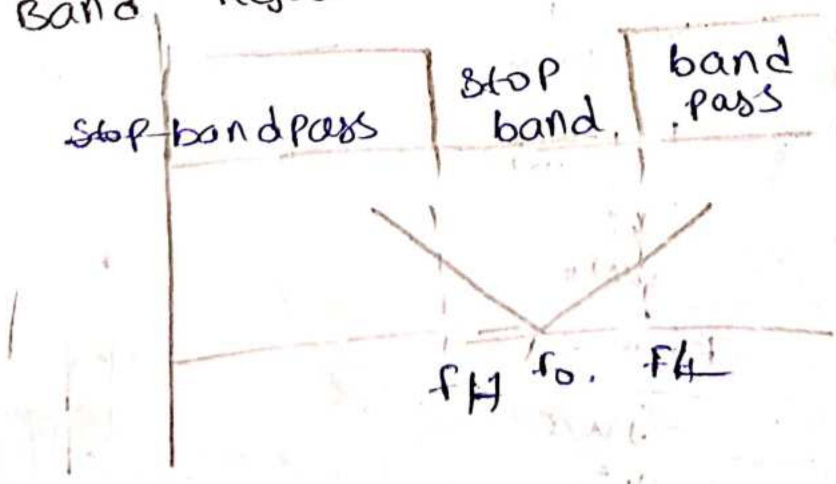
LPF which allows low frequency.

3. Band Pass filter



It allows particular band pass.

4. Band Reject filter.



All Pass filter.

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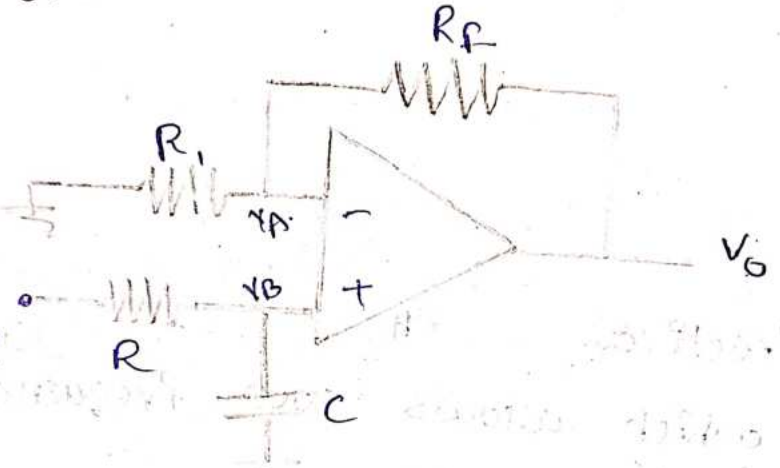
First order LPF



input is given to the non-inverting terminal,

$$\frac{V_o}{V_i} = 1 + R_f/R_1$$

$$V_o = V_i(1 + R_f/R_1).$$

$$V_o = V_B(1 + R_f/R_1)$$

By voltage divide rule.

$$V_B = \frac{V_i \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

$$V_B = \frac{\frac{V_i}{j\omega C}}{\frac{Rj\omega C + 1}{j\omega C}}$$

$$\boxed{V_B = \frac{V_i}{1 + j\omega RC}.}$$

sub V_B value in V_o.

$$V_o = (1 + R_f/R_1)\left(\frac{V_i}{1+j\omega RC}\right)$$

$$V_o = \frac{A_o \cdot V_i}{1+j2\pi fRC}$$

$$\frac{V_o}{V_i} = \frac{A_o}{1+j(f/f_H)}$$

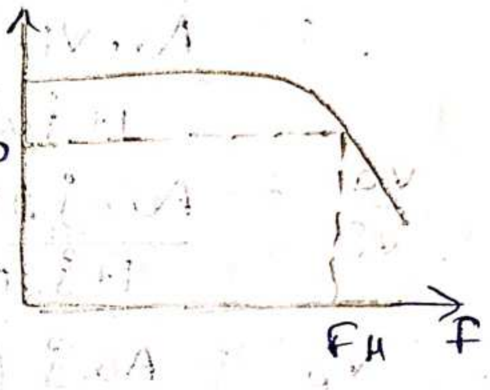
$$\left|\frac{V_o}{V_i}\right| = \frac{A_o}{\sqrt{1+(f/f_H)^2}}$$

$f = 0$, $G = A_o$

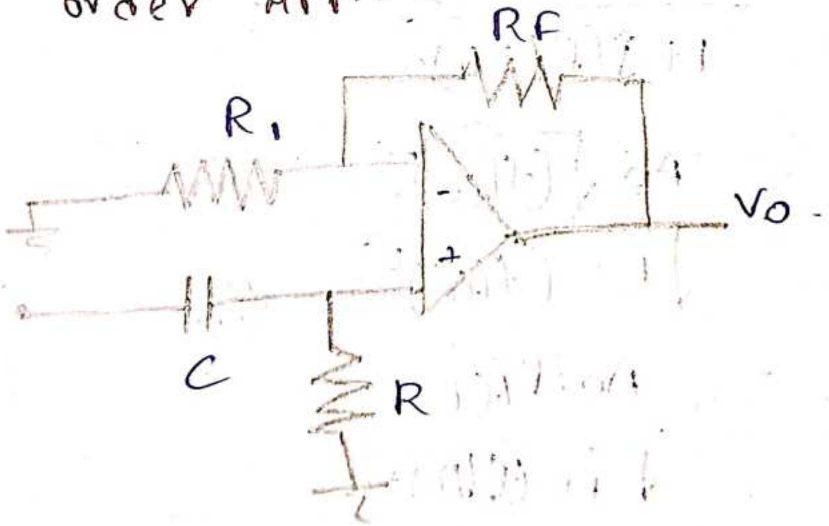
$f < f_H$, $G = A_o$

$f = f_H$, $G < \frac{A_o}{\sqrt{2}}$

$f > f_H$, $G < \frac{A_o}{\sqrt{2}} \downarrow$.



first order HPF.



$$\frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_1}\right)$$

$$V_o = V_i(1 + R_f/R_1)$$

$$V_o = (1 + R_f/R_1)V_B$$

By voltage divide rule,

$$V_B = \frac{V_i \cdot R}{C+R} = \frac{V_i \cdot R}{R + \frac{1}{j\omega C}}$$

$$V_B = \frac{\frac{V_i \cdot R}{R j\omega C + 1}}{j\omega C}$$

$$\boxed{V_B = \frac{V_i \cdot j\omega RC}{1 + j\omega RC}}$$

Sub V_B value in V_0.

$$V_0 = \left(\frac{V_i \cdot j\omega RC}{1 + j\omega RC}\right)\left(1 + R_f/R_1\right)$$

$$= \frac{A_0 \cdot V_i \cdot j\omega RC}{1 + j\omega RC}$$

$$\frac{V_0}{V_i} = \frac{A_0 \cdot j 2\pi f RC}{1 + j 2\pi f RC}$$

$$\frac{V_0}{V_i} = \frac{A_0 \cdot j\,(f/f_L)}{1 + j(f/f_L)}$$

$$\left|\frac{V_0}{V_i}\right| = \frac{A_0 \sqrt{(f/f_L)^2}}{\sqrt{1 + (f/f_L)^2}}$$

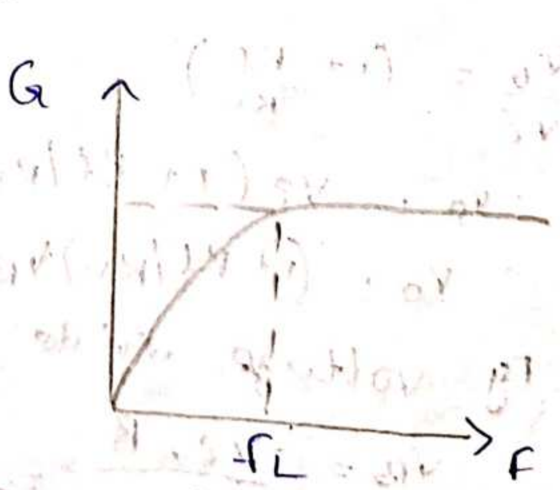
$$\left|\frac{V_0}{V_i}\right| = \frac{A_0 \cdot f/f_L}{\sqrt{1 + (f/f_L)^2}}$$

$f = 0$, $G = 0$

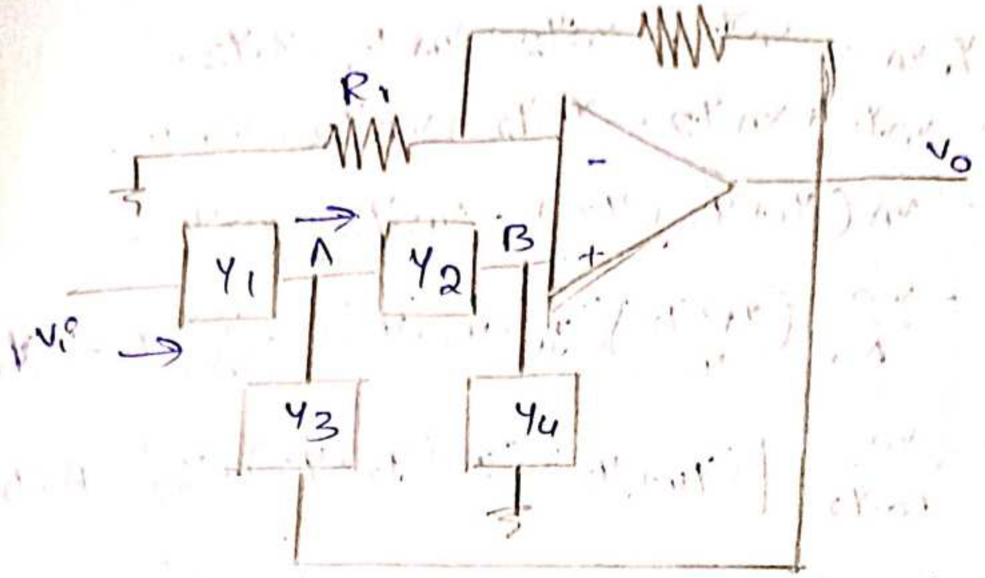
$f < f_L$, $G \uparrow$

$f = f_L$, $G = \frac{A_0}{\sqrt{2}}$

$f > f_L$, $G = A_0$.



second order



Apply KCL at Node A,

$$Y_1(V_i - V_A) = (V_A - V_B)Y_2 + (V_A - V_o)Y_3 \quad \text{--- (1)}$$

Apply KCL at Node B,

$$V_B Y_4 = (V_A - V_B)Y_2$$

$$V_B Y_4 = V_A Y_2 - V_B Y_2$$

$$V_B Y_4 + V_B Y_2 = V_A Y_2$$

$$V_B[Y_2 + Y_4] = V_A Y_2$$

$$V_A = \frac{V_B[Y_2 + Y_4]}{Y_2}$$

$$V_A = V_B\left[\frac{Y_4 + Y_2}{Y_2}\right]$$

$$\frac{V_o}{V_B} = (1 + R_f/R_1)$$

$$V_o = V_i\,(1 + R_f/R_1)$$

$$V_o = V_B(1 + R_f/R_1)$$

$$V_o = V_B \cdot A_o$$

$$\frac{V_o}{V_B} = A_o \quad \text{--- (2)}$$

$$V_B = \frac{V_o}{A_o}$$

$$V_A = \frac{V_o}{A_o}\left(\frac{Y_4 + Y_2}{Y_2}\right) \rightarrow (3)$$

from equation 1,

$$V_i Y_1 - Y_1 V_A = V_A Y_2 - V_B Y_2 + V_A Y_3 - V_o Y_3.$$

$$V_i Y_1 = V_A Y_1 + V_A Y_2 - V_B Y_2 + V_A Y_3 - V_o Y_3.$$

$$V_i Y_1 = V_A (Y_1 + Y_2 + Y_3) - V_B Y_2 - V_o Y_3.$$

$$= \frac{V_o}{A_o}\left(\frac{Y_4 + Y_2}{Y_2}\right)(Y_1 + Y_2 + Y_3) - \frac{V_o}{A_o} Y_2 - Y_3 V_o.$$

$$V_i Y_1 = \frac{V_o}{A_o . Y_2}\left[(Y_4 + Y_2)(Y_1 + Y_2 + Y_3) - Y_2^2 - A_o Y_2 Y_3\right]$$

$$= \frac{V_o}{A_o . Y_2}\left[Y_4 Y_1 + Y_4 Y_2 + Y_4 Y_3 + Y_2 Y_1 + Y_2^2 + Y_2 Y_3 - Y_2^2 - A_o Y_2 Y_3\right]$$

$$V_i Y_1 = \frac{V_o}{A_o . Y_2}\left[Y_2 Y_1 + Y_4 (Y_1 + Y_2 + Y_3) + Y_2 Y_3 (1 - A_o)\right]$$

$$\boxed{\frac{V_o}{V_i} = \frac{A_o . Y_1 . Y_2}{Y_2 Y_1 + Y_4 (Y_1 + Y_2 + Y_3) + Y_2 Y_3 (1 - A_o)}}$$

23/9/24 Second order LPF :

$$\frac{V_o}{V_i} = \frac{A_o . Y_1 . Y_2}{Y_2 Y_1 + Y_4 [Y_1 + Y_2 + Y_3] + Y_2 Y_3 (1 - A_o)}$$

$$= \frac{A_o \left(\frac{1}{R}\right)\left(\frac{1}{R}\right)}{\left(\frac{1}{R}\right)\left(\frac{1}{R}\right) + SC\left[\frac{1}{R} + \frac{1}{R} + SC\right] + \frac{1}{R}(SC)[1 - A_o]}$$

$$= \frac{\frac{A_o}{R^2}}{\frac{1}{R^2} + SC\left[\frac{R + R + SCR}{R}\right] + \frac{1}{R}[SC](1 - A_o)}$$

$$\frac{\frac{A_o}{R^2}}{1 + \frac{R^2 SC\left[\frac{2R + SCR}{R^2}\right] + RSC(1 - A_o)}{}}$$

$$\frac{A_o}{1 + RSC[2R + 1.}$$

$$= \frac{A_0 (1/R^2)}{\frac{1}{R^2} + SC\left(\frac{1}{R} + \frac{1}{R} + SC\right) + \left(\frac{1}{R}\right)(SC)(1-A_0)}$$

$$= \frac{A_0}{1 + SCR^2\left[\frac{2}{R} + SC\right] + \frac{R^2}{R} SC(1-A_0)}$$

$$= \frac{A_0}{1 + SCR^2\left(\frac{2 + SCR}{R}\right) + RSC(1-A_0)}$$

$$= \frac{A_0}{1 + SCR(2 + SCR) + SCR(1-A_0)}$$

$$= \frac{A_0}{1 + SCR(2 + 1 + SCR - A_0)}$$

$$= \frac{A_0}{1 + SCR(3 + SRC - A_0)}$$

$$\frac{V_o}{V_i} = \frac{A_0}{1 + (SCR)^2 + SCR(3 - A_0)}$$

$RC = \frac{1}{\omega_n}$, $\alpha = 3 - A_0$, $S = j\omega$.

$\omega_n \rightarrow$ upper cut off frequency.

$$\frac{V_o}{V_i} = \frac{A_0}{1 + \left(\frac{S}{\omega_n}\right)^2 + \frac{S}{\omega_n}(\alpha)}$$

This resembles $S_n^2 + S_n \alpha + 1$ [second order characteristic equation]

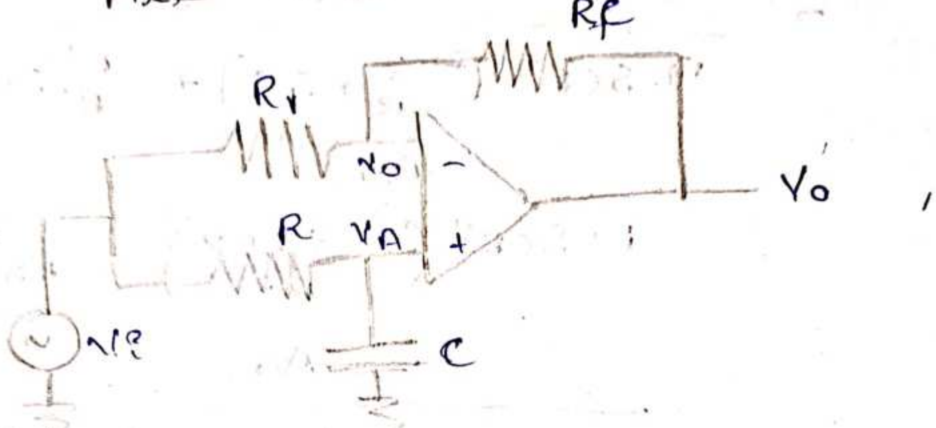
Second order HPF.

$$\frac{V_o}{V_i} = \frac{A_0 \cdot Y_1 Y_2}{Y_1 Y_2 + Y_4 [Y_1 + Y_2 + Y_3] + Y_2 Y_3 (1 - A_0)}$$

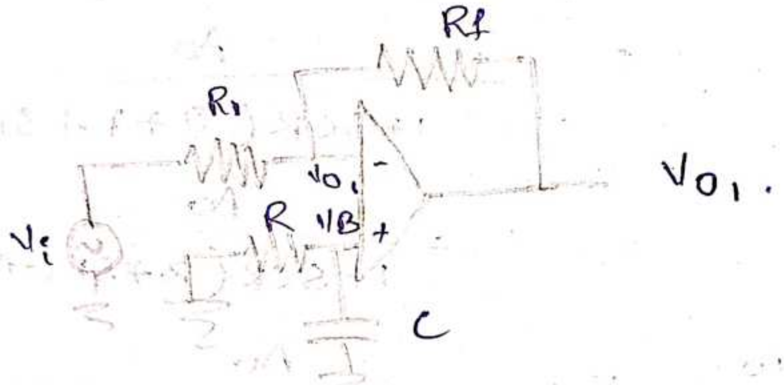
$Y_1 = Y_2 = SC$, $Y_3 = Y_4 = \frac{1}{R}$

$$\frac{V_o}{V_i} = \frac{A_o (sC)^2}{(sC)^2 + (1-A_o)\frac{sC}{R} + \frac{1}{R}\left(2sC + \frac{1}{R}\right)}$$

All Pass filter.



case-i,



$$\frac{V_{o1}}{V_i} = \frac{-R_f}{R_1}$$

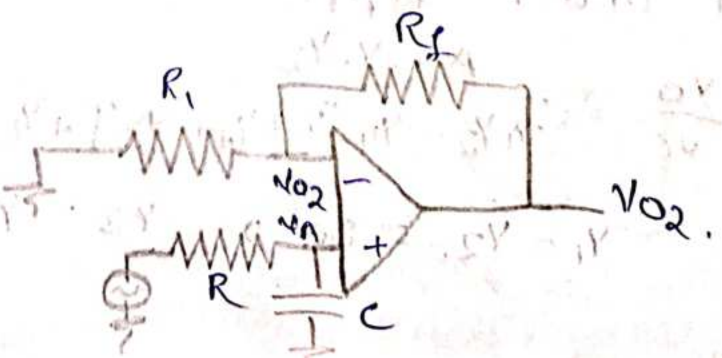
let $R_f = R_1$

$$\frac{V_{o1}}{V_i} = \frac{-\cancel{R_1}}{\cancel{R_1}}$$

$$\frac{V_{o1}}{V_i} = -1$$

$$\boxed{V_{o1} = -V_i}$$

case-ii,



$$\frac{V_{o2}}{V_A} = \left(1+\frac{R_f}{R_1}\right).$$

$$R_f = R_1$$

$$\frac{V_{o2}}{V_A} = \left(1+\frac{R_1}{R_1}\right)$$

$$\frac{V_{o2}}{V_A} = (1+1)$$

$$V_{o2} = 2V_A.$$

By voltage divide rule.

$$V_A = \frac{V_i\left(\frac{1}{j\omega C}\right)}{R+\frac{1}{j\omega C}}$$

$$= \frac{\frac{V_i}{\cancel{j\omega C}}}{\frac{Rj\omega C+1}{\cancel{j\omega C}}}$$

$$= \frac{V_i}{1+j\omega RC}$$

$$\boxed{V_{o2} = \frac{2V_i}{1+j\omega RC}}$$

$$V_o = V_{o1}+V_{o2}.$$

$$= -V_i + \frac{2V_i}{1+j\omega RC}$$

$$= V_i\left(-1+\frac{2}{1+j\omega RC}\right)$$

$$\frac{V_o}{V_i} = \frac{-1-j\omega RC+2}{1+j\omega RC}$$

$$= \frac{1-j\omega RC}{1+j\omega RC} = \frac{1-j2\pi fRC}{1+j2\pi fRC}$$

$$\left|\frac{V_o}{V_i}\right| = \frac{\sqrt{1+(f/f_a)^2}}{\sqrt{1+(f/f_a)^2}} \qquad 2\pi RC = f_a$$

$$\boxed{\left|\frac{V_o}{V_i}\right| = 1}$$

Band Pass filter.

Based on the bandwidth, band pass filters are classified into 2 types.

1. Narrow Band Pass Filter.
2. Wide Band Pass filter.

* Relation between frequency and quality factor,

$$Q_0 = \frac{f_0}{B.W}$$

Q_0 is $\frac{1}{\alpha}$ [B.W].

$Q_0 < 10 \rightarrow$ WBPF.

$Q_0 > 10 \rightarrow$ NBPF.

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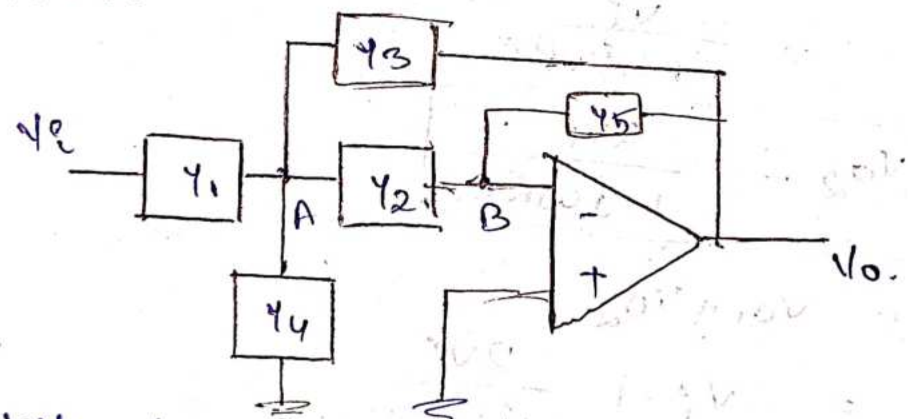
* NBPF.



KCL at node A.

$(V_i - V_A)Y_1 = (V_A - V_B)Y_2 + V_A \cdot Y_4 + (V_A - V_0)Y_3.$

$V_i Y_1 - V_A \cdot Y_1 = V_A Y_2 - V_B Y_2 + V_A \cdot Y_4 + V_A \cdot Y_3 - V_0 Y_3.$

$V_i Y_1 = V_A(Y_1 + Y_2 + Y_3 + Y_4) - V_0 V_3 - \underbrace{V_B Y_2}_{0}.$

$V_i Y_1 = V_A(Y_1 + Y_2 + Y_3 + Y_4) - V_0 Y_3 \longrightarrow (1)$

KCL at node B

$(V_A - V_B)Y_2 = (V_B - V_0)Y_5.$

$V_A Y_2 - V_B Y_2 = V_B Y_5 - V_0 Y_5.$

$V_A Y_2 = \underbrace{V_B(Y_2 + Y_5)}_{0} - V_0 Y_5$

$$V_A = \frac{-V_0 Y_5}{Y_2} \rightarrow (2).$$

sub (2) in (1).

$$V_i Y_1 = -\frac{V_0 Y_5}{Y_2}(Y_1+Y_2+Y_3+Y_4) - V_0 Y_3.$$

$$V_i Y_1 = \frac{-V_0}{Y_2}\left[Y_5(Y_1+Y_2+Y_3+Y_4) + Y_3 Y_2\right].$$

$$\frac{V_0}{V_i} = \frac{-Y_1 Y_2}{Y_5(Y_1+Y_2+Y_3+Y_4)+Y_3Y_2}$$

$Y_1 = G_1$; $Y_2 = SC_2$; $Y_3 = SC_3$; $Y_4 = G_4$; $Y_5 = G_5$;

$$\frac{V_0}{V_i} = \frac{-G_1 SC_2.}{G_5(G_1+SC_2+SC_3+G_4)+SC_2SC_3}$$

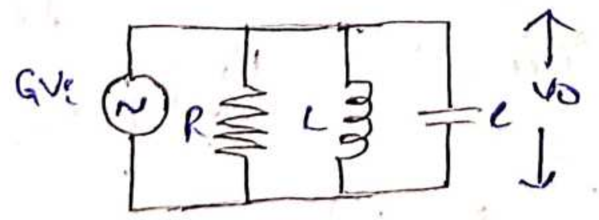
$$= \frac{-G_1 SC_2}{SC_2SC_3 + G_5(G_1+G_4) + G_5(SC_2+SC_3)}$$

$$= \frac{-G_1 \cancel{SC_2}}{\cancel{SC_2}\left[SC_3 + \frac{G_5(G_1+G_4)}{SC_2} + G_5\frac{(SC_2+SC_3)}{SC_2}\right]}$$

$$= \frac{-G_1}{SC_3 + \frac{G_5(G_1+G_4)}{SC_2} + \frac{G_5 \cancel{S}(C_2+C_3)}{\cancel{S}C_2}}$$

$$\frac{V_0}{V_i} = \frac{-G_1}{SC_3 + \frac{G_5(G_1+G_4)}{SC_2} + \frac{G_5(C_2+C_3)}{C_2}}$$

above equation is equivalant parallel ckt.



$$\frac{V_o}{V_i} = \frac{G'}{SC + 1/SL + G_1}$$

$$G_1 = \frac{G_5(C_2+C_3)}{C_2.}$$

$$\frac{1}{\cancel{s}L} = \frac{G_5(G_1+G_4)}{\cancel{s}C_2}$$

$$\frac{1}{L} = \frac{G_5(G_1+G_4)}{C_2}$$

$$\cancel{s}L = \cancel{s}C_3 \Rightarrow C = C_3.$$

Resonant frequency or central frequency.

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

$$\omega_0^2 = \frac{1}{LC}.$$

$$= \frac{G_5(G_1+G_4)}{C_2C_3}.$$

$$\omega_0^2 = \frac{G_5(G_1+G_4)}{C_2C_3}.$$

second order equation is

$$\frac{V_o}{V_i} = \frac{-A_0 \cdot \omega_0 \alpha s}{s^2 + s\omega_0\alpha s + \omega_0^2}$$

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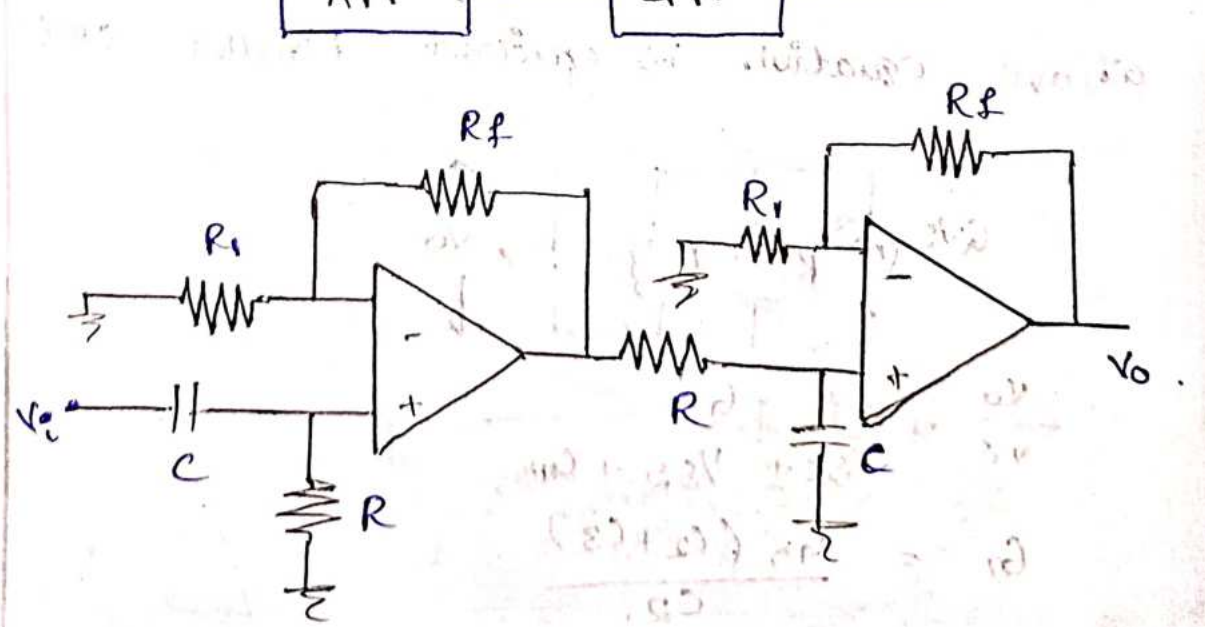
* WBPF.

$$V_i \longrightarrow \boxed{HPF} \longrightarrow \boxed{LPF} \longrightarrow V_o$$



$$\frac{V_o}{V_i} = \frac{A_{OH}\,(f/f_L)}{\sqrt{1+(f/f_L)^2}} \cdot \frac{A_{OL}}{\sqrt{1+(f/f_H)^2}}$$

$$\frac{V_o}{V_i} = \frac{A_o\,(f/f_L)}{\sqrt{\left[1+(f/f_L)^2\right]\left[1+(f/f_H)^2\right]}}$$

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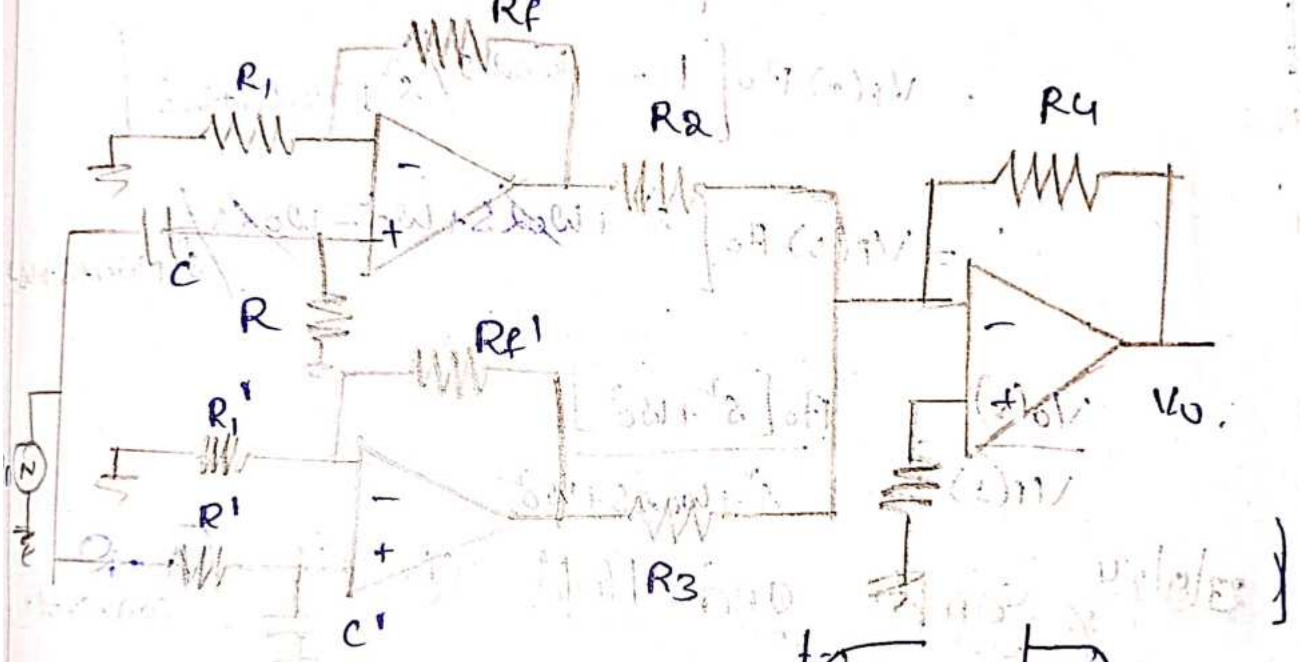
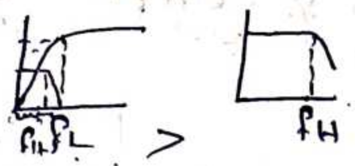
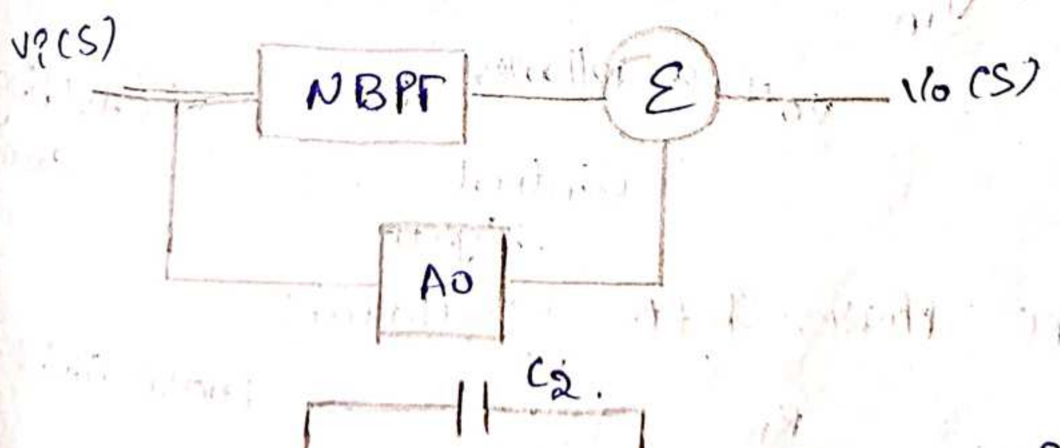
Wide Band Reject filter.

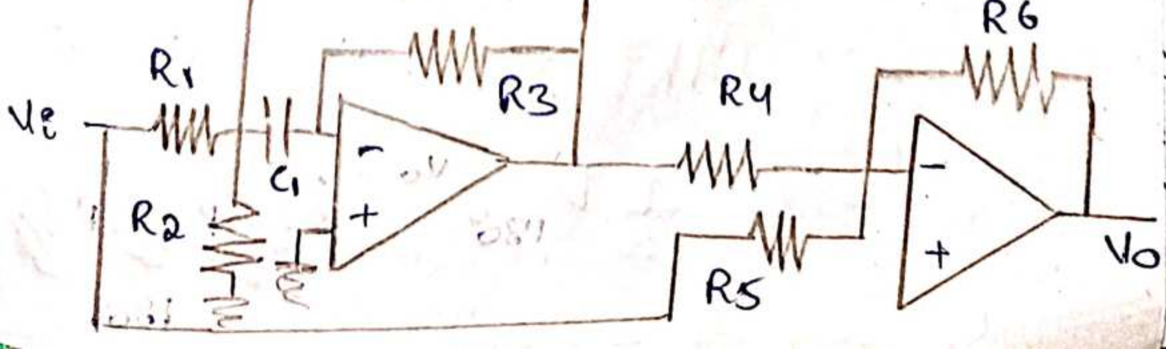




1. $f_L > f_H$.
2. HPF, LPF → unity gain.

Narrow Band Reject filter:





$$\frac{V_o}{V_i} = \frac{-A_o \omega_o \alpha s}{s^2 + s\omega_o \alpha s + \omega_o^2} \quad \text{NBPF}$$

$$V_o(s) = V_i(s) A_o + V_i(s) \left[\frac{-A_o \omega_o \alpha s}{s^2 + \omega_o \alpha s + \omega_o^2} \right].$$

unity gain

$$= V_i(s) A_o - \frac{V_i(s) A_o \omega_o \alpha s}{s^2 + \omega_o \alpha s + \omega_o^2}$$

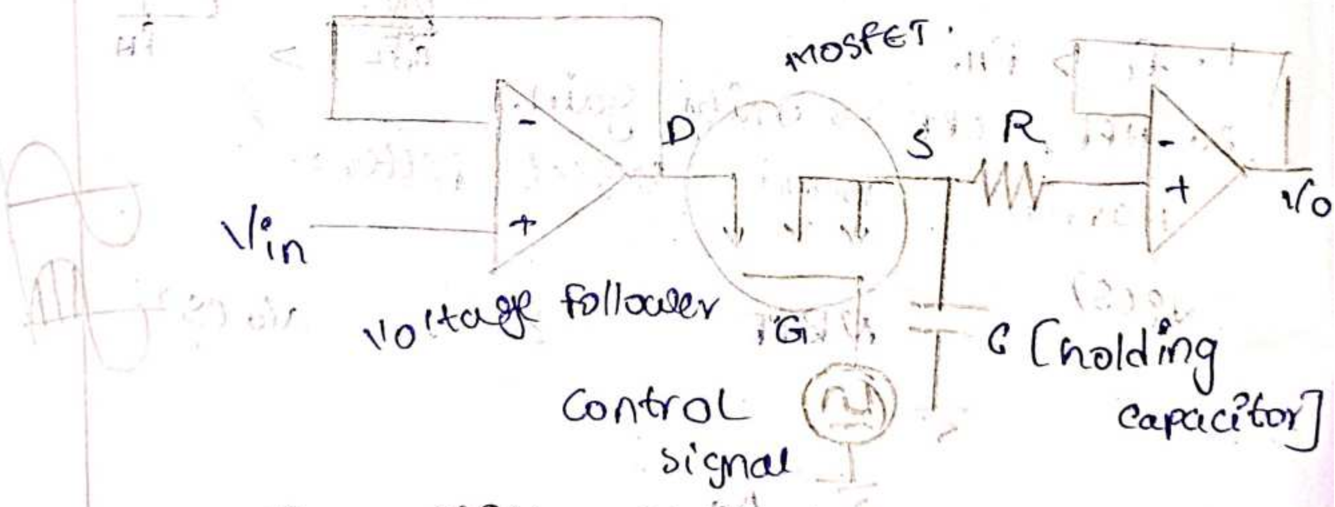
$$= V_i(s) \cancel{A_o} \left[\cancel{V_i} \right.$$

$$= V_i(s) A_o \left[1 - \omega_o \alpha s / s^2 + \omega_o \alpha s + \omega_o^2 \right]$$

$$= V_i(s) A_o \left[s^2 + \cancel{\omega_o \alpha s} + \omega_o^2 - \cancel{\omega_o \alpha s} / s^2 + \omega_o \alpha s + \omega_o^2 \right]$$

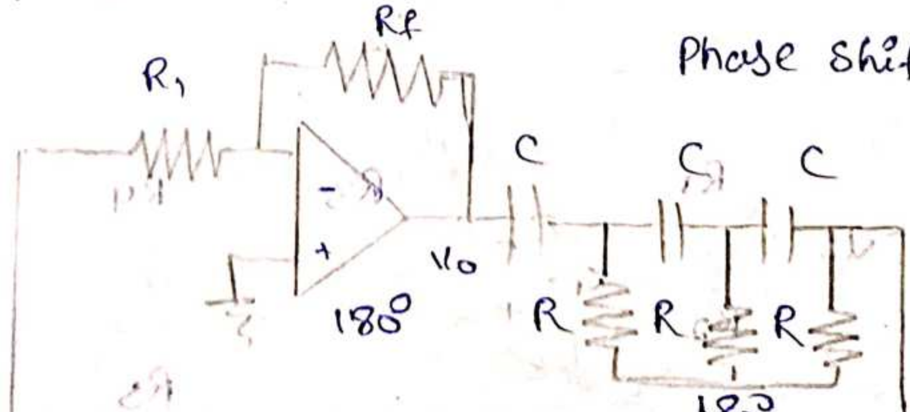
$$\frac{V_o(s)}{V_i(s)} = \frac{A_o [s^2 + \omega_o^2]}{s^2 + \omega_o \alpha s + \omega_o^2}$$

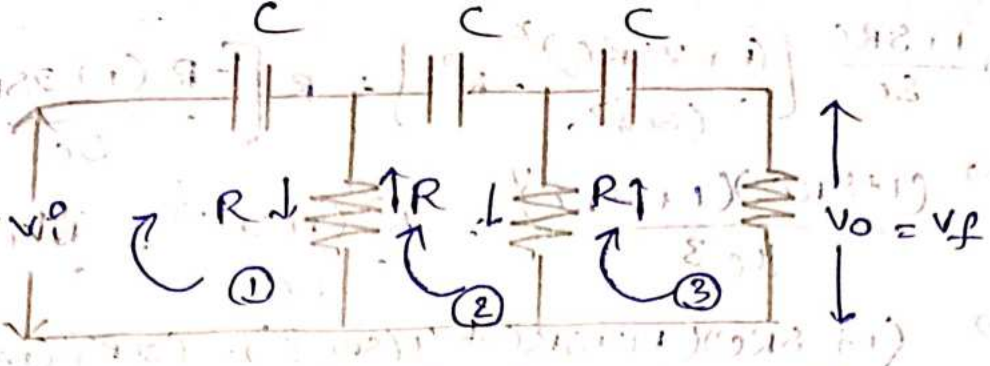
27/9/24 * Sample and Hold circuit [A-D converter]



* RC Phase-Shift oscillator:-

Phase shift = 0.





$$I_1\left(R+\frac{1}{j\omega C}\right) - I_2 R = V_i \rightarrow ①$$

$$-I_1 R + I_2\left(2R+\frac{1}{j\omega C}\right) - I_3 R = 0 \rightarrow ②$$

$$-I_2 R + I_3\left(2R+\frac{1}{j\omega C}\right) = 0 \rightarrow ③$$

$$\begin{bmatrix} R+\frac{1}{j\omega C} & -R & 0 \\ -R & 2R+\frac{1}{j\omega C} & -R \\ 0 & -R & 2R+\frac{1}{j\omega C} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_i \\ 0 \\ 0 \end{bmatrix}$$

$I_3 = D3/D.$

$$D = R+\frac{1}{j\omega C}\left(\left(2R+\frac{1}{j\omega C}\right)^2 - R^2\right) - R\left[-R\left(2R+1/j\omega C\right)\right]$$

$j\omega = s.$

$$D = R+\frac{1}{SC}\left[\left(2R+\frac{1}{SC}\right)^2 - R^2\right] - R\left[-R\left(2R+\frac{1}{SC}\right)\right]$$

$$= R+\frac{1}{SC}\left[\left(\frac{2RSC+1}{SC}\right)^2 - R^2\right] - R\left[-R\left(\frac{2RSC+1}{SC}\right)\right]$$

$$= R+\frac{(2RSC)^2+1+4RSC-R^2SC}{SC^3} + R^2\frac{(2RSC+1)}{SC}$$

$$= R+\frac{4R^2SC^2+4RSC-R^2SC+1}{SC^3} + \frac{R^2(2RSC)+R^2}{SC}$$

$$= R+\frac{\cancel{SC}\left[4R^2SC^2+4R-R^2+\frac{1}{SC}\right]}{\cancel{SC^3}} + \frac{R^2(2RSC)+R^2}{SC}$$

$$= \frac{RSC+4R^2SC^2-\cancel{R^2}+4R+\frac{1}{SC}+2R^3SC^2+R^2SC}{SC^2}$$

$$= \frac{RSC+4R^2SC^2+4R+\frac{1}{SC}+2R^3SC^2-R^2+R^2SC}{SC^2}$$

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$$\Rightarrow \frac{1+SRC}{SC}\left[\frac{(1+2SRC)^2}{(SC)^2}-R^2\right]+R\left[\frac{-R(1+2SRC)}{SC}\right]$$

$$\Rightarrow \frac{(1+SRC)(1+2SRC)^2}{SC^3}-\frac{R^2(1+SRC)}{SC}-\frac{R^2(1+2SRC)}{SC}$$

$$\Rightarrow \frac{(1+SRC)(1+4SRC+4(SRC)^2)-(SCR)^2(1+SRC)}{SC^3}-(SCR)^2(1+2SRC)$$

$$\Rightarrow \frac{\left[1+4S\cancel{RC}+4(SRC)^2+S\cancel{RC}+4(SRC)^2+4(SRC)^3-(SRC)^2+(SRC)^3-(SCR)^2-2(SRC)^3\right]}{(SC)^3}$$

$$D=\frac{1+5SRC+6(SRC)^2+(SRC)^3}{(SC)^3}$$

$$D_3=\begin{bmatrix}\overset{+}{\frac{1+SRC}{SC}} & \overset{-}{-R} & \overset{+}{V_i}\\ -R & \frac{1+2SRC}{SC} & 0\\ 0 & -R & 0\end{bmatrix}$$

$$D_3=V_iR^2$$

$$I_3=\frac{D_3}{D}=\frac{V_iR^2(SC)^3}{1+5SRC+6(SRC)^2+(SRC)^3}$$

$$V_o=I_3\cdot R$$

$$=\frac{V_iR^3(SC)^3}{1+5SRC+6(SRC)^2+(SRC)^3}$$

$$=\frac{V_i(RSC)^3}{1+5SRC+6(SRC)^2+(SRC)^3}$$

$$\beta=\frac{V_o}{V_i}=\frac{(RSC)^3}{1+5SRC+6(SRC)^2+(SRC)^3}$$

Replace $s = j\omega$; $s^2 = -\omega^2$, $s^3 = -j\omega^3$.

$$\beta = \frac{-j\omega^3 R^3 C^3}{1 + 5j\omega RC - 6\omega^2R^2C^2 - j\omega^3R^3C^3}$$

Divide and multiply numerator & denominator with $-j\omega^3R^3C^3$.

$$\beta = \frac{\frac{-j\omega^3R^3C^3}{-j\omega^3R^3C^3}}{\frac{-1}{j\omega^3R^3C^3} + \frac{5j\omega RC}{-j\omega^3R^3C^3} - \frac{6\omega^2R^2C^2}{-j\omega^3R^3C^3} \frac{-j\omega^3R^3C^3}{-j\omega^3R^3C^3}}$$

$$= \frac{1}{j/\omega^3R^3C^3 - 5/\omega^2R^2C^2 - 6j/\omega RC + 1}$$

let $\alpha = \frac{1}{\omega RC}$.

$$\beta = \frac{1}{j\alpha^3 - 5\alpha^2 - 6j\alpha + 1}$$

$$= \frac{1}{1 - 6j\alpha - 5\alpha^2 + j\alpha^3}$$

$$\beta = \frac{1}{(1-5\alpha^2) - j\alpha(6-\alpha^2)}$$

$j\alpha(6-\alpha^2) = 0$

$6 - \alpha^2 = 0$

$6 = \alpha^2$

$$\boxed{\alpha = \sqrt{6}}$$

$\frac{1}{\omega RC} = \sqrt{6}$

$\frac{1}{2\pi f RC} = \sqrt{6}$

$$\boxed{f = \frac{1}{2\pi RC\sqrt{6}}}$$

$$\beta = \frac{1}{(1-5\alpha^2)}$$

$$\beta = \frac{1}{1-5(6)}$$

$$= \frac{1}{1-30}$$

$$= \frac{1}{-29}.$$

$$|\beta| = \left|\frac{1}{-29}\right|$$

$$= \frac{1}{29}.$$

To have oscillations $|A\beta| \geq 1$.

$$|A||\beta| \geq 1$$

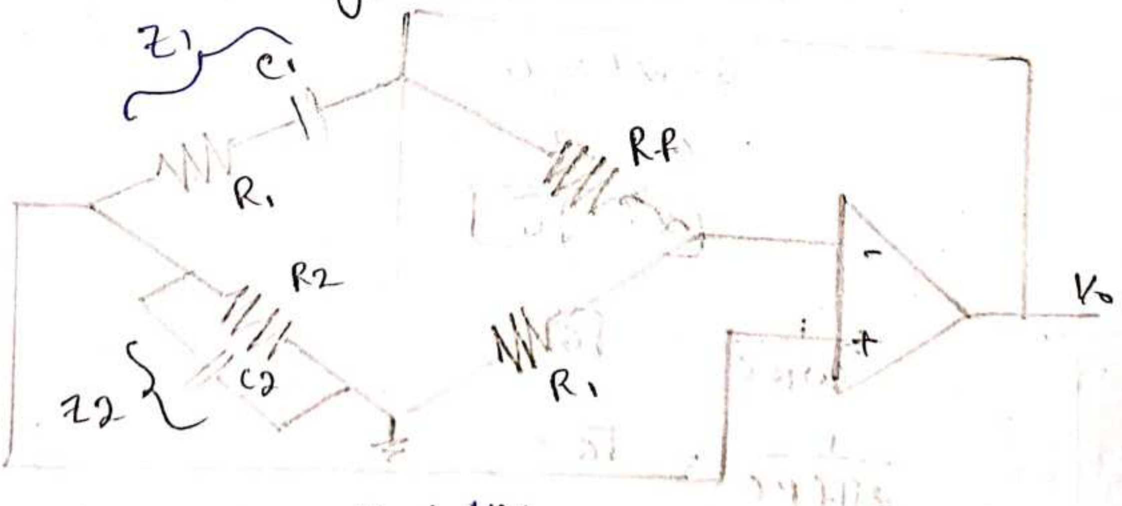
$$|A| \geq \frac{1}{|\beta|}$$

$$\geq \frac{1}{\frac{1}{29}}$$

$$|A| \geq 29$$

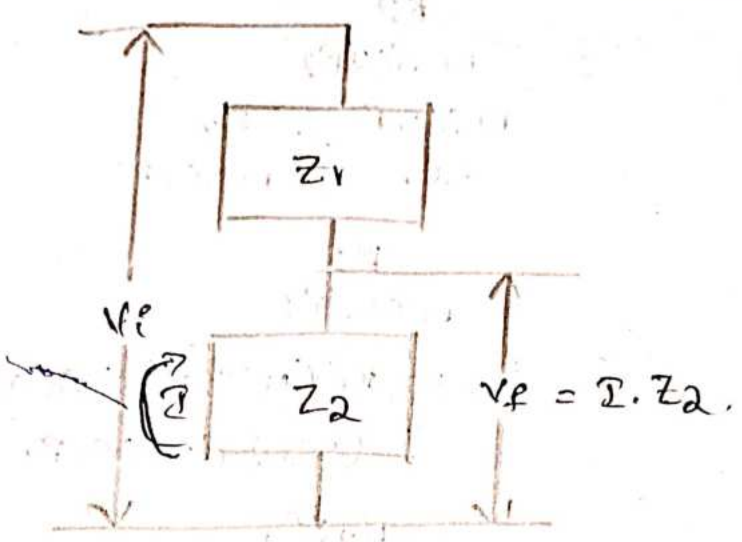
when op-amp gain is greater than or equal to 29 then only it produce oscillations

3/10/24 wien Bridge oscillator



frequency stability purpose - R_1, R_2

Gain varying purpose R_f, R_1.



$$I = \frac{V_{in}}{Z_1 + Z_2}.$$

$$V_o = V_f = I \cdot Z_2.$$

$$V_o = \frac{V_{in} Z_2}{Z_1 + Z_2}.$$

$$\frac{V_o}{V_{in}} = \frac{Z_2}{Z_1 + Z_2}.$$

$$Z_1 = R_1 + \frac{1}{j\omega C_1}$$

$$= \frac{R_1 j\omega C_1 + 1}{j\omega C_1}.$$

$$Z_2 = \frac{R_2 \cdot \frac{1}{j\omega C_2}}{R_2 + \frac{1}{j\omega C_2}}$$

$$= \frac{\frac{R_2}{\cancel{j\omega C_2}}}{\frac{R_2 j\omega C_2 + 1}{\cancel{j\omega C_2}}}$$

$$Z_2 = \frac{R_2}{1 + R_2 j\omega C_2}.$$

Replace $j\omega C$ with SC.

$$Z_1 = \frac{1 + SC_1 R_1}{SC_1}.$$

$$Z_2 = \frac{R_2}{1 + SC_2 R_2}.$$

$$= \frac{\frac{R_2}{1+SC_2R_2}}{\frac{1+SC_1R_1}{SC_1} + \frac{R_2}{1+SC_2R_2}}$$

$$= \frac{\frac{R_2}{1+SC_2R_2}}{\frac{(1+SC_1R_1)(1+SC_2R_2)+SC_1R_2}{(SC_1)(1+SC_2R_2)}}$$

$$= \frac{R_2SC_1}{(1+SC_1R_1)(1+SC_2R_2)+SC_1R_2}$$

$$= \frac{SC_1R_2}{1+SC_2R_2+SC_1R_1+S^2C_1C_2R_1R_2+SC_1R_2}$$

$$= \frac{SC_1R_2}{1+S(C_1R_1+C_2R_2+C_1R_2)+S^2C_1C_2R_1R_2}$$

Replace s with $j\omega$ and $s^2 = -\omega^2$

$$= \frac{j\omega C_1R_2}{1+j\omega(C_1R_1+C_2R_2+C_1R_2)-\omega^2C_1C_2R_1R_2}$$

$$\beta = \frac{j\omega C_1R_2}{(1-\omega^2C_1C_2R_1R_2)+j\omega(C_1R_1+C_2R_2+C_1R_2)}$$

Rationalise the above equation.

$$\beta = \frac{j\omega C_1R_2\left([1-\omega^2C_1C_2R_1R_2)-j\omega(C_1R_1+C_2R_2+C_1R_2)\right]}{(1-\omega^2C_1C_2R_1R_2)^2+j\omega(C_1R_1+C_2R_2+C_1R_2)\left[(1-\omega^2C_1C_2R_1R_2)-j\omega(C_1R_1+C_2R_2+C_1R_2)\right]}$$

$$= \frac{j\omega C_1R_2\left[[1-\omega^2C_1C_2R_1R_2]-j\omega(R_1C_1+R_2C_2+R_2C_1)\right]}{(1-\omega^2C_1C_2R_1R_2)^2+\omega^2(R_1C_1+R_2C_2+R_2C_1)^2}$$

$$\beta = \frac{\omega^2C_1R_2[R_1C_1+R_2C_2+R_2C_1]+j\omega C_1R_2(1-\omega^2C_1C_2R_1R_2)}{(1-\omega^2C_1C_2R_1R_2)^2+\omega^2(R_1C_1+R_2C_2+R_2C_1)^2}$$

$$j\omega C_1 R_2 \left[1-\omega^2 C_1 C_2 R_1 R_2\right] = 0$$

$$1 = \omega^2 C_1 C_2 R_1 R_2$$

$$\omega^2 = \frac{1}{C_1 C_2 R_1 R_2}$$

$$\omega = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}}$$

$$\omega = \frac{1}{\sqrt{C_1 C_2 R_1 R_2}}$$

let $C_1 = C_2 = C$, $R_1 = R_2 = R$.

$$\omega = \frac{1}{\sqrt{C^2 R^2}}$$

$$2\pi f = \frac{1}{RC}$$

$$\boxed{f = \frac{1}{2\pi RC}}$$

Replace R_1, R_2 with R and $C_1 C_2$ with C in equation ①.

~~$\beta = j\omega C R\left[1-\omega^2 C\right.$~~

$$\beta = \frac{\omega^2 C R\left[RC+RC+RC\right] - j\omega CR\left(1-\omega^2 C^2 R^2\right)}{\left(1-\omega^2 C^2 R^2\right)^2 + \omega^2\left(RC+RC+RC\right)}$$

$$= \frac{\omega^2 RC\left[3RC\right] - j\omega RC\left[1-\omega^2 (RC)^2\right]}{1-\omega^2\left[RC\right]^2 + \omega^2\left[3RC\right]^2}$$

$$f = \frac{1}{2\pi RC}$$

$$2\pi f = \frac{1}{RC}$$

$$\omega = \frac{1}{RC}$$

$$\beta = \frac{\frac{1}{(RC)^2} RC(3RC) - j\left(\frac{RC}{RC}\right)\left[1-\frac{\cancel{R^2C}}{RC^2}\right]}{1-\left(\frac{RC}{RC}\right)^2 + \left(\frac{RC}{RC}\right)^2 3^2}$$

$$= \frac{3\cancel{RC^2}}{\cancel{RC^2}} \Big/ 3^2.$$

$= \frac{3}{9}$ 	$|A||\beta| \geq 1$

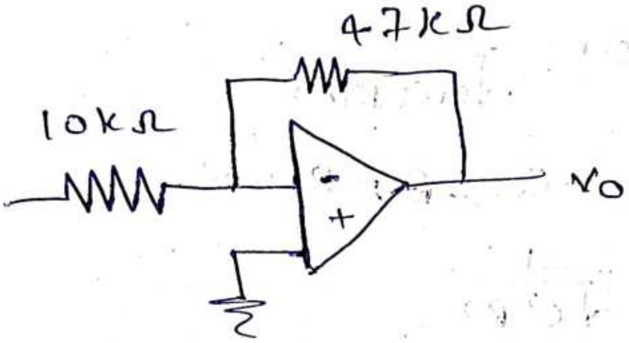
$|A| \geq 1/|\beta|$

$\beta = \frac{1}{3}$. 	$|A| \geq 3$.

$A = 3$

5/10/24

Determine voltage gain of OP-amp as shown below.



$$\frac{V_o}{V_i} = -(R_f/R_1)$$

$$= -\left(\frac{47}{10}\right)$$

$$= -4.7$$

$\boxed{* \text{Gain} = 4.7.}$

A Sine wave 0.5 of peak voltage is applied to an interting amplifier using $R_1 = 10k$ and $R_f = 50k\Omega$. It uses supply voltage of ±12V. Determine output and sketch the wave form. If the amplitude of sine wave is increase to 5 V. what will be the output. Is it practically possible. Sketch the wave form.

$$G = \frac{V_o}{V_{in}} = \frac{-R_f}{R_1}$$

$$= \frac{-50}{10}$$

$$G = 5$$

i, $V_o = V_{in}(5)$

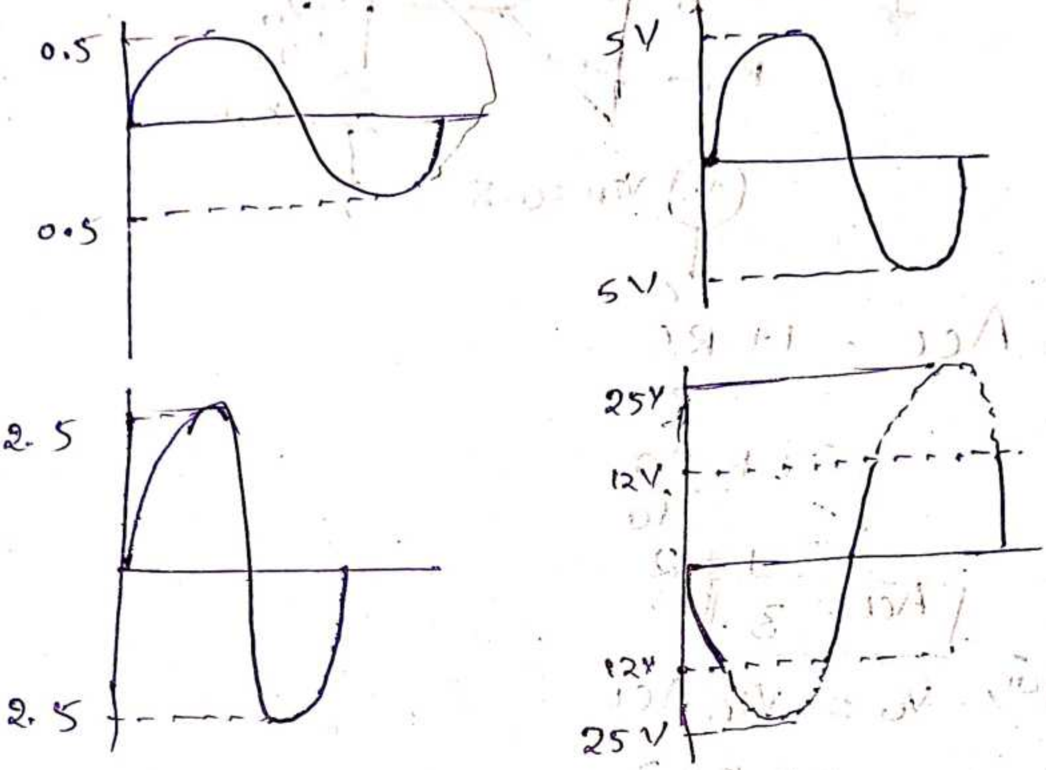
$= 0.5\,(5)$

$= 2.5V.$

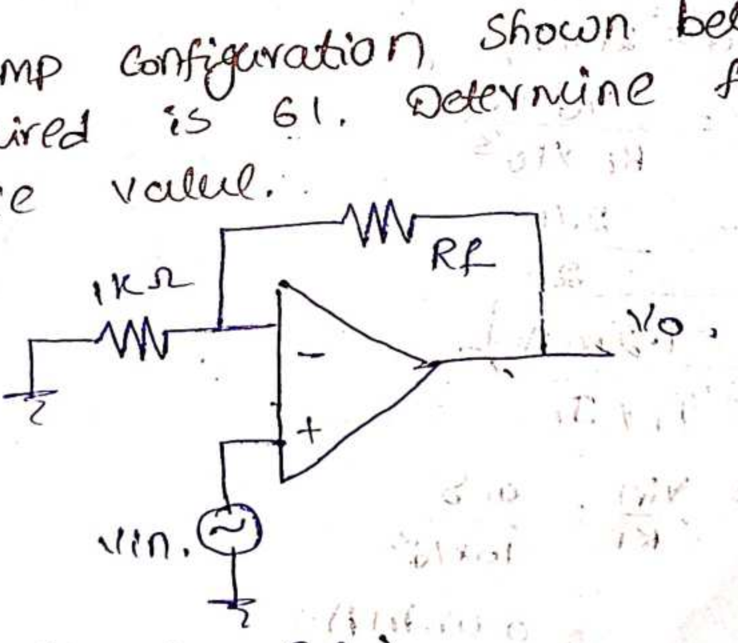
ii, $V_o = V_{in}(5)$

$= 5(5)$

$= 25\,V$



or op-amp configuration shown below, the gain required is 61. Determine feedback resistance value.



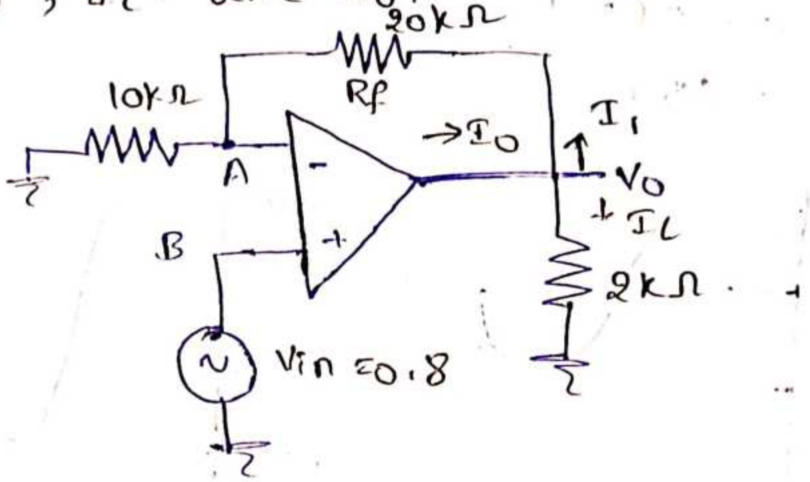
$$G = \left(1 + \frac{R_f}{R_1}\right)$$

$$61 = \left(1 + \frac{R_f}{1k}\right)$$

$$\frac{R_f}{1k\Omega} = 61 - 1$$

$$\boxed{R_f = 60\,k\Omega}$$

For non-inverting amplifier shown below. Calculate A_{cl}, V_o, I_L and I_o



i, $A_{cl} = 1 + \frac{R_f}{R_1}$

$= 1 + \frac{20}{10}$

$= 1 + 2$

$\boxed{A_{cl} = 3}$

ii, $V_o = V_{in} A_{cl}$

$= 0.8 \times 3$

$\boxed{V_o = 2.4V}$

iii, $I_L = \frac{V_o}{R_L \times 10^3}$

$= \frac{2.4}{2}$

$\boxed{I_L = 1.2mA}$

iv, $I_o = I_1 + I_L$

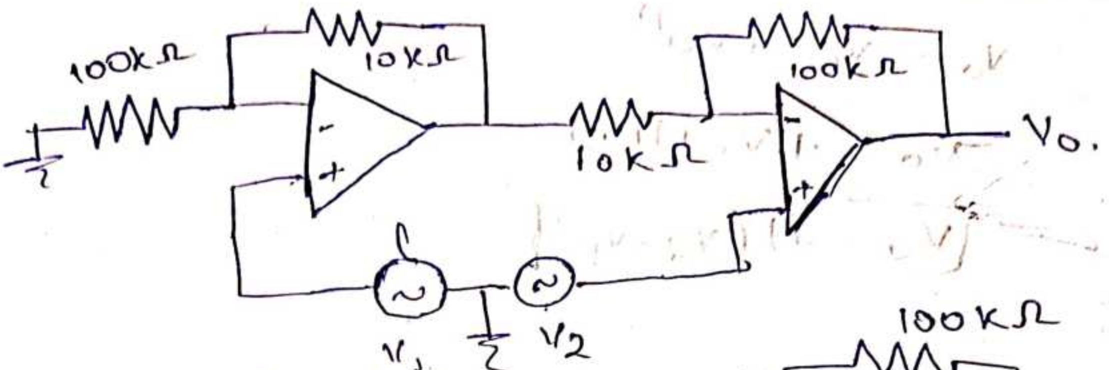
$I_1 = \frac{V_{in}}{R_1} = \frac{0.8}{10 \times 10^3}$

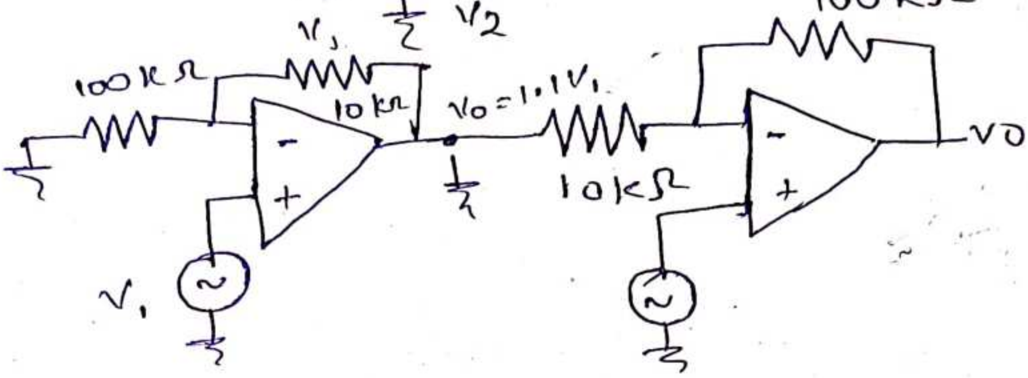
$= 0.08mA$

$I_o = 0.08mA + 1.2mA$

$\boxed{I_o = 1.28mA}$

Find output voltage Vo in terms of V_1 and V_2 of circuit shown below for OP-amp





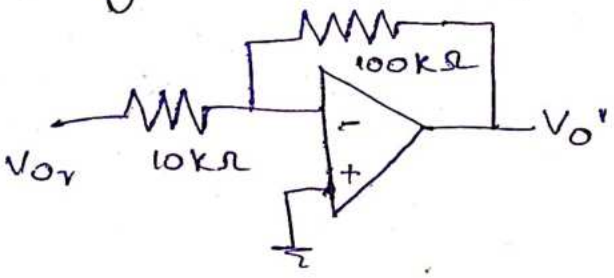
$$V_o = \left(1 + \frac{R_f}{R_1}\right)V_1$$

$$= \left(1 + \frac{10}{100}\right)V_1$$

$$= (1 + 0.1)V_1$$

$$= 1.1V_1.$$

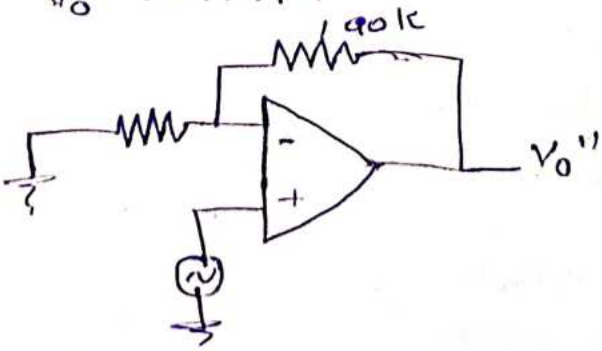
By using superposition Theorem,



$$V_o' = \frac{-R_f}{R_1} V_o'$$

$$= -\frac{10\cancel{0}}{1\cancel{0}}(1.1V_1)$$

$$V_o' = -11V_1.$$



$$V_o'' = \left(1 + \frac{10\cancel{0}}{1\cancel{0}}\right)V_2$$

$$V_o'' = 11V_2$$

$$V_o = V_o' + V_o''$$

$$V_o = -11V_1 + 11V_2$$

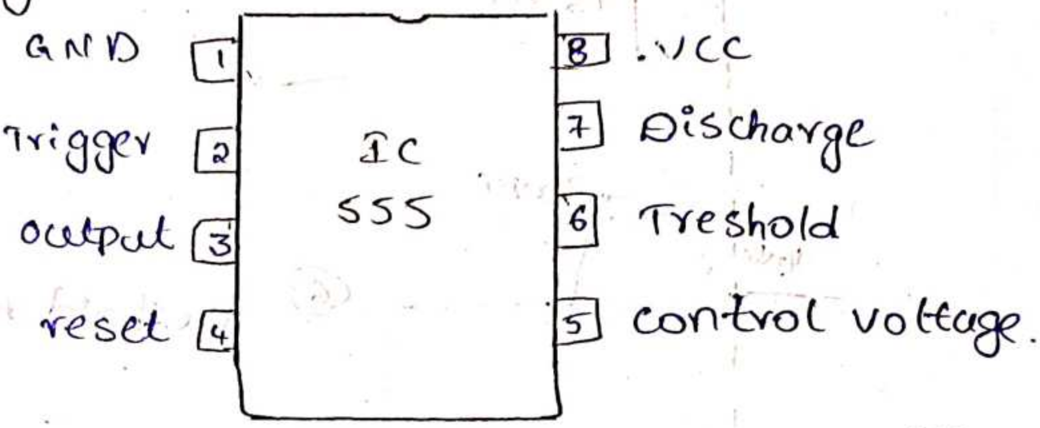
$$\boxed{V_o = 11(V_2 - V_1)}$$

UNIT - 4

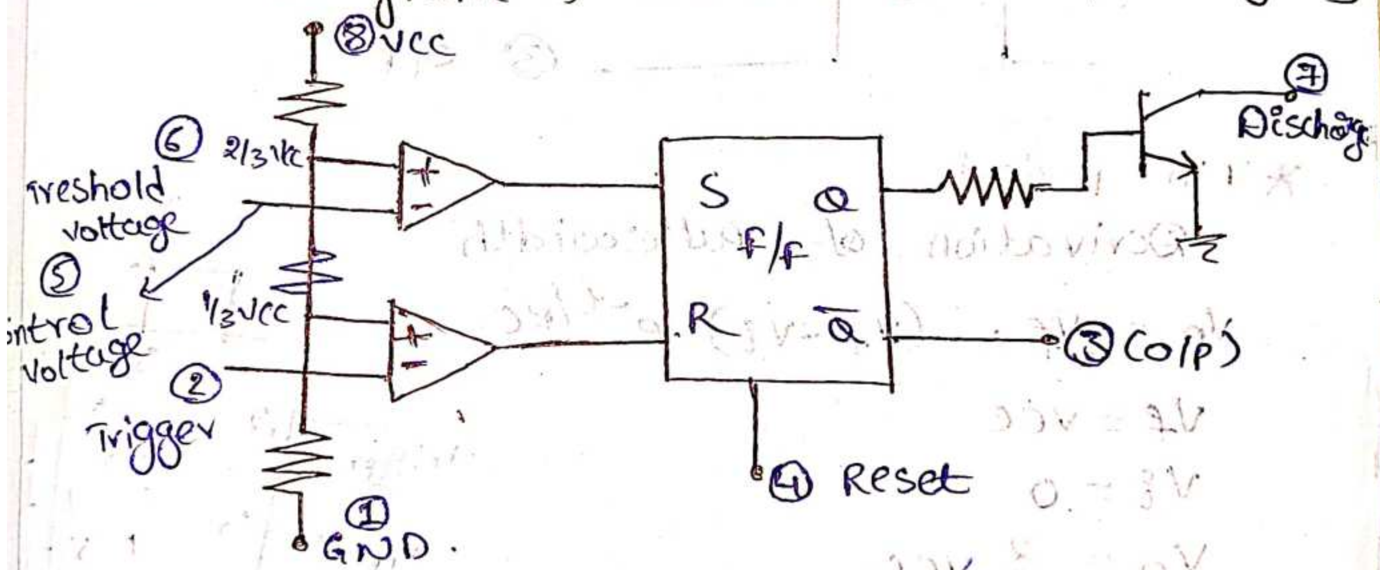
21/10/24

Timers and Phase Locked Loops.

* Pin Diagram

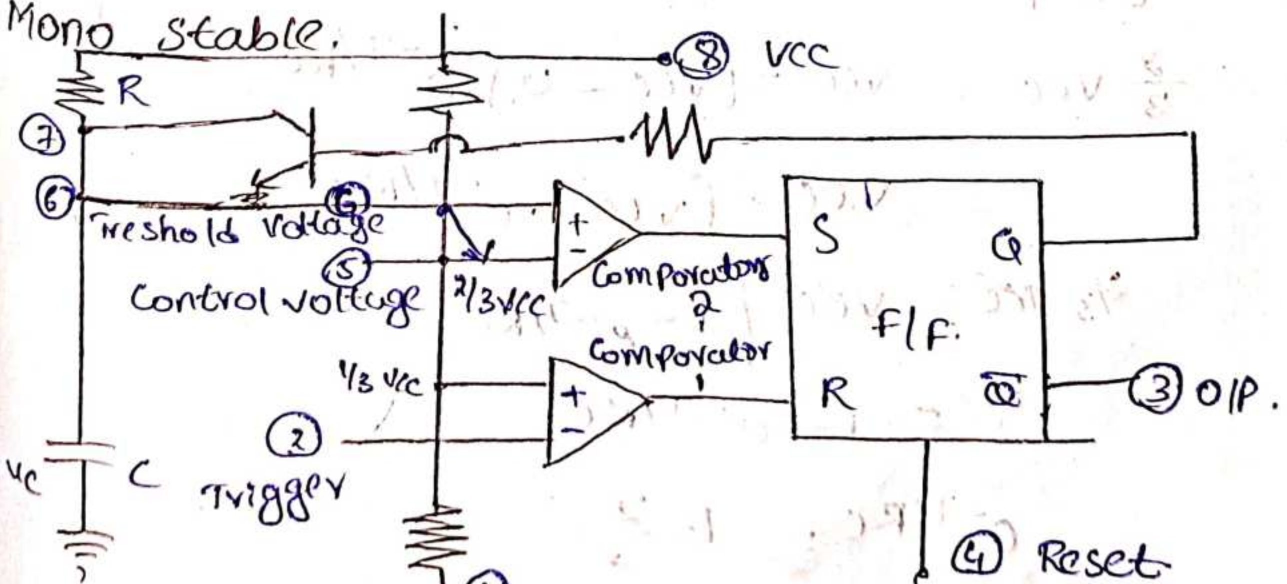


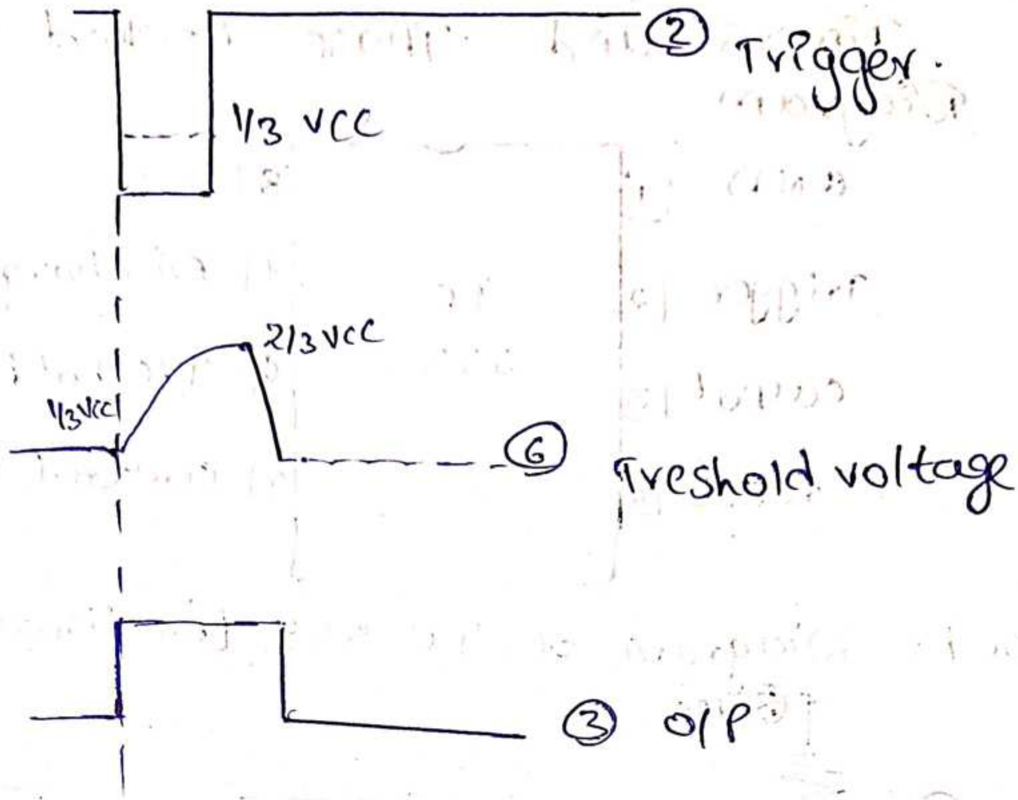
* Block Diagram of IC 555 [Functional Diagram]



* Applications

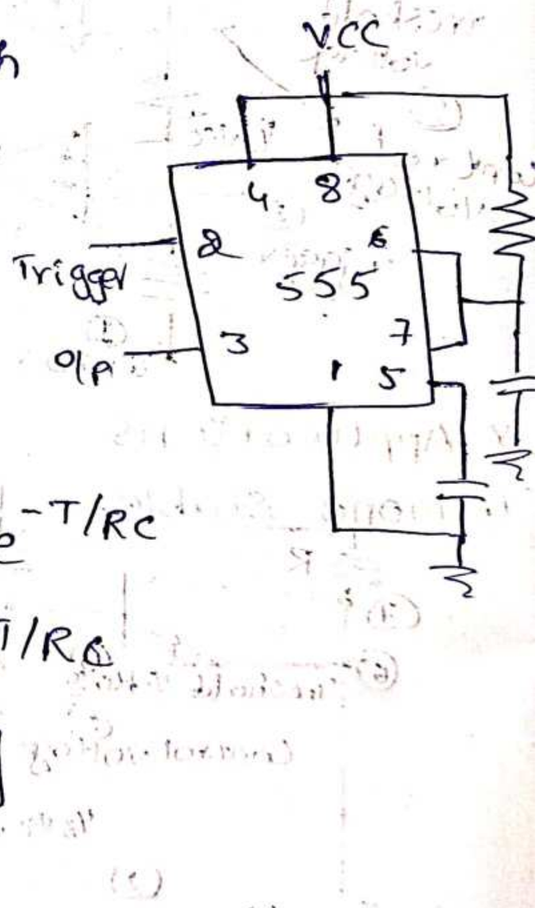
1. Mono stable.





* Time period

Derivation of pulsewidth



$$V_0 = V_f - (V_f - V_i) \cdot e^{-t/RC}$$

$V_f = VCC$

$V_i = 0$

$V_0 = \frac{2}{3} VCC$

$t = T$

$$\frac{2}{3} VCC = VCC - [VCC - 0] \cdot e^{-T/RC}$$

$$= VCC - [VCC] \cdot e^{-T/RC}$$

$$2/3\, \cancel{VCC} = \cancel{VCC} \left[1 - e^{-T/RC}\right]$$

$$\frac{2}{3} = 1 - e^{-T/RC}$$

$$e^{-T/RC} = 1 - \frac{2}{3}$$

$$e^{-T/RC} = \frac{1}{3}$$

$$e^{T/RC} = 3$$

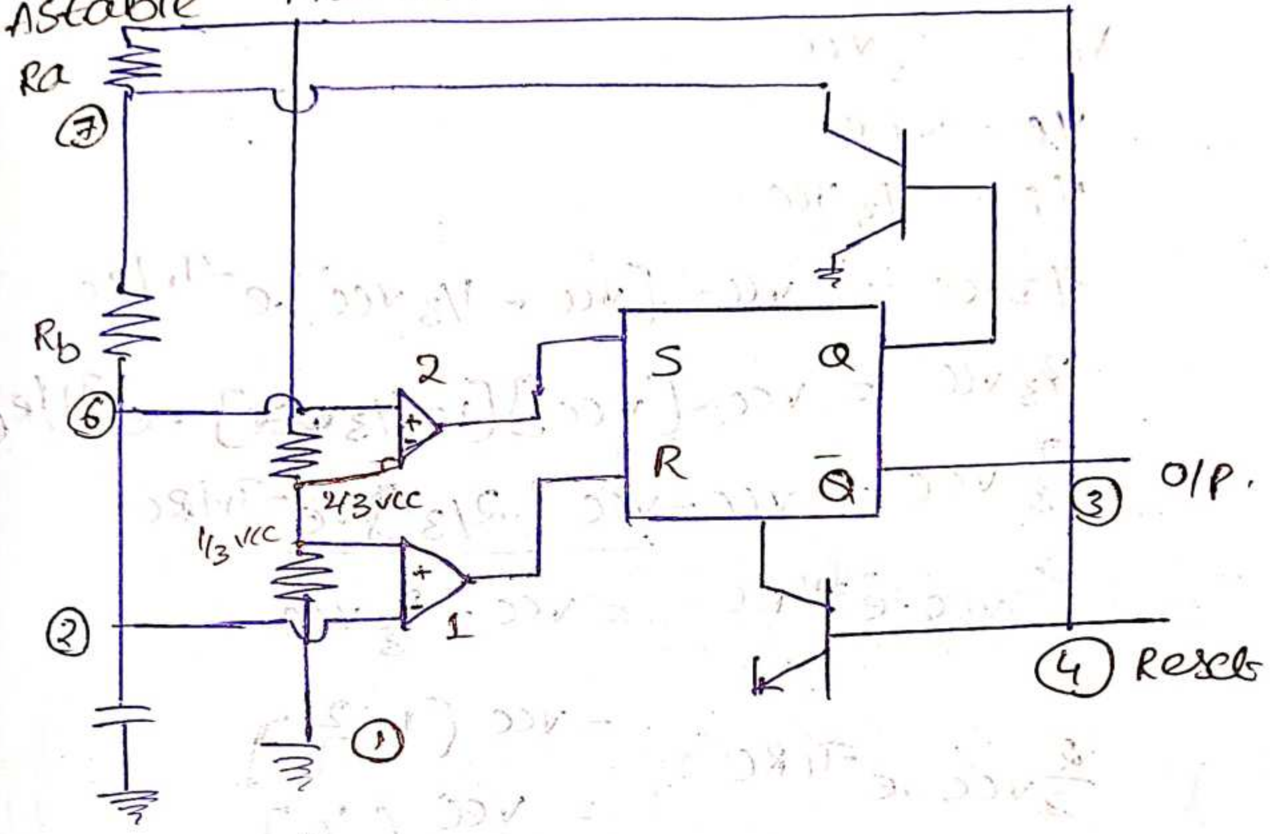
$$T/RC = \ln(3)$$

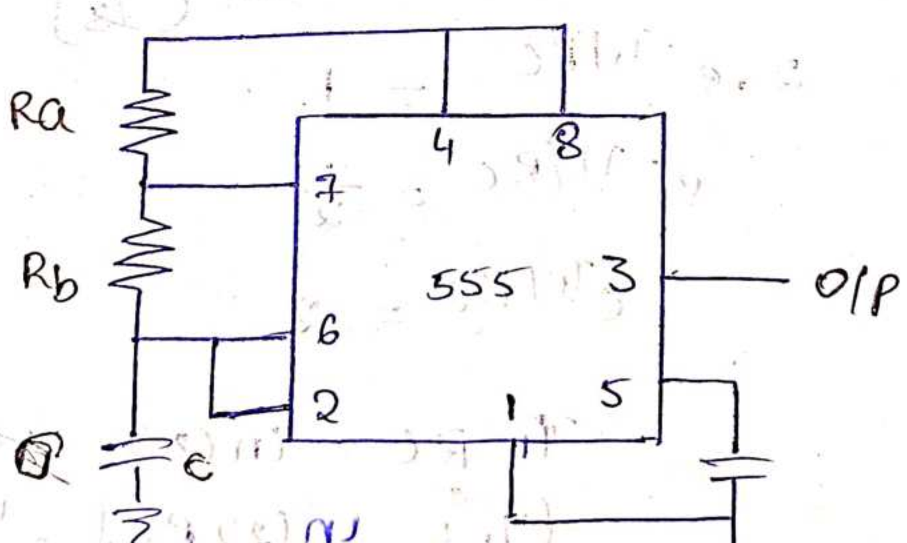
$T/RC = 1.1$

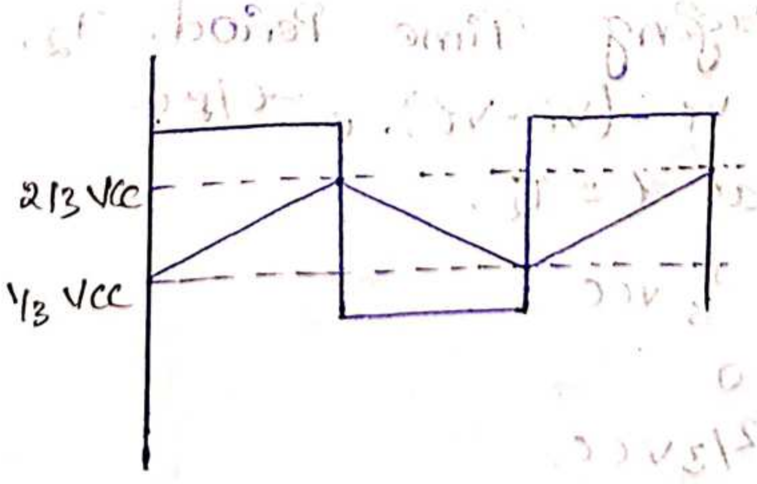
$$\boxed{T = 1.1RC}$$

10/2/24

Astable Multivibrators.







* charging time period. T_1

$$V_0 = V_f - (V_f - V_i) \cdot e^{-t/RC}$$

at $t = T_1$.

$V_0 = \frac{2}{3} Vcc$

$V_f = Vcc$

$V_i = 1/3\, Vcc$

$$2/3\, Vcc = Vcc - [Vcc - 1/3\, Vcc] \cdot e^{-T_1/RC}.$$

$$2/3\, Vcc = Vcc - [Vcc][1 - 1/3] \cdot e^{-T_1/RC}.$$

$$\frac{2}{3} Vcc = Vcc - \underline{Vcc\, (2/3) \cdot e^{-T_1/RC}}.$$

$$\frac{2}{3} Vcc \cdot e^{-T_1/RC} = Vcc - \frac{2}{3} Vcc.$$

$$= Vcc \left(1 - \frac{2}{3}\right)$$

$$\frac{2}{\cancel{3}} \cancel{Vcc} \cdot e^{-T_1/RC} = \cancel{Vcc}\, (1/\cancel{3})$$

$$2 \cdot e^{-T_1/RC} = 1.$$

$$e^{-T_1/RC} = \frac{1}{2}.$$

$$e^{T_1/RC} = 2.$$

$$T_1/RC = \ln(2).$$

$$\boxed{T_1 = \ln(2) RC} = \text{"} 0.693 (R_a + R_b\text{"}$$

* Discharging Time Period. T_2.

$$V_0 = V_f - (V_f - V_i) \cdot e^{-t/RC}.$$

at $t = T_2$.

$V_0 = \frac{1}{3} Vcc$

$V_f = 0$

$V_i = 2/3\, Vcc.$

$$\frac{1}{3}V_{CC} = 0 - \left[0 - \frac{2}{3}V_{CC}\right] \cdot e^{-T_2/RC}$$

$$\frac{1}{3}V_{CC} = \frac{2}{3}V_{CC} \cdot e^{-T_2/RC}$$

$$2 \cdot e^{-T_2/RC} = 1$$

$$e^{-T_2/RC} = \frac{1}{2}$$

$$e^{T_2/RC} = 2.$$

$$T_2/RC = \ln(2)$$

$$T_2 = \ln(2) RC$$

$$\boxed{T_2 = 0.693 R_b C.}$$

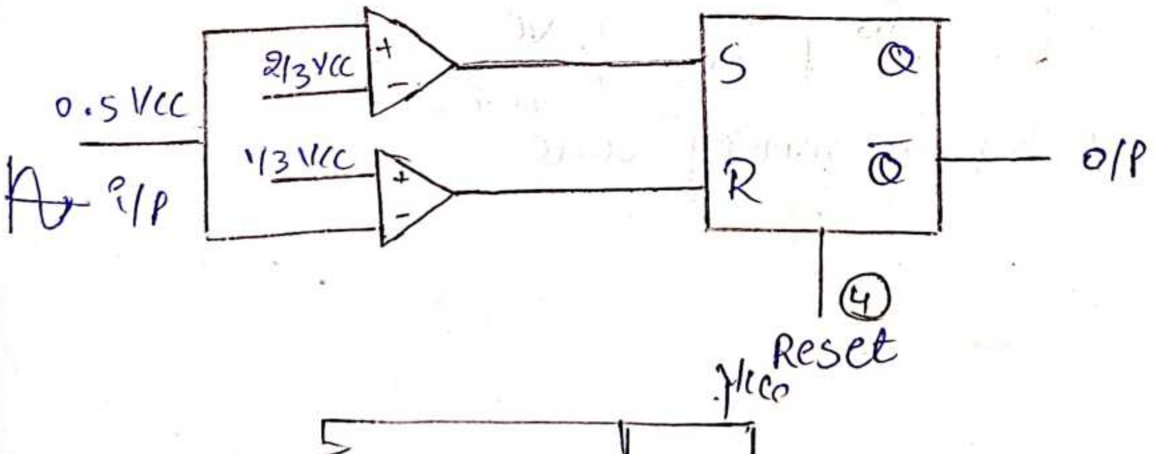
$$T = T_1 + T_2.$$

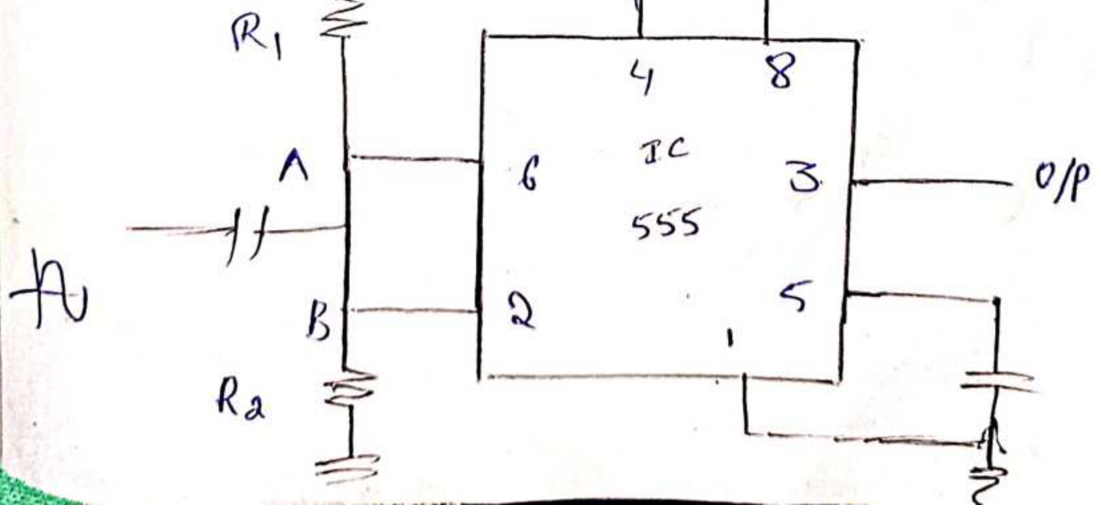
$$= 0.693(R_a+R_b)C + 0.693 R_b C.$$

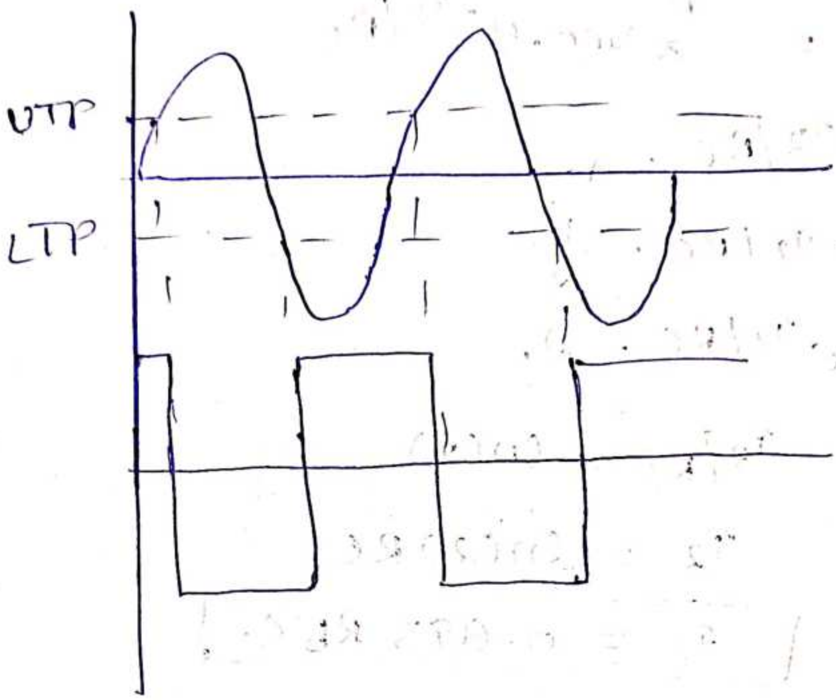
$$\boxed{T = 0.693[R_a+2R_b]C}$$

10/24

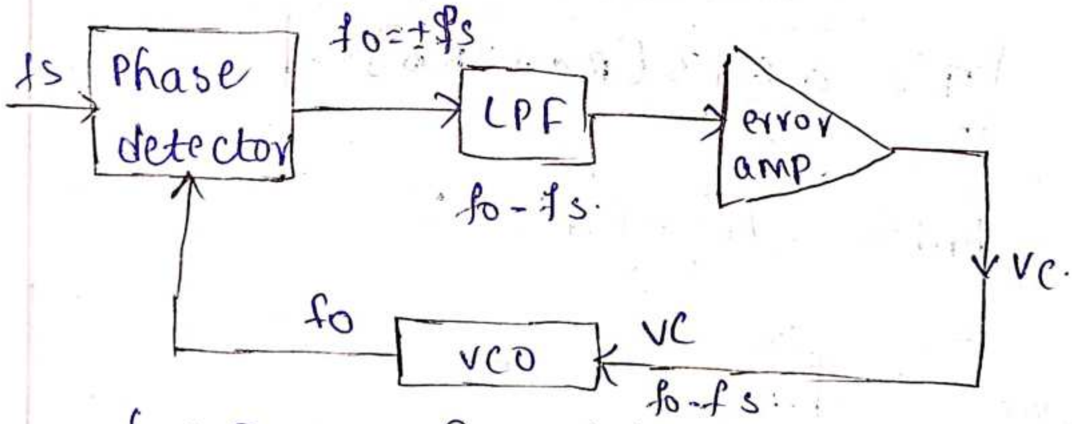
Schmitt Trigger







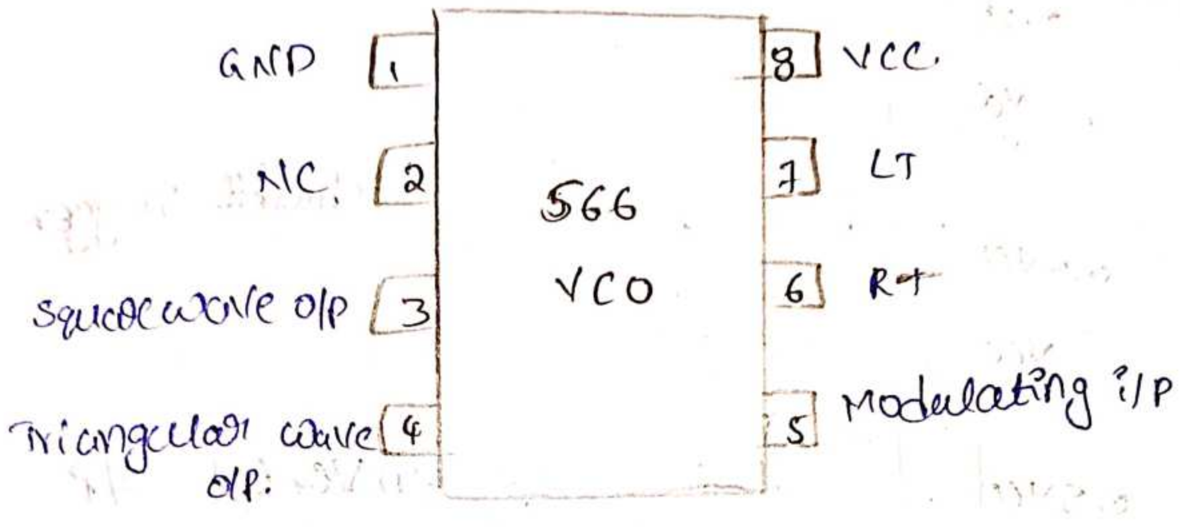
Phase locked loop.

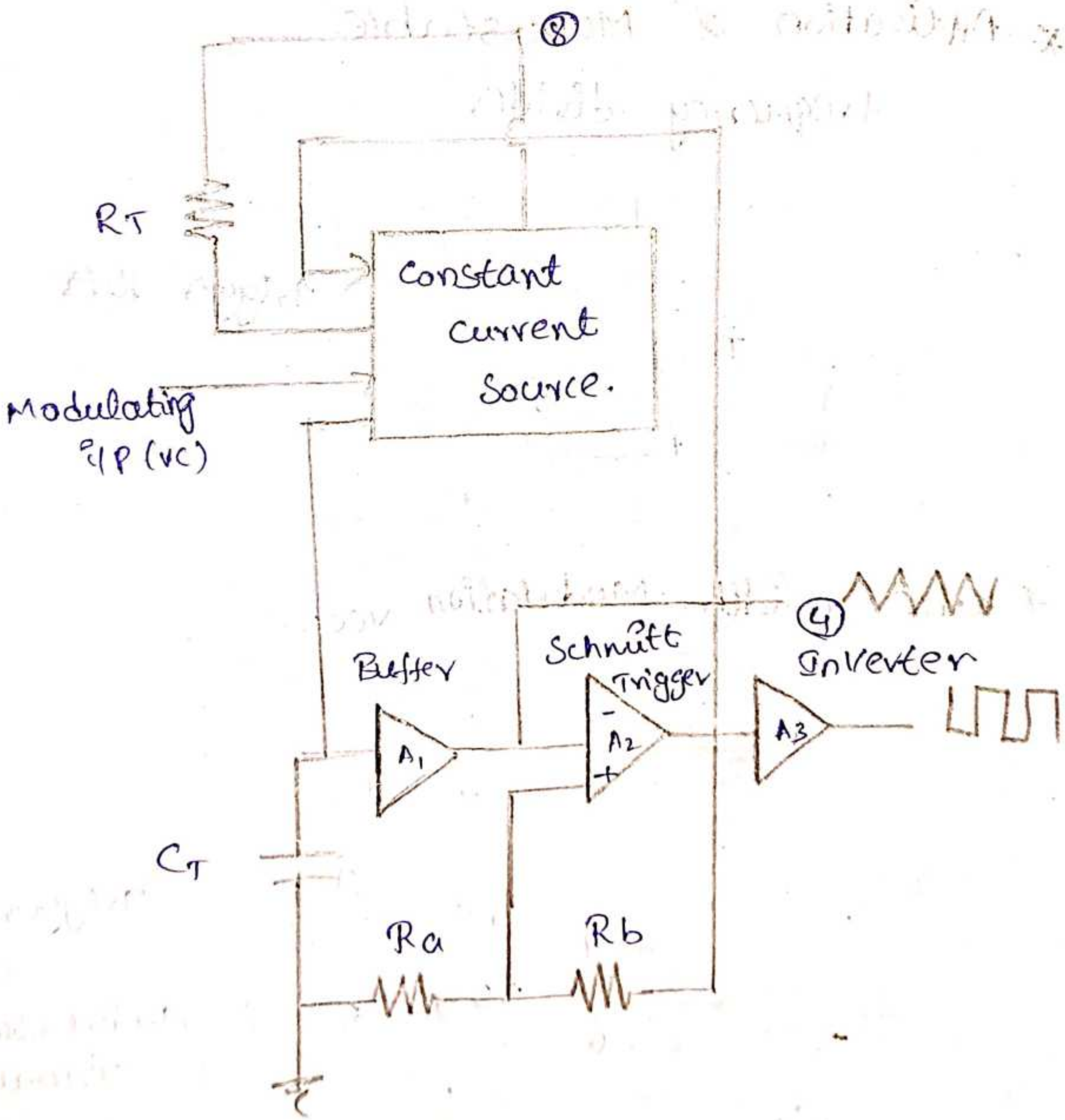


fo → Free running state

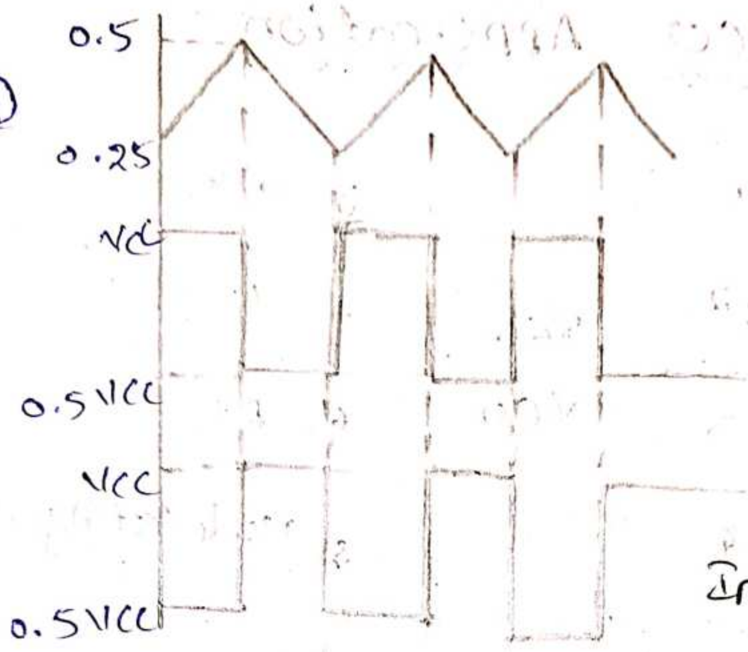
10/24.

* VCO Application.





(4)

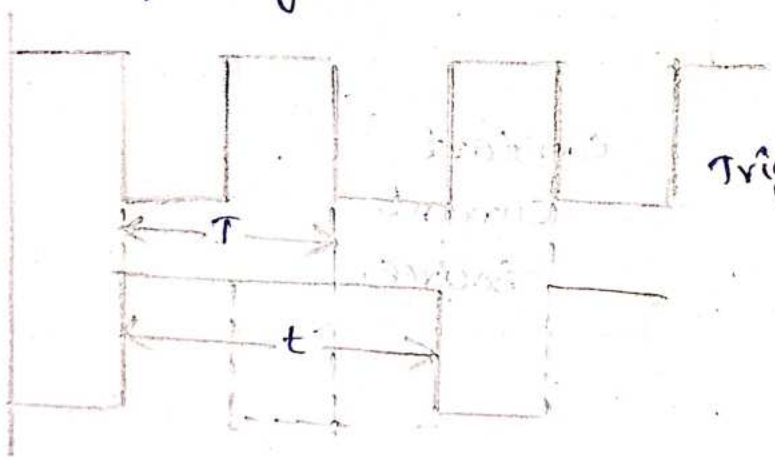


Schmitt Trigger o/p.

Inverted O/P

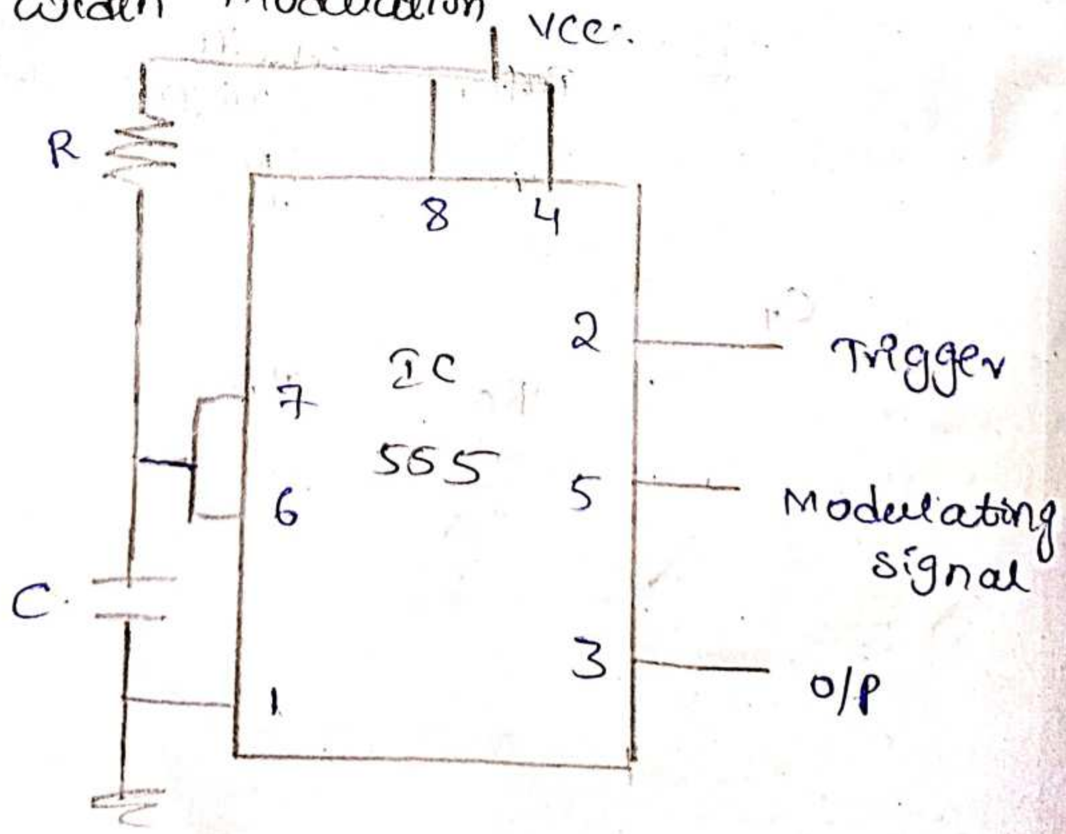
* Application of Monostable.

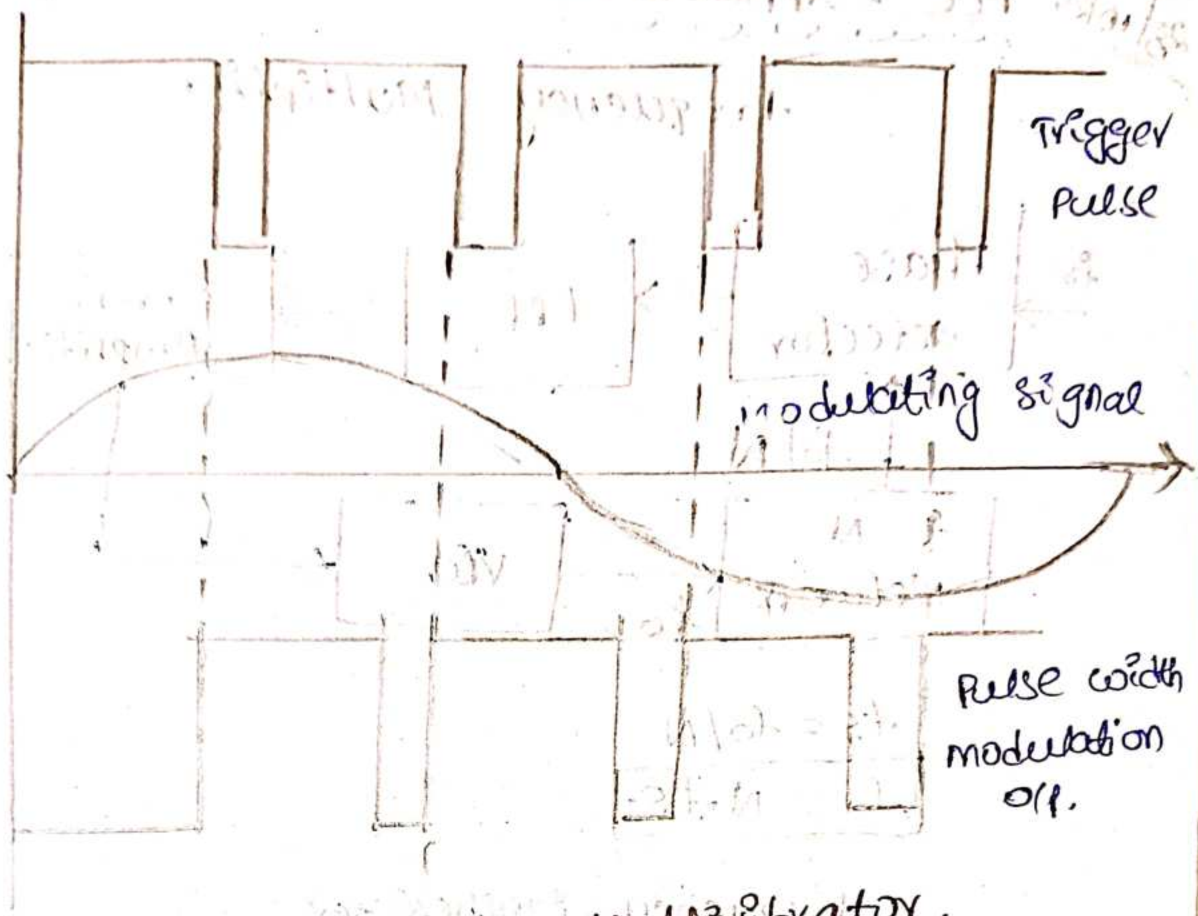
frequency divider



Trigger Pulse.

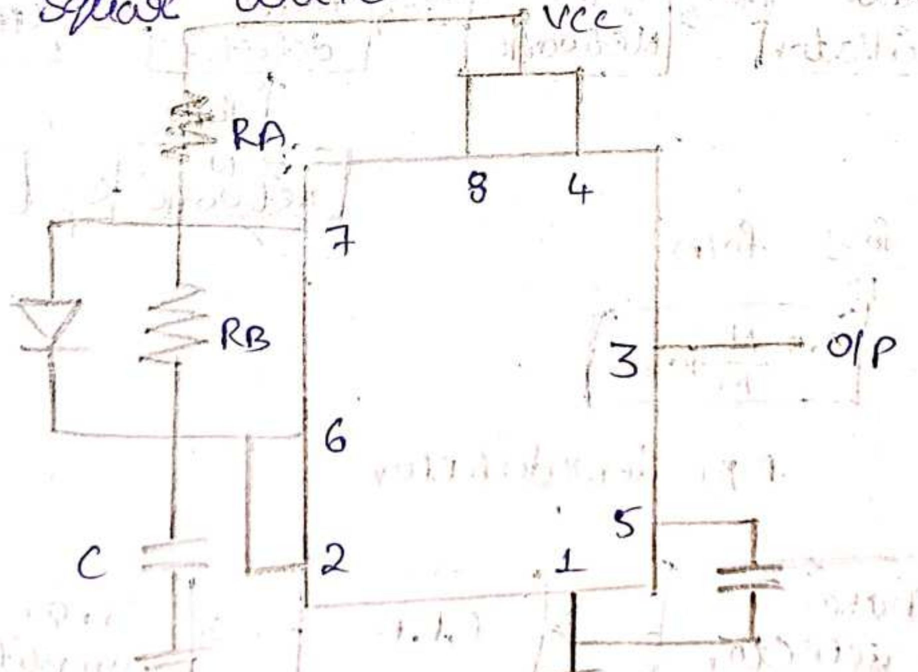
* Pulse Width Modulation

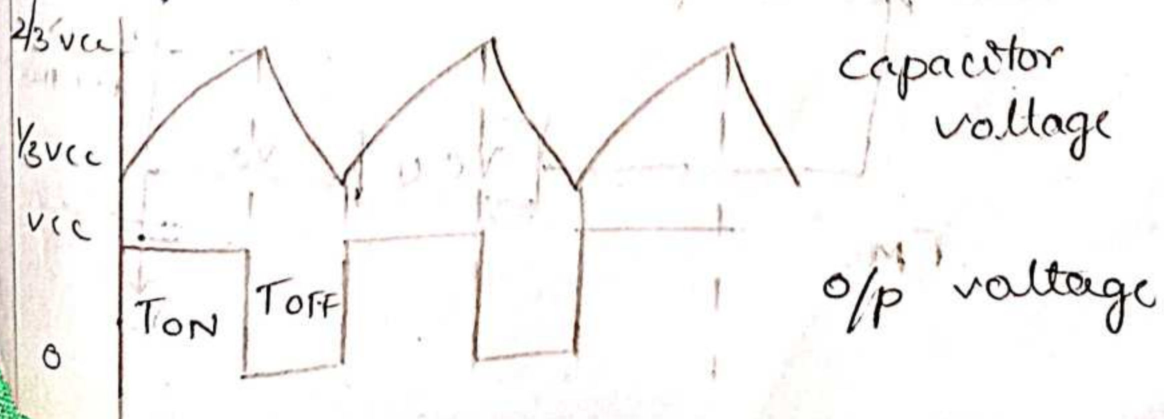




* Application of Astable Multivibrator.

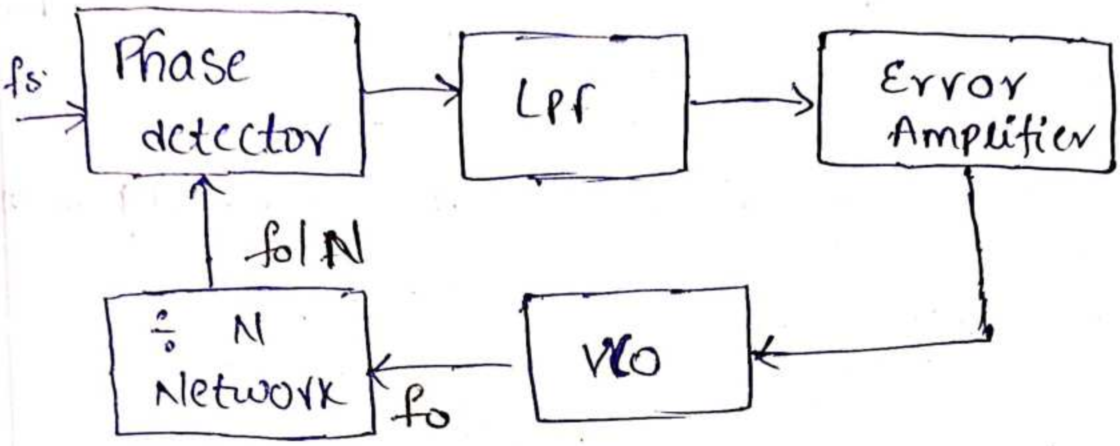
Square wave Generator.





28/10/24 PLL - Application

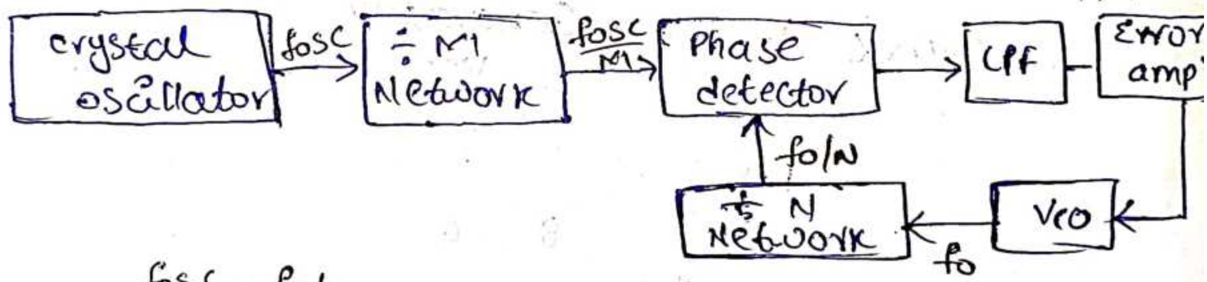
Frequency Multiplier



$fs = fo/N$

$\boxed{fo = N \cdot fs}$

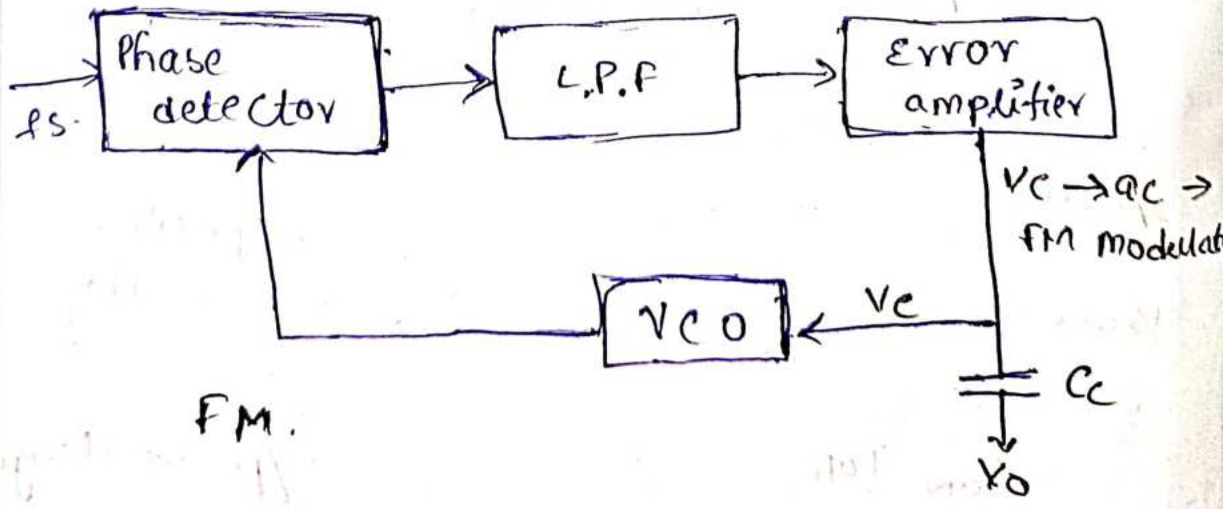
Frequency synthesizer.



$\frac{fosc}{M} = fo/N$

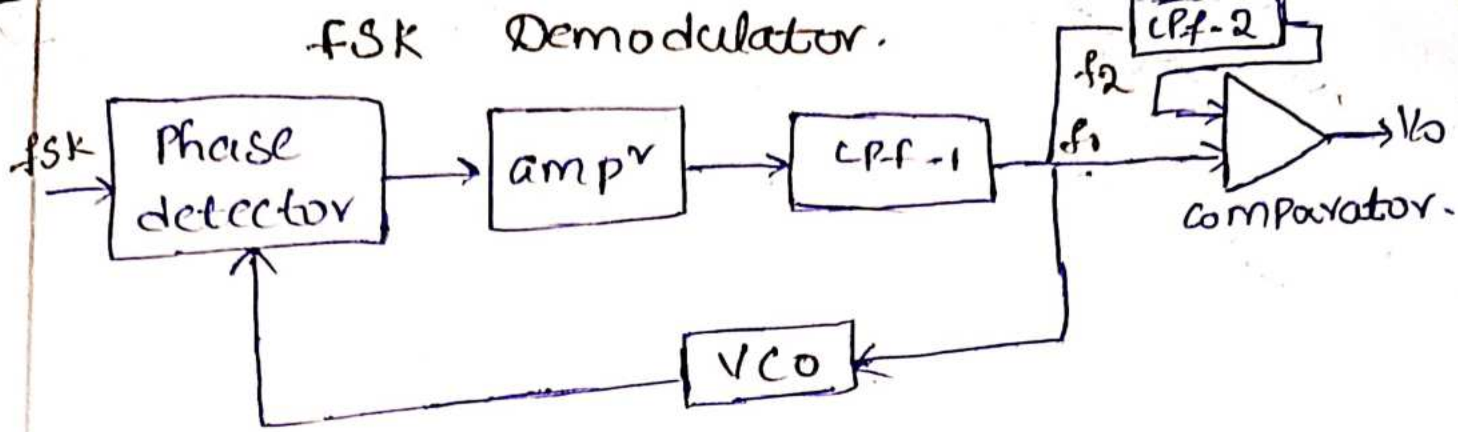
$\boxed{fo = \frac{N}{M} fosc}$

FM demodullator.



FM.

FSK Demodulator.



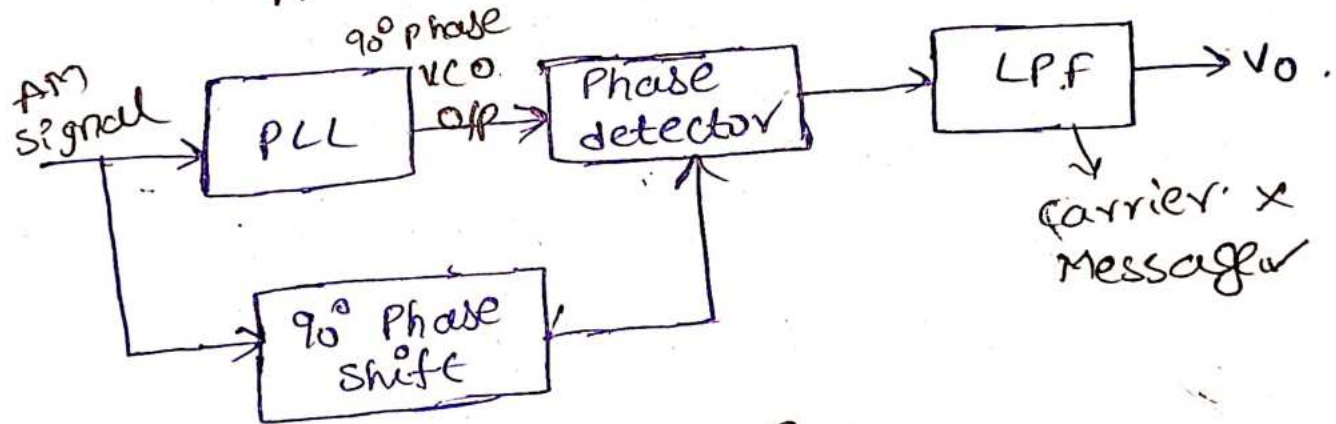
$VC_1 = f_1 - f_0$

$VC_2 = f_2 - f_0$

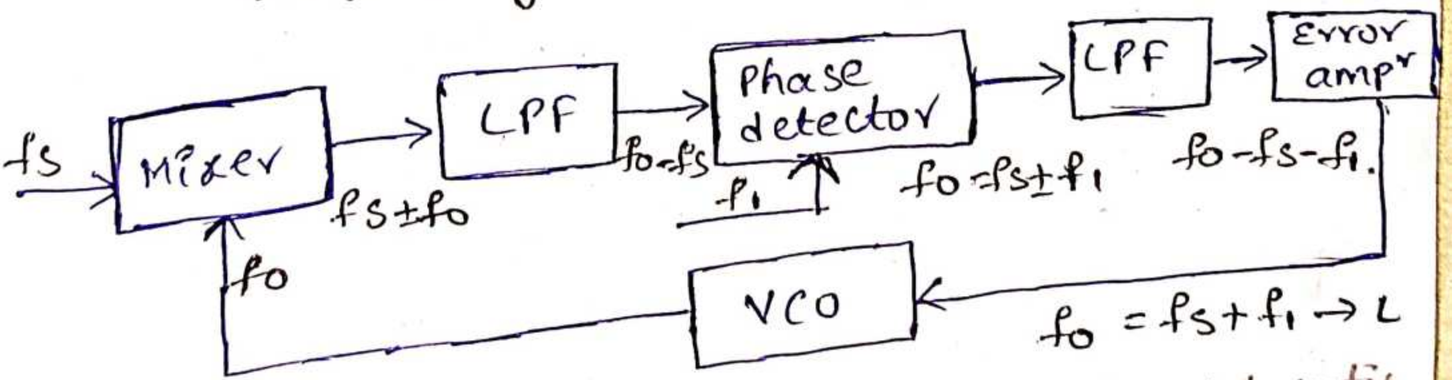
$\Delta VC = VC_2 - VC_1$

$\boxed{\Delta VC = f_2 - f_1}$

AM Demodulator.



Frequency translation.



loop is repeated until VCO is equal to, $f_0 = f_s + f_1$.

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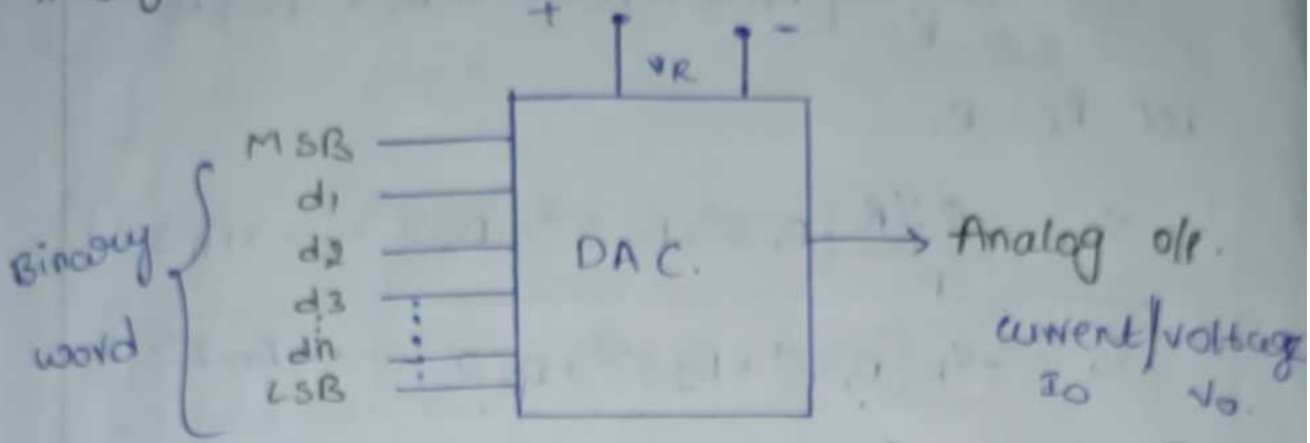
Digital to Analog Converter & Analog to Digital Converter

* Digital to Analog Convertor:



$$V_o = K \cdot V_{FS}\left[D_1 \cdot 2^{-1} + D_2\, 2^{-2} + D_3 \cdot 2^{-3} + \dots + D_n\, 2^{-n}\right] \rightarrow (1)$$

$V_o \rightarrow$ output voltage.

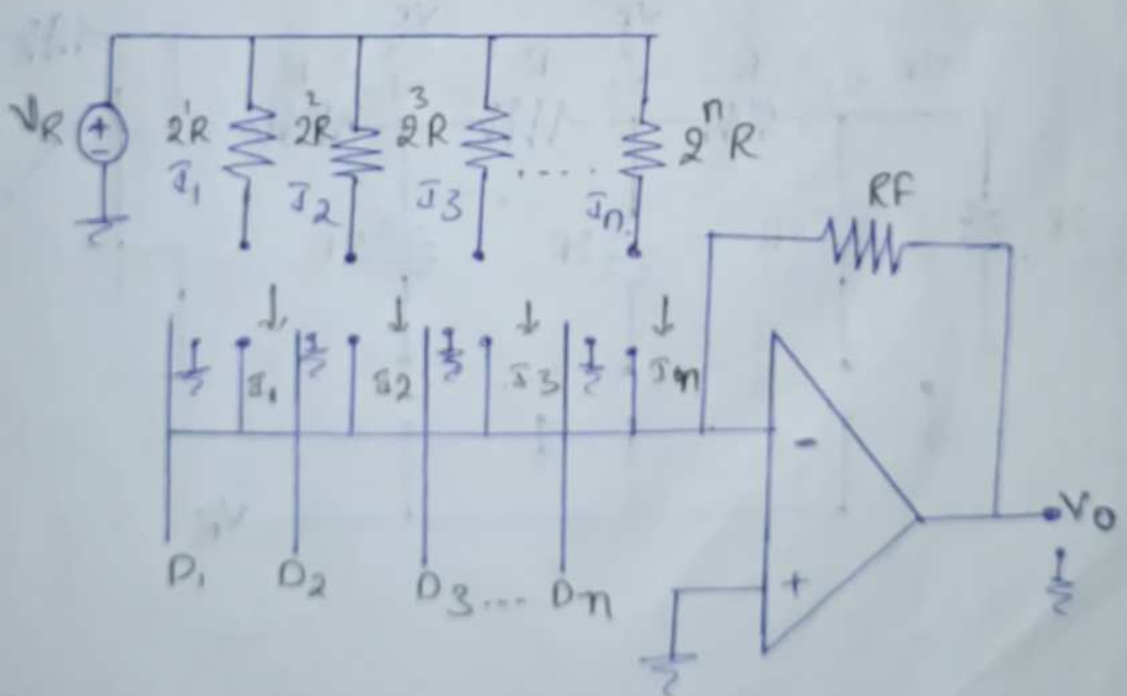
$K \rightarrow$ scaling factor $K = 1$.

$V_{FS} \rightarrow$ full scale voltage.

$D_1 \rightarrow$ MSB.

$D_n \rightarrow$ LSB.

* Binary weighted Resistor.



$I_0 = I_1 + I_2 + I_3 + \dots I_n$

$V_0 = -I_0 \cdot R_f$. $\quad I_0 = \frac{V_0}{R_f}$

$$V_o = -R_f\left[I_1 + I_2 + I_3 + \cdots + I_n\right]$$

$$= -R_f\left[D_1 \cdot \frac{V_R}{2R} + D_2 \cdot \frac{V_R}{2^2 R} + D_3 \cdot \frac{V_R}{2^3 R} + \cdots D_n \cdot \frac{V_R}{2^n R}\right]$$

$$= \frac{-R_f \cdot V_R}{R}\left[D_1 \cdot 2^{-1} + D_2 \cdot 2^{-2} + D_3 \cdot 2^{-3} + \cdots + D_n \cdot 2^{-n}\right]$$

let $R_f = R$.

$$V_o = \frac{-R \cdot V_R}{R}\left[D_1 \cdot 2^{-1} + D_2 \cdot 2^{-2} + D_3 \cdot 2^{-3} + \cdots + D_n \cdot 2^{-n}\right]$$

$$\boxed{V_o = -V_R\left[D_1 \cdot 2^{-1} + D_2 \cdot 2^{-2} + D_3 \cdot 2^{-3} + \cdots + D_n \cdot 2^{-n}\right]}$$

Advantages :-

1. simple Construction.
2. fast Convertion.

Disadvantages :-

1. Require large range of resistors.
2. Require low switch resistor in transistor.
3. More cost.

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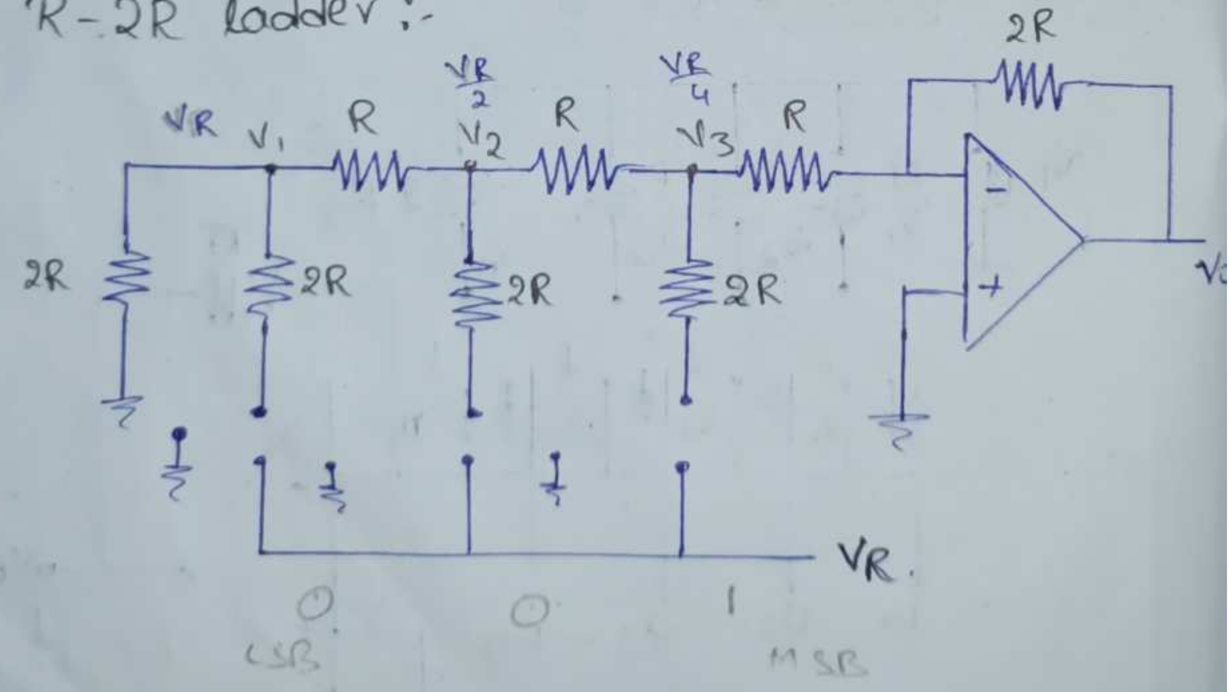
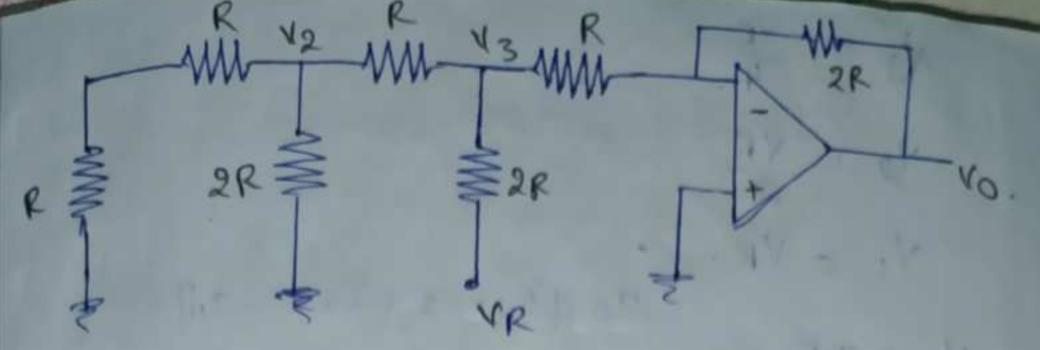
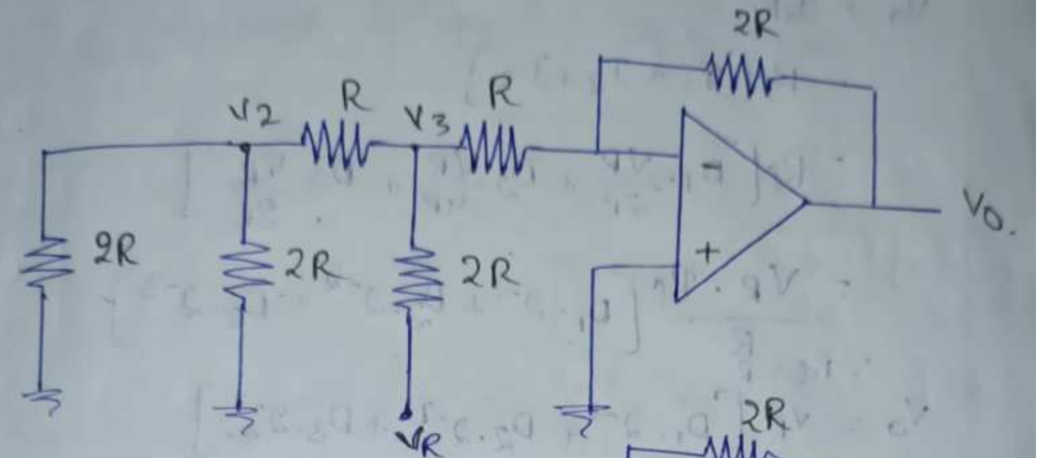
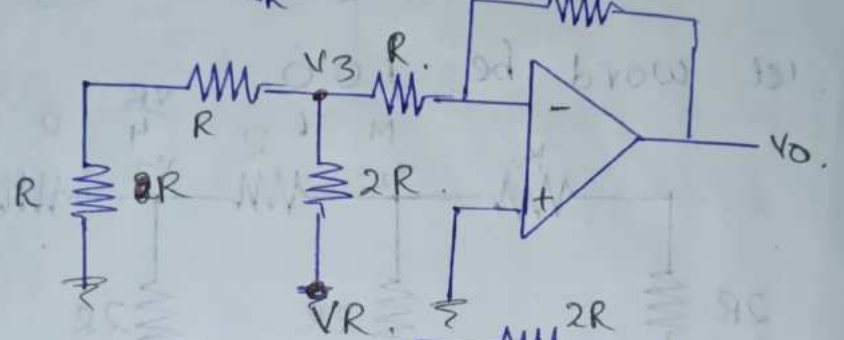
* R-2R ladder :-

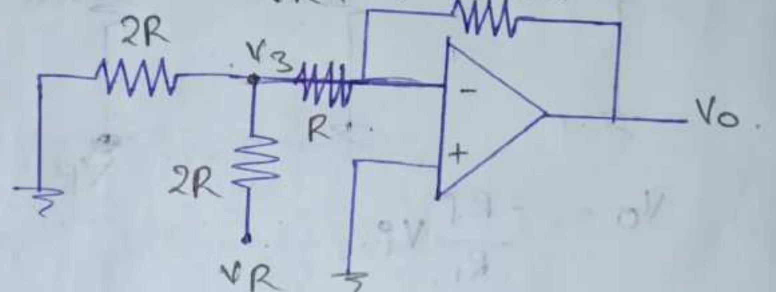


1 0 0 = 4.









By voltage division rule,

$$V_3 = \frac{V_R \cdot \frac{2R}{3}}{2R + \frac{2R}{3}}$$

$$= \frac{\frac{2R \cdot V_R}{\cancel{3}}}{\frac{6R+2R}{\cancel{3}}}$$

$$= \frac{\cancel{2}RV_R}{_4\cancel{8}R}$$

$2R \parallel R$

$$\frac{2R \cdot R}{2R+R} = \frac{2R^{\cancel{2}}}{3\cancel{R}}$$

$$= \frac{2R}{3}.$$

$V_3 = \frac{V_R}{4}$

$V_2 = \frac{V_R}{2}$

$V_1 = V_R$

$I_0 = [I_1 + I_2 + I_3 + \dots I_n]$

$V_0 = I_0 R_f$

$= R_f [I_1 + I_2 + I_3]$

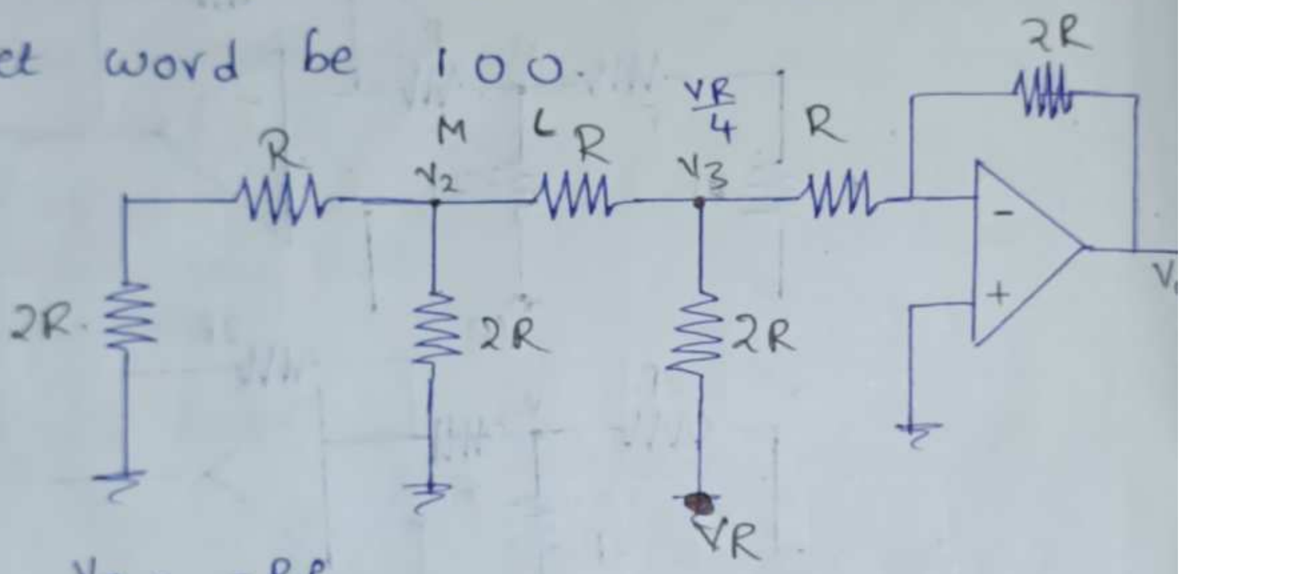
$= R_f \left[D_1 \cdot \frac{V_R}{2R} + D_2 \frac{V_R}{4R} + D_3 \frac{V_R}{8R} \right]$

$= \frac{V_R \cdot R_f}{R} \left[D_1 \cdot 2^{-1} + D_2 \cdot 2^{-2} + D_3 \cdot 2^{-3} \right]$

$\because R_f = R$

$V_0 = V_R \left[D_1 \cdot 2^{-1} + D_2 \cdot 2^{-2} + D_3 \cdot 2^{-3} + \dots \right]$

let word be 100.



$V_0 = \frac{-R_f}{R_1} V_i$

$= \frac{-2R}{\frac{V_R}{4}}$

$= \frac{-2R}{R} \times \frac{V_R}{4}$

$\boxed{V_0 = \frac{-V_R}{2}}$

$V_0 = V_R \left[D_1 \cdot 2^{-1} + D_2 \cdot 2^{-2} + D_3 \cdot 2^{-3} \right]$

$\frac{V_R}{} = V_R \left[0 + 0 + (1) 2^{-3} \right]$

$\frac{1}{2} = \frac{1}{8}$

$\Rightarrow$ "$\frac{1}{4}$" [$\because$ Inverting Terminal]

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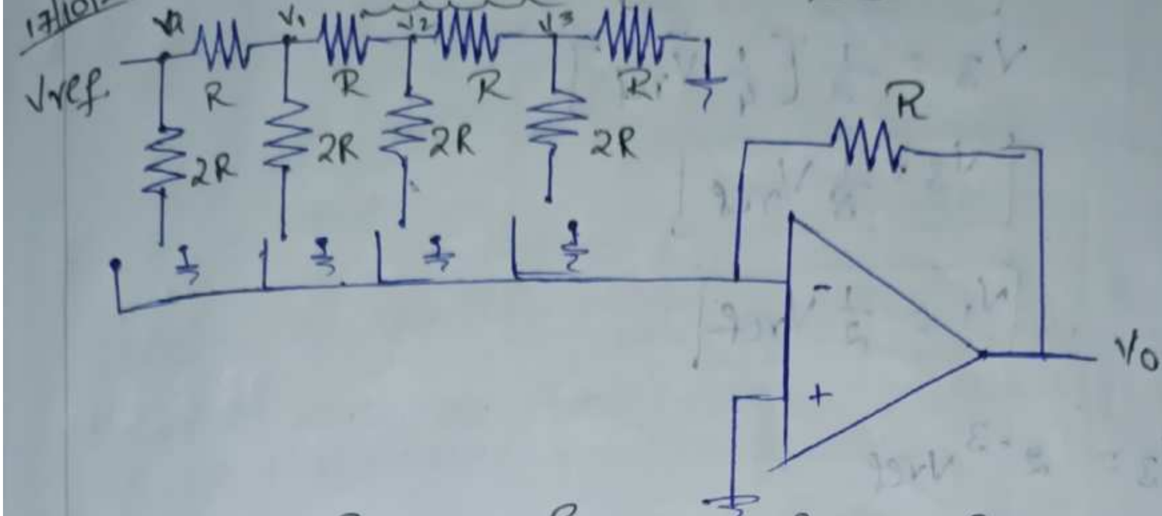
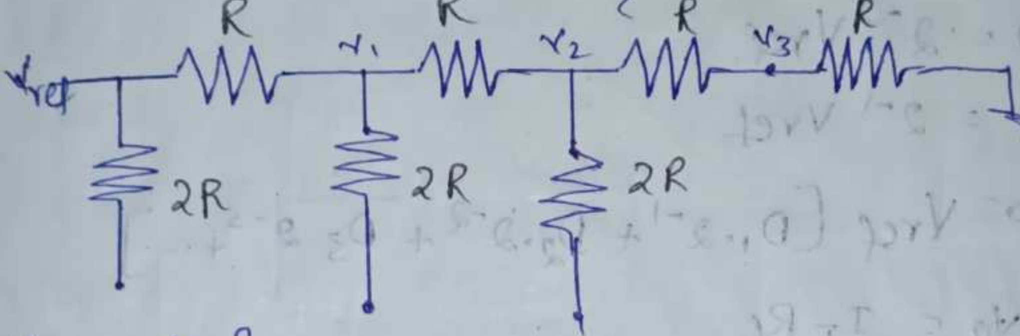
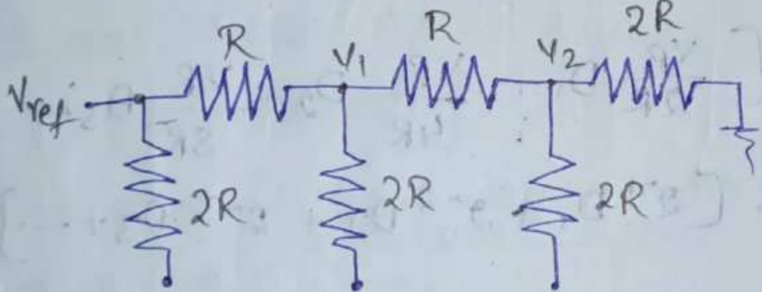
"Inverted R-2R" Ladder



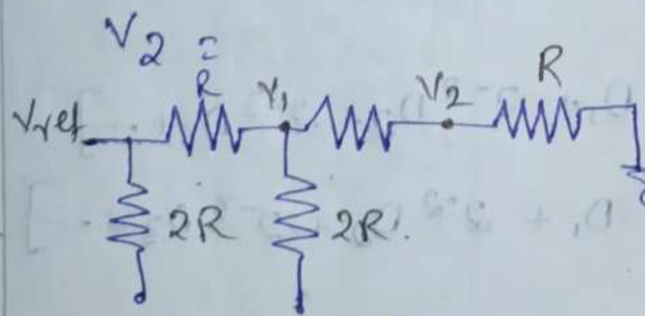


$V_3 = \frac{V_2 . R}{2R}$

$\therefore V_3 = \frac{1}{2} V_2$.



$V_2 =$



$V_2 = \frac{V_1 \cancel{R}}{2\cancel{R}}$

$\therefore V_2 = \frac{1}{2} V_1$

Sub v_1 value in v_2.

$$V_2 = \frac{1}{2}\left[\frac{1}{2} V_{ref}\right]$$

$$\boxed{V_2 = \frac{1}{4} V_{ref}}$$

$$V_3 = \frac{1}{2}\left[\frac{1}{4} V_{ref}\right]$$

$$\boxed{V_3 = \frac{1}{8} V_{ref}}$$

$$\boxed{V_1 = \frac{1}{2} V_{ref}}$$

$V_3 = 2^{-3} V_{ref}$

$V_2 = 2^{-2} V_{ref}$

$V_1 = 2^{-1} V_{ref}$

$V_0 = V_{ref}\left[D_1 \cdot 2^{-1} + D_2 \cdot 2^{-2} + D_3 \cdot 2^{-3} + \cdots\right]$

$V_0 = I_T R_f$

$I_T = I_1 + I_2 + I_3 + \cdots$

$V_0 = \left[I_1 + I_2 + I_3 + \cdots\right] R_f$

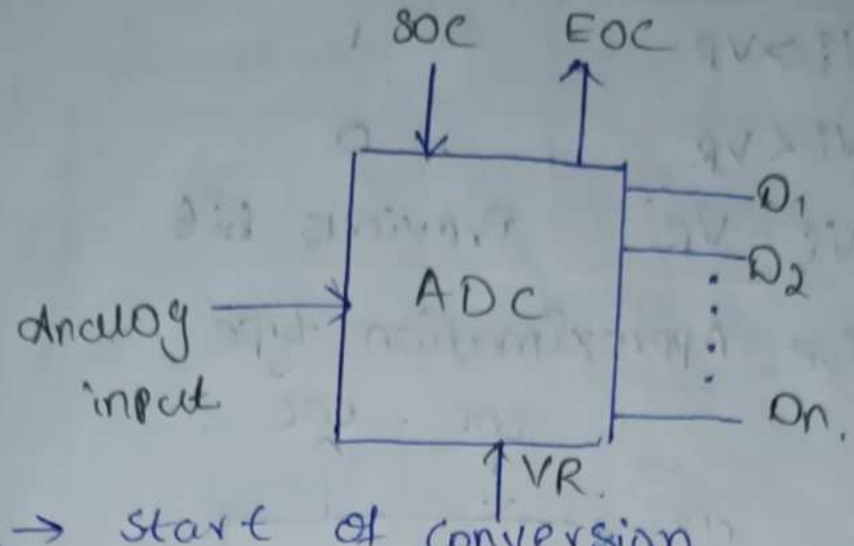
$V_0 = R_f\left[\frac{V_R}{2R} D_1 + \frac{V_R}{4R} D_2 + \frac{V_R}{8R} D_3 + \cdots\right]$

$V_0 = \frac{V_R R_f}{R}\left[2^{-1} D_1 + 2^{-2} D_2 + 2^{-3} D_3 + \cdots\right]$

if $R_f = R$

$V_0 = \frac{V_R \cdot R}{R}\left[2^{-1} D_1 + 2^{-2} D_2 + 2^{-3} D_3 + \cdots\right]$

$V_0 = V_R\left[2^{-1} D_1 + 2^{-2} D_2 + 2^{-3} D_3 + \cdots\right]$.



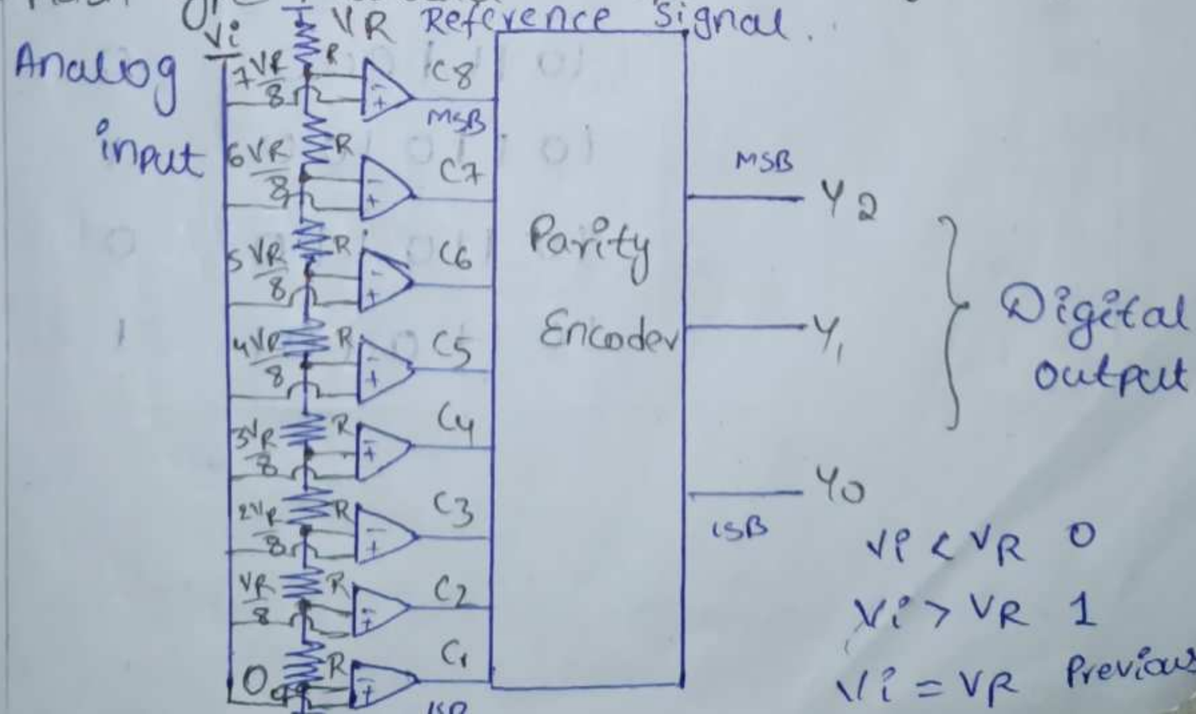
SOC → Start of conversion.
EOC → End of Conversion.
VR → Reference Voltage.

ADC is classified into 2 types.

1. Direct type.
→ Flash type / Parallel comporator type ADC
→ Counter type.
→ Successive approximation type.
→ Servo / tracking type.

2. Integrating type
→ Dual slope ADC
→ charge Balancing

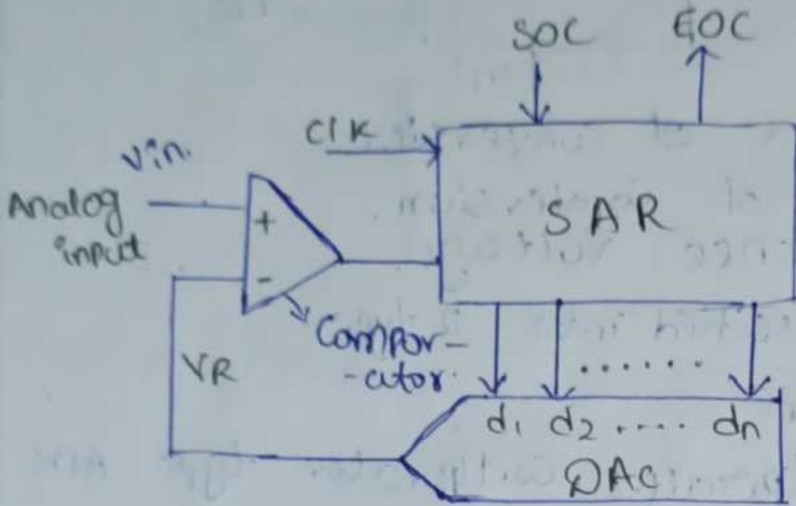
1. Flash type / Parallel comporator type ADC.



$V_i < V_R$ 0
$V_i > V_R$ 1
$V_i = V_R$ Previous.

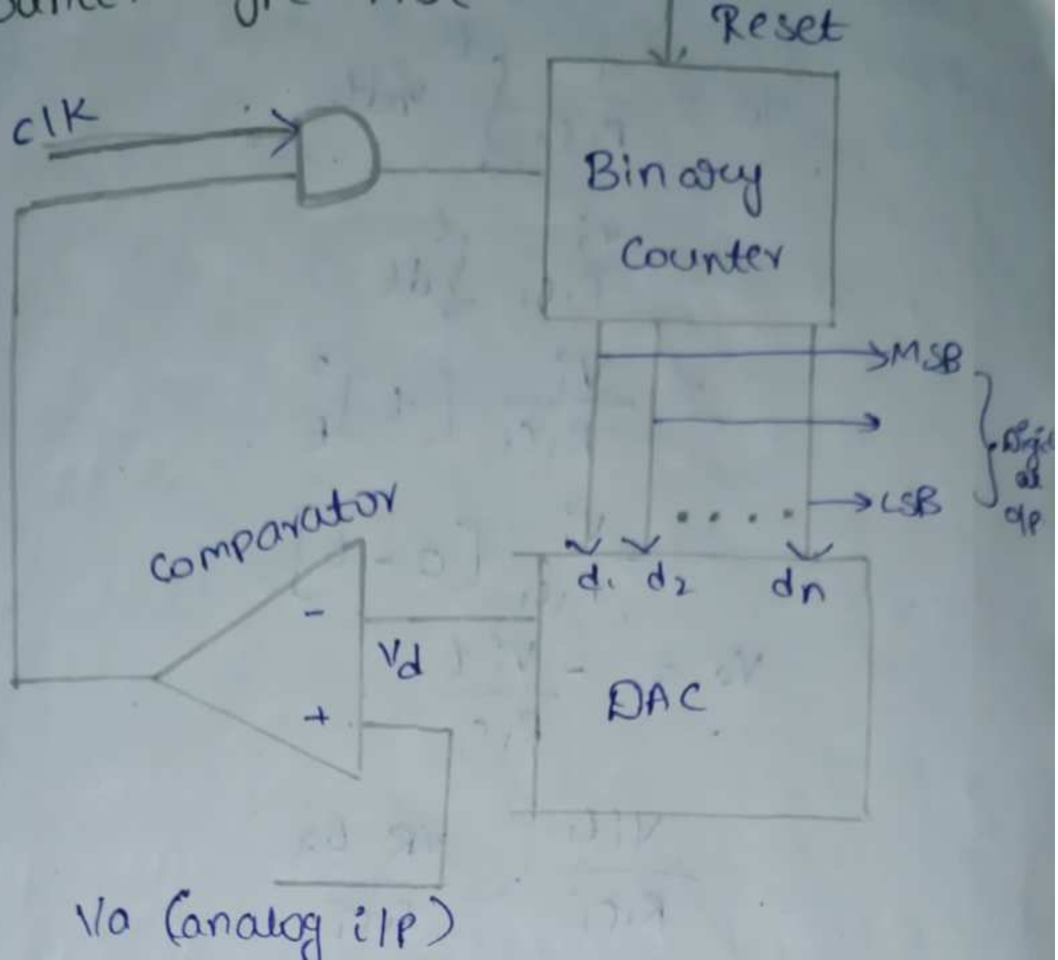
Analog input	Digital output
$V_i > V_R$	1
$V_i < V_R$	0
$V_i = V_R$	Previous bit

2. Successive Approximation type

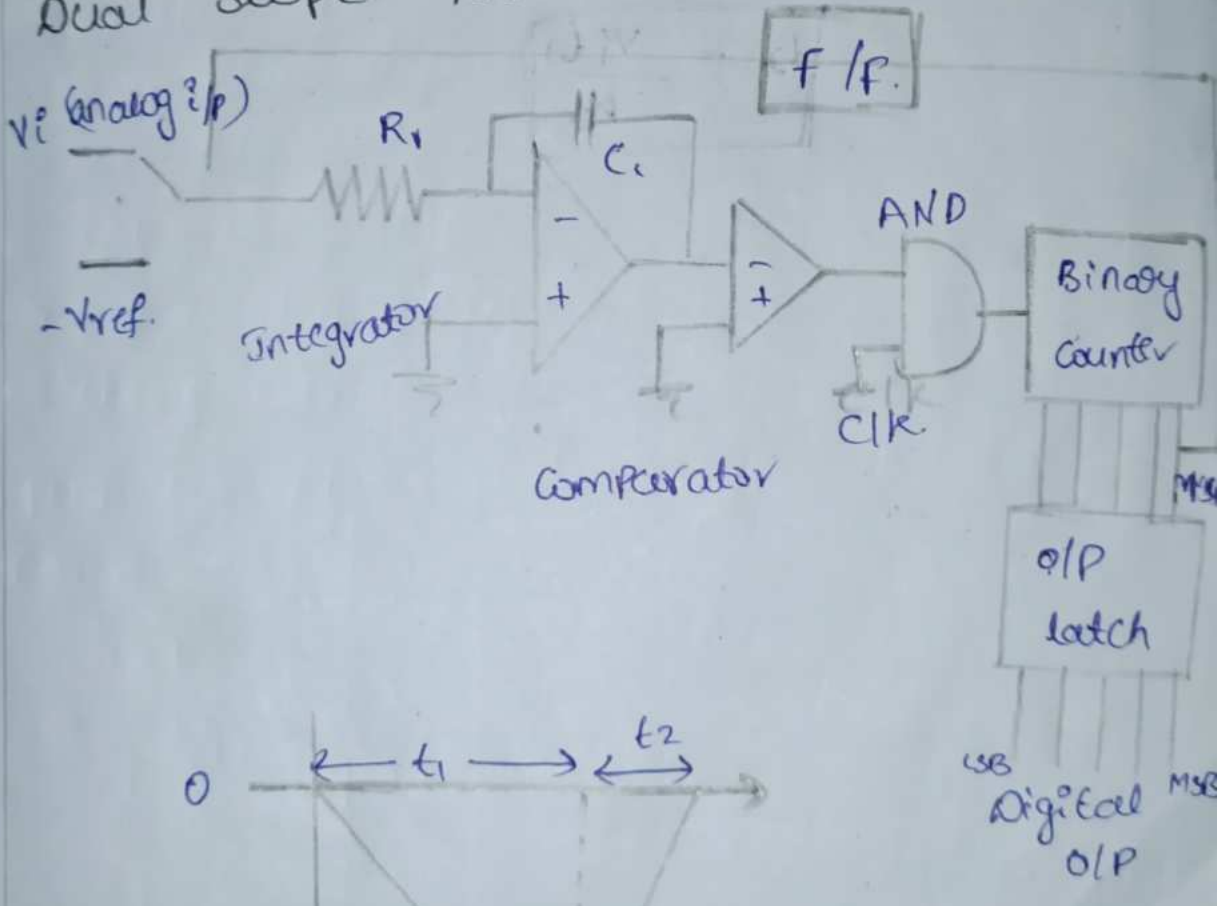


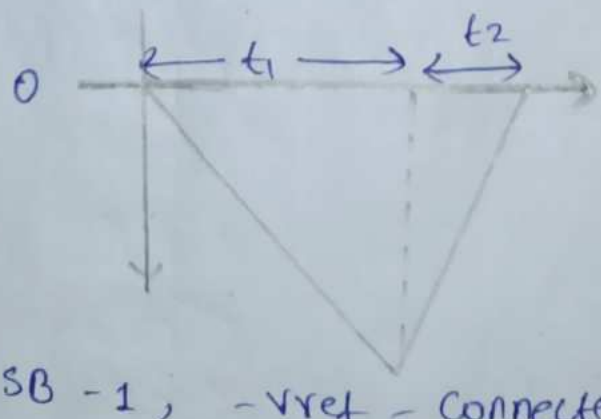
Digital Equivalent signal	S.A.R output	Comparator output
10110101	1000 0000	1
	11000000	0
	10100000	1
	10110000	1
	10111000	0
	10110100	1
	10110110	0
	10110101	1

3. Counter type ADC



Dual slope ADC





MSB - 1, - Vref - Connected to R_1.

$$V_0 = \frac{1}{R_1 C_1} \int_0^t v_i dt$$

$$= \frac{v_i}{R_1 C_1} \int_0^t dt$$

$$= \frac{V_i}{R_1 C_1} \left[t \right]_t^0$$

$$= \frac{V_i}{R_1 C_1} \left[0 - t \right]$$

$$V_0 = -\frac{V_i t}{R_1 C_1}$$

$$\frac{V_i t_1}{R_1 C_1} = \frac{V_R t_2}{\cancel{R_1 C_1}}$$

$$v_i t_i = V_R t_2$$

$$\boxed{t_2 = \frac{V_i t_1}{V_R}}$$